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Ultragenyx Pharmaceutical Inc.*

**UNITED STATES DISTRICT COURT
DISTRICT OF NEW JERSEY**

ULTRAGENYX PHARMACEUTICAL INC.,

Plaintiff,

v.

NAVINTA LLC, AUROBINDO PHARMA
LIMITED, AUROBINDO PHARMA USA,
INC., ESJAY PHARMA PRIVATE LIMITED,
and ESJAY PHARMA LLC,

Defendants.

Civil Action No.

(Electronically Filed)

COMPLAINT FOR PATENT INFRINGEMENT

Plaintiff Ultragenyx Pharmaceutical Inc. (“Ultragenyx” or “Plaintiff”), by its undersigned attorneys, brings this Complaint against Defendants Navinta LLC (“Navinta”); Aurobindo Pharma Limited (“APL”) and Aurobindo Pharma USA, Inc. (“Aurobindo USA”) (APL and Aurobindo USA, together, “Aurobindo”); and Esjay Pharma Private Limited (“EPPL”) and Esjay Pharma LLC (“Esjay Pharma”) (EPPL and Esjay Pharma, together, “Esjay”) (Navinta, Aurobindo, and Esjay, collectively, “Defendants”), and hereby alleges as follows:

NATURE OF THE ACTION

1. This is an action for patent infringement under the patent laws of the United States, 35 U.S.C. § 100, *et seq.*, arising from the Defendants' filing of their respective Abbreviated New Drug Application ("ANDA"), Nos. 219480 ("Navinta's ANDA"), 219711 ("Aurobindo's ANDA"), and 219512 ("Esjay's ANDA"), with the U.S. Food and Drug Administration ("FDA") seeking approval to commercially manufacture, use, offer for sale, sell, and/or import purported generic versions of triheptanoin oral liquid, 100% w/w (DOJOLVI® (triheptanoin) Oral Liquid) prior to the expiration of U.S. Patent No. 12,551,461 ("the '461 patent").

THE PARTIES

2. Plaintiff Ultragenyx is a pharmaceutical company focused on treating rare diseases. Ultragenyx is involved in the research and development of novel products for the treatment of rare and ultra-rare genetic diseases for which there are typically no approved treatments and there is a high unmet medical need.

3. Ultragenyx is a corporation organized and existing under the laws of the State of Delaware with its principal place of business at 60 Leveroni Court, Novato, California 94949.

4. On information and belief, Defendant Navinta LLC is a company organized and existing under the laws of the State of New Jersey with a place of business at 1499 Lower Ferry Rd., Ewing, New Jersey 08618.

5. On information and belief, Defendant APL is a corporation organized and existing under the laws of India, having a principal place of business at Galaxy Floors 22-24 Plot No 1 Survey No 83/1 Hyderabad Knowledge City Raidurg Panmaktha Ranga Reddy District, Hyderabad, Telangana, 500 032, India.

6. On information and belief, Defendant Aurobindo USA is a corporation organized and existing under the laws of the State of Delaware with a principal place of business at 279 Princeton Hightstown Road, East Windsor, New Jersey 08520.

7. On information and belief, Aurobindo USA is a wholly owned subsidiary of APL.

8. On information and belief, Defendant Esjay Pharma is a company organized and existing under the laws of the State of Florida, having a principal place of business at 50 Millstone Road, Building 300, Suite No. 140, East Windsor, NJ 08520.

9. On information and belief, Defendant EPPL is a corporation organized and existing under the laws of India, having a principal place of business at Plot No G28 & 29, 12th Main Road, SIPCOT Industrial Park, Sriperumbudur Taluk, Irungattukottai, Tamil Nadu – 602 117, India.

10. On information and belief, Esjay Pharma is a subsidiary of EPPL.

THE PATENT-IN-SUIT

11. Ultragenyx is the owner and assignee of the '461 patent, entitled "Compositions of triglycerides and uses thereof" (attached hereto as Exhibit A).

12. On February 17, 2026, the '461 patent was duly and legally issued to Ultragenyx as assignee of inventors John Klopp, Gabrielle Morris, Emil Kakkis, and Steven Jungles.

13. Ultragenyx is the owner of all right, title, and interest in the '461 patent.

14. Ultragenyx submitted the '461 patent for listing in the FDA publication, "Approved Drug Products with Therapeutic Equivalence Evaluations" (the "Orange Book") on February 19, 2026.

THE DOJOLVI® DRUG PRODUCT

15. Ultragenyx holds an approved New Drug Application ("NDA") under Section 505(a) of the Federal Food, Drug, and Cosmetic Act ("FDCA"), 21 U.S.C. § 355(a), for triheptanoin oral liquid (NDA No. 213687), which it sells under the trade name DOJOLVI®.

DOJOLVI[®] is an FDA-approved medication for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

16. Pursuant to 21 U.S.C. § 355(b)(1) and attendant FDA regulations, the '461 patent is listed in the Orange Book with respect to DOJOLVI[®].

17. The claims of the '461 patent cover, *inter alia*, pharmaceutical compositions of triheptanoin.

18. The FDA's Orange Book for DOJOLVI[®] currently lists the expiration of the '461 patent as November 14, 2034.

JURISDICTION AND VENUE

19. Plaintiff incorporates each of the preceding paragraphs 1–18 as if fully set forth herein.

20. This is a civil action for patent infringement arising under the patent laws of the United States, 35 U.S.C. § 100, et seq., including 35 U.S.C. § 271, for infringement of the '461 patent. This Court has jurisdiction over the subject matter of this action pursuant to 28 U.S.C. §§ 1331, 1338(a), 2201, and 2202.

21. Venue is proper in this Judicial District pursuant to 28 U.S.C. §§ 1391 and 1400(b).

a. Personal Jurisdiction: Navinta

22. This Court has personal jurisdiction over Navinta by virtue of, *inter alia*, its systematic and continuous contacts with the State of New Jersey. On information and belief, Navinta is a company organized and existing under the laws of the State of New Jersey, and Navinta's principal place of business is in Ewing, New Jersey. On information and belief, Navinta is registered with the State of New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0600139277. On information and belief,

Navinta purposely has conducted and continues to conduct business in this Judicial District. By virtue of its physical presence in New Jersey, this Court has personal jurisdiction over Navinta.

23. This Court has personal jurisdiction over Navinta because Navinta is a New Jersey company with a principal place of business in Ewing, New Jersey.

24. On information and belief, Navinta is registered with the State of New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0600139277.

25. Navinta is also subject to personal jurisdiction in New Jersey because, *inter alia*, it has purposely availed itself of the benefits and protections of New Jersey's laws such that it should reasonably anticipate being sued in this Judicial District. On information and belief, Navinta develops, manufactures, imports, markets, distributes, uses, offers to sell, and/or sells generic drugs throughout the United States, including in New Jersey, and therefore transacts business within New Jersey related to Plaintiff's claims, and/or has engaged in systematic and continuous business contacts within New Jersey.

26. By a letter dated August 14, 2024 ("Navinta Notice Letter"), Navinta notified Plaintiff that Navinta had submitted Navinta's ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w) ("Navinta's ANDA Product"), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Navinta's ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the '461 patent.

27. Navinta has committed an act of infringement in this Judicial District by filing ANDA No. 219480 with the intent to make, use, sell, offer for sale, and/or import Navinta's ANDA Product in or into this Judicial District, prior to the expiration of the '461 patent.

28. Navinta has taken the costly, significant step of applying to the FDA for approval to engage in future activities, including the marketing of the Navinta's ANDA Product, which will be purposefully directed at New Jersey and elsewhere in the United States.

29. On information and belief, if Navinta's ANDA is approved, Navinta will manufacture, market, promote, sell, offer for sale, import, use, and/or distribute Navinta's ANDA Product within the United States, including in New Jersey. On information and belief, Navinta's ANDA Product will be prescribed by physicians practicing in New Jersey, dispensed by pharmacies located within New Jersey, and used by patients in New Jersey. Each of these activities would have a substantial effect within New Jersey and would constitute infringement of the '461 patent in the event that Navinta's ANDA Product is approved before the expiration of the '461 patent.

30. Navinta did not contest personal jurisdiction or venue in this Judicial District in *Ultragenyx Pharmaceutical Inc., et al. v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

31. Venue is proper in this Court as to Navinta pursuant to 28 U.S.C. § 1400(b) because Navinta is organized and exists under the laws of the State of New Jersey and therefore resides in this Judicial District.

b. Personal Jurisdiction: Aurobindo

32. This Court has personal jurisdiction over APL by virtue of, *inter alia*, its systematic and continuous contacts with the State of New Jersey. APL itself, and through its wholly owned subsidiary Aurobindo USA, purposely availed itself of the benefits and protections of New Jersey's laws such that it should reasonably anticipate being sued in this Judicial District. On information and belief, APL itself, and through its wholly owned subsidiary Aurobindo USA, develops, manufactures, imports, markets, distributes, uses, offers to sell, and/or sells generic drugs

throughout the United States, including in New Jersey, and therefore transacts business within New Jersey related to Plaintiff's claims, and/or has engaged in systematic and continuous business contacts within New Jersey. In addition, APL is subject to personal jurisdiction in New Jersey because, on information and belief, it controls and dominates Aurobindo USA, and therefore the activities of Aurobindo USA in this jurisdiction are attributed to APL.

33. On information and belief, Aurobindo USA has a principal place of business in East Windsor, New Jersey and also operates a distribution center at 6 Wheeling Road, Dayton, New Jersey 08810.

34. By a letter dated August 22, 2024 ("Aurobindo Notice Letter"), Aurobindo notified Plaintiff that Aurobindo had submitted Aurobindo's ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w) ("Aurobindo's ANDA Product"), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Aurobindo's ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the '461 patent.

35. APL, in concert with Aurobindo USA, has committed an act of infringement in this Judicial District by filing ANDA No. 219711 with the intent to make, use, sell, offer for sale, and/or import Aurobindo's ANDA Product in or into this Judicial District, prior to the expiration of the '461 patent.

36. APL's Annual Report from 2024-25 states that it is the "#1 largest generic pharma company in the US by Rx dispensed" and that the US accounted for a "47%" "Share of Consolidated Revenue." See Aurobindo Annual Report 2024-25 ("Aurobindo Annual Report"), at 8, 15 *available at* <https://www.aurobindo.com/api/uploads/annualreports/AurobindoPharmaIR2025-website.pdf>

(last visited April 3, 2026). The Aurobindo Annual Report states that Aurobindo has an “oral facility in Dayton, New Jersey which will commercialise [sic] in FY26” and which “is designed to cater the growing demand for oral pharmaceutical products in the US.” *Id.* at 75, 97. On information and belief, through its own actions and through the actions of its agents, affiliates, and subsidiaries, APL filed 31 ANDAs with the US FDA” during fiscal year 2024-25 ” *Id.* at 101.

37. This Court has personal jurisdiction over Aurobindo USA because Aurobindo USA is a corporation with a principal place of business in East Windsor, New Jersey and at least one other regular and established place of business in Dayton, New Jersey.

38. Aurobindo USA is also subject to personal jurisdiction in New Jersey because, *inter alia*, it has purposely availed itself of the benefits and protections of New Jersey’s laws such that it should reasonably anticipate being sued in this Judicial District. On information and belief, Aurobindo USA develops, manufactures, imports, markets, distributes, uses, offers to sell, and/or sells generic drugs throughout the United States, including New Jersey, and therefore transacts business within New Jersey related to Plaintiff’s claims, and/or has engaged in systematic and continuous business contacts within New Jersey.

39. Aurobindo USA’s website states that its “global headquarters” is “Aurobindo Pharma Limited” in Telangana, India. *See* Corporate Offices, <https://www.aurobindousa.com/company/corporate-offices/> (last visited April 3, 2026).

40. On information and belief, APL and Aurobindo USA acted in concert to prepare and submit Aurobindo’s ANDA.

41. Aurobindo has taken the costly, significant step of applying to the FDA for approval to engage in future activities, including the marketing of Aurobindo’s ANDA Product, which will be purposefully directed at New Jersey and elsewhere in the United States.

42. On information and belief, Aurobindo has systematic and continuous contacts with New Jersey; has established distribution channels for drug products in New Jersey; regularly and continuously conducts business in New Jersey, including by selling drug products in New Jersey, either directly or indirectly through its subsidiaries, agents, or affiliates; has purposefully availed itself of the privilege of doing business in New Jersey; and derives substantial revenue from the sale of drug products in New Jersey.

43. On information and belief, if Aurobindo's ANDA is approved, Aurobindo will manufacture, market, promote, sell, offer for sale, import, use, and/or distribute Aurobindo's ANDA Product within the United States, including in New Jersey, consistent with Aurobindo's practices for the marketing and distribution of other generic pharmaceutical products. On information and belief, Aurobindo regularly does business in New Jersey, and its practices with other generic pharmaceutical products have involved placing those products into the stream of commerce for distribution throughout the United States, including in New Jersey. On information and belief, Aurobindo's generic pharmaceutical products are used and/or consumed within and throughout the United States, including in New Jersey. On information and belief, Aurobindo's ANDA Product will be prescribed by physicians practicing in New Jersey, dispensed by pharmacies located within New Jersey, and used by patients in New Jersey. Each of these activities would have a substantial effect within New Jersey and would constitute infringement of the '461 patent in the event that Aurobindo's ANDA Product is approved before the '461 patent expires.

44. On information and belief, APL is registered with the State of New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0100904116.

45. On information and belief, Aurobindo USA is registered with the State of New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0100921223.

46. On information and belief, Aurobindo derives substantial revenue from generic pharmaceutical products that are used and/or consumed within New Jersey, and that are manufactured by APL and/or for which Aurobindo USA is/are the named applicant(s) on approved ANDAs. On information and belief, various products for which APL and/or Aurobindo USA is/are the named applicant(s) on approved ANDAs are available at retail pharmacies in New Jersey.

47. On information and belief, Aurobindo has consented to jurisdiction and venue in this Judicial District in one or more prior cases arising out of the filing of its ANDAs, and/or has filed counterclaims in such cases. *See, e.g., Merck Sharp & Dohme B.V. v. Aurobindo Pharma USA, Inc., et al.*, No. 3:20-cv-02576 (D.N.J. Mar. 10, 2020); *Chesi USA, Inc., et al. v. Aurobindo Pharma USA, Inc., et al.*, No. 3:19-cv-18756-FLW-LHG (D.N.J. Oct. 7, 2019); *Actelion Pharm. Ltd. v. Aurobindo Pharma USA, Inc., et al.*, No. 3:19-cv-15437-FLW-LHG (D.N.J. July 16, 2019); *Celgene Corp. v. Aurobindo Pharma Ltd., et al.*, No. 2:19-cv-05799-ES-MAH (D.N.J. Feb. 14, 2019); *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

48. Additionally, this Court has personal jurisdiction over APL because the requirements of Federal Rule of Civil Procedure 4(k)(2)(A) are met as (a) Plaintiff's claims arise under federal law; (b) APL is a foreign defendant not subject to general personal jurisdiction in the courts of any state; and (c) APL has sufficient contacts in the United States as a whole, including, but not limited to, by participating in the preparation and submission of Aurobindo's ANDA, and/or manufacturing and/or selling pharmaceutical products distributed throughout the United

States, including in this Judicial District, such that this Court's exercise of jurisdiction over APL satisfies due process.

49. Venue is proper in this Court as to APL because APL is a foreign entity who may be sued in any Judicial District, including in the District of New Jersey. 28 U.S.C. § 1391(c)(3); *see also* 28 U.S.C. § 1400(b).

50. Venue is proper in this Judicial District pursuant to 28 U.S.C. § 1400(b) as to Aurobindo USA, because, on information and belief, Aurobindo USA has a regular and established place of business in New Jersey, and because, on information and belief, Aurobindo USA has committed or aided, abetted, contributed to, and/or participated in the commission of, acts of infringement of the asserted patent that will lead to foreseeable harm and injury to Plaintiff by preparing or assisting in preparing Aurobindo's ANDA in New Jersey and/or with the intention of seeking to market Aurobindo's ANDA Product nationwide, including within New Jersey.

c. Personal Jurisdiction: Esjay

51. This Court has personal jurisdiction over Esjay Pharma because, *inter alia*, it has purposely availed itself of the benefits and protections of New Jersey's laws such that it should reasonably anticipate being sued in this Judicial District. On information and belief, Esjay Pharma itself, and through its parent EPPL, develops, manufactures, imports, markets, distributes, uses, offers to sell, and/or sells generic drugs throughout the United States, including in New Jersey, and therefore transacts business within New Jersey related to Plaintiff's claims, and/or has engaged in systematic and continuous business contacts within New Jersey.

52. On information and belief, Esjay Pharma has a principal place of business in East Windsor, NJ.

53. This Court has personal jurisdiction over EPPL because, on information and belief, it controls and dominates Esjay Pharma, and therefore the activities of Esjay Pharma in this jurisdiction are attributed to EPPL.

54. By a letter dated August 27, 2024 (“Esjay First Notice Letter”), Esjay notified Plaintiff that Esjay had submitted Esjay’s ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w) (“Esjay’s ANDA Product”), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Esjay’s ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the ’461 patent.

55. By a letter dated September 23, 2024 (“Second Esjay Notice Letter”), Esjay notified Plaintiff that Esjay had submitted Esjay’s ANDA to the FDA for Esjay’s ANDA Product, seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Esjay’s ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the ’461 patent.

56. By a letter dated March 11, 2026 (“Third Esjay Notice Letter”), Esjay notified Plaintiff that Esjay had submitted Esjay’s ANDA to the FDA for Esjay’s ANDA Product, seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Esjay’s ANDA Product in or into the United States, including New Jersey, prior to the expiration of the ’461 patent.

57. Esjay Pharma, in concert with EPPL, has committed an act of infringement in this Judicial District by filing ANDA No. 219512 with the intent to make, use, sell, offer for sale, and/or import Esjay’s ANDA Product in or into this Judicial District, prior to the expiration of the ’461 patent.

58. This Court has personal jurisdiction over Esjay Pharma because Esjay Pharma has a principal place of business in East Windsor, New Jersey.

59. On information and belief, Esjay Pharma and EPPL acted in concert to prepare and submit Esjay's ANDA.

60. Esjay has taken the costly, significant step of applying to the FDA for approval to engage in future activities, including the marketing of Esjay's ANDA Product, which will be purposefully directed at New Jersey and elsewhere in the United States.

61. On information and belief, Esjay has systematic and continuous contacts with New Jersey; has established distribution channels for drug products in New Jersey; regularly and continuously conducts business in New Jersey, including by selling drug products in New Jersey, either directly or indirectly through its subsidiaries, agents, or affiliates; has purposefully availed itself of the privilege of doing business in New Jersey; and derives substantial revenue from the sale of drug products in New Jersey.

62. On information and belief, if Esjay's ANDA is approved, Esjay will manufacture, market, promote, sell, offer for sale, import, use, and/or distribute Esjay's ANDA Product within the United States, including in New Jersey. On information and belief, Esjay regularly does business in New Jersey, and its practices with other generic pharmaceutical products have involved placing those products into the stream of commerce for distribution throughout the United States, including in New Jersey. On information and belief, Esjay's generic pharmaceutical products are used and/or consumed within and throughout the United States, including in New Jersey. On information and belief, Esjay's ANDA Product will be prescribed by physicians practicing in New Jersey, dispensed by pharmacies located within New Jersey, and used by patients in New Jersey. Each of these activities would have a substantial effect within New Jersey and would constitute

infringement of the '461 patent in the event that Esjay's ANDA Product is approved before the '461 patent expires.

63. On information and belief, Esjay Pharma is registered with the State of New Jersey's Division of Revenue and Enterprise Services as a business operating in New Jersey under Business ID No. 0450051802.

64. On information and belief, Esjay derives substantial revenue from generic pharmaceutical products that are used and/or consumed within New Jersey, and that are manufactured by EPPL and/or for which Esjay Pharma is/are the named applicant(s) on approved ANDAs. On information and belief, various products for which EPPL and/or Esjay Pharma is/are the named applicant(s) on approved ANDAs are available at retail pharmacies in New Jersey.

65. EPPL did not contest personal jurisdiction or venue in this Judicial District in *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

66. Esjay Pharma did not contest personal jurisdiction or venue in this Judicial District in *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

67. Additionally, this Court has personal jurisdiction over EPPL because the requirements of Federal Rule of Civil Procedure 4(k)(2)(A) are met as (a) Plaintiff's claims arise under federal law; (b) EPPL is a foreign defendant not subject to general personal jurisdiction in the courts of any state; and (c) EPPL has sufficient contacts in the United States as a whole, including, but not limited to, by participating in the preparation and submission of Esjay's ANDA, and/or manufacturing and/or selling pharmaceutical products distributed throughout the United

States, including in this Judicial District, such that this Court's exercise of jurisdiction over EPPL satisfies due process.

68. Venue is proper in this Court as to EPPL because EPPL is a foreign entity who may be sued in any Judicial District, including in the District of New Jersey. 28 U.S.C. § 1391(c)(3); *see also* 28 U.S.C. § 1400(b).

69. Venue is proper in this Judicial District pursuant to 28 U.S.C. § 1400(b) as to Esjay Pharma because, on information and belief, Esjay Pharma has a regular and established place of business in New Jersey, and because, on information and belief, Esjay Pharma has committed or aided, abetted, contributed to, and/or participated in the commission of, acts of infringement of the asserted patent that will lead to foreseeable harm and injury to Plaintiff by preparing or assisting in preparing Esjay's ANDA in New Jersey and/or with the intention of seeking to market Esjay's ANDA Product nationwide, including within New Jersey.

**ACTS GIVING RISE TO THIS SUIT: DEFENDANTS' ANDAS AND NOTICES OF
PARAGRAPH IV CERTIFICATIONS**

a. Navinta (ANDA No. 219480)

70. On information and belief, Navinta has submitted or caused the submission of Navinta's ANDA to the FDA under 21 U.S.C. § 355(j), to obtain approval to engage in the commercial manufacture, use, offer to sell, or sale within the United States or importation into the United States of Navinta's ANDA Product, as purported generic versions of DOJOLVI®, prior to the expiration of the '461 patent.

71. On information and belief, the FDA has not yet approved Navinta's ANDA.

72. In the Navinta Notice Letter, Navinta notified Plaintiff that Navinta had submitted Navinta's ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale,

and/or importation of Navinta's ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748. Plaintiff timely commenced an action in this Court within forty-five (45) days pursuant to 21 U.S.C. § 355(j)(5)(B)(iii), which remains pending. *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

73. Thereafter, on February 17, 2026, Plaintiff's '461 patent issued, and Plaintiff submitted that patent for listing in the Orange Book on February 19, 2026.

74. In the Navinta Notice Letter, Navinta notified Plaintiff of the submission of Navinta's ANDA to the FDA. The purpose of this submission was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Navinta's ANDA Product prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the '461 patent.

75. In the Navinta Notice Letter, Navinta acknowledged that the Reference Listed Drug for Navinta's ANDA is DOJOLVI[®]. DOJOLVI[®] is indicated for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

76. In the Navinta Notice Letter, Navinta also stated that Navinta's ANDA Product is a triheptanoin oral liquid.

77. On information and belief, Navinta has supplemented its ANDA to the FDA to include a certification pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) asserting that the '461 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Navinta's ANDA Product.

78. By filing its ANDA, Navinta has represented to the FDA that Navinta's ANDA Product is bioequivalent to Plaintiff's DOJOLVI®.

79. On information and belief, Navinta, through its own actions and through the actions of its agents, affiliates, and subsidiaries, prepared and submitted Navinta's ANDA, and intends to further prosecute Navinta's ANDA. On information and belief, if the FDA approves Navinta's ANDA, Navinta will manufacture, import, distribute, promote, market, offer for sale, or sell Navinta's ANDA Product within the United States, or will import Navinta's ANDA Product into the United States. On information and belief, if the FDA approves Navinta's ANDA, Navinta will actively induce or contribute to the manufacture, use, offer for sale, sale, or importation of Navinta's ANDA Product in or into the United States.

b. Aurobindo (ANDA No. 219711)

80. On information and belief, Aurobindo has submitted or caused the submission of Aurobindo's ANDA to the FDA under 21 U.S.C. § 355(j), to obtain approval to engage in the commercial manufacture, use, offer to sell, or sale within the United States or importation into the United States of Aurobindo's ANDA Product, as purported generic versions of DOJOLVI®, prior to the expiration of the '461 patent.

81. On information and belief, the FDA has not yet approved Aurobindo's ANDA.

82. In the Aurobindo Notice Letter, Aurobindo notified Plaintiff that Aurobindo had submitted Aurobindo's ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Navinta's ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748. Plaintiff timely commenced an action in this Court within forty-five (45) days pursuant to 21 U.S.C. § 355(j)(5)(B)(iii), which remains

pending. *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

83. Thereafter, on February 17, 2026, Plaintiff's '461 patent issued, and Plaintiff submitted that patent for listing in the Orange Book on February 19, 2026.

84. In the Aurobindo Notice Letter, Aurobindo notified Plaintiff of the submission of Aurobindo's ANDA to the FDA. The purpose of this submission was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Aurobindo's ANDA Product prior to the expiration of U.S. Patent No. 8,697,748 and, therefore, prior to the expiration of the '461 patent.

85. In the Aurobindo Notice Letter, Aurobindo acknowledged that the Reference Listed Drug for Aurobindo's ANDA is DOJOLVI®. DOJOLVI® is indicated for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

86. In the Aurobindo Notice Letter, Aurobindo also stated that Aurobindo's ANDA Product is a triheptanoin oral liquid.

87. On information and belief, Aurobindo has supplemented its ANDA to the FDA to include a certification pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) asserting that the '461 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Aurobindo's ANDA Product.

88. By filing its ANDA, Aurobindo has represented to the FDA that Aurobindo's ANDA Product is bioequivalent to Plaintiff's DOJOLVI®.

89. On information and belief, Aurobindo, through its own actions and through the actions of its agents, affiliates, and subsidiaries, prepared and submitted Aurobindo's ANDA, and

intends to further prosecute Aurobindo's ANDA. On information and belief, if the FDA approves Aurobindo's ANDA, Aurobindo will manufacture, import, distribute, promote, market, offer for sale, or sell Aurobindo's ANDA Product within the United States, or will import Aurobindo's ANDA Product into the United States. On information and belief, if the FDA approves Aurobindo's ANDA, Aurobindo will actively induce or contribute to the manufacture, use, offer for sale, sale, or importation of Aurobindo's ANDA Product in or into the United States.

c. Esjay (ANDA No. 219512)

90. On information and belief, Esjay has submitted or caused the submission of Esjay's ANDA to the FDA under 21 U.S.C. § 355(j), to obtain approval to engage in the commercial manufacture, use, offer to sell, or sale within the United States or importation into the United States of Esjay's ANDA Product, as purported generic versions of DOJOLVI[®], prior to the expiration of the '461 patent.

91. On information and belief, the FDA has not yet approved Esjay's ANDA.

92. In the First Esjay Letter and the Second Esjay Letter, Esjay notified Plaintiff that Esjay had submitted Esjay's ANDA to the FDA for purported generic versions of triheptanoin oral liquid (100% w/w), seeking FDA approval to engage in the commercial manufacture, use, sale, offer for sale, and/or importation of Esjay's ANDA Product in or into the United States, including New Jersey, prior to the expiration of U.S. Patent No. 8,697,748. Plaintiff timely commenced an action in this Court within forty-five (45) days pursuant to 21 U.S.C. § 355(j)(5)(B)(iii), which remains pending. *Ultragenyx Pharmaceutical Inc., et al v. Navinta LLC, et al.*, No. 2:24-cv-09483-ES-MAH (D.N.J. Sept. 26, 2024).

93. Thereafter, on February 17, 2026, Plaintiff's '461 patent issued, and Plaintiff submitted that patent for listing in the Orange Book on February 19, 2026.

94. In the Third Esjay Notice Letter, Esjay notified Plaintiff that, as part of its ANDA, Esjay had filed a Paragraph IV Certification pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) asserting that the '461 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Esjay's ANDA Product. The purpose of this submission was to obtain approval under the FDCA to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Esjay's ANDA Product prior to the expiration of the '461 patent.

95. In the First, Second, and Third Esjay Notice Letters, Esjay acknowledged that the Reference Listed Drug for Esjay's ANDA is DOJOLVI[®]. DOJOLVI[®] is indicated for the treatment of adult and pediatric patients with molecularly confirmed long-chain fatty acid oxidation disorders (LC-FAOD).

96. In the First, Second, and Third Esjay Notice Letters, Esjay also stated that Esjay's ANDA Product is a triheptanoin oral liquid.

97. On information and belief, Esjay submitted its ANDA to the FDA containing certifications pursuant to 21 U.S.C. § 355(j)(2)(A)(vii)(IV) asserting that the '461 patent is invalid, unenforceable, and/or will not be infringed by the manufacture, use, offer for sale, sale, and/or importation of Esjay's ANDA Product.

98. By filing its ANDA, Esjay has represented to the FDA that Esjay's ANDA Product is bioequivalent to Plaintiff's DOJOLVI[®].

99. On information and belief, Esjay, through its own actions and through the actions of its agents, affiliates, and subsidiaries, prepared and submitted Esjay's ANDA, and intends to further prosecute Esjay's ANDA. On information and belief, if the FDA approves Esjay's ANDA, Esjay will manufacture, import, distribute, promote, market, offer for sale, or sell Esjay's ANDA Product within the United States, or will import Esjay's ANDA Product into the United States. On

information and belief, if the FDA approves Esjay's ANDA, Esjay, through their own actions and through the actions of their agents, affiliates, and subsidiaries, will actively induce or contribute to the manufacture, use, offer for sale, sale, or importation of Esjay's ANDA Product in or into the United States.

COUNT I – INFRINGEMENT BY NAVINTA OF THE '461 PATENT

100. Plaintiff incorporates each of the preceding paragraphs 1–99 as if fully set forth herein.

101. Navinta's submission of Navinta's ANDA with a Paragraph IV Certification for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Navinta's ANDA Product in or into the United States before the expiration of the '461 patent is an act of infringement of the '461 patent under 35 U.S.C. § 271(e)(2)(A).

102. There is a justiciable controversy between Plaintiff and Navinta as to the infringement of the '461 patent.

103. Unless enjoined by this Court, upon FDA approval of Navinta's ANDA, Navinta will infringe one or more claims of the '461 patent under 35 U.S.C. § 271(a), including at least claim 1, by making, using, offering to sell, selling, and/or importing Navinta's ANDA Product in the United States.

104. Unless enjoined by this Court, upon FDA approval of Navinta's ANDA, Navinta will induce infringement of one or more claims of the '461 patent under 35 U.S.C. § 271(b), including at least claim 1, by making, using, offering to sell, selling, and/or importing Navinta's ANDA Product in the United States. On information and belief, upon FDA approval of Navinta's ANDA, Navinta will intentionally encourage acts of direct infringement with knowledge of the '461 patent and knowledge that its acts are encouraging infringement.

105. Unless enjoined by this Court, upon FDA approval of Navinta's ANDA, Navinta will contributorily infringe one or more claims of the '461 patent under 35 U.S.C. § 271(c), including at least claim 1, by making, using, offering to sell, selling, and/or importing Navinta's ANDA Product in the United States. On information and belief, Navinta has had and continues to have knowledge that Navinta's ANDA Product is especially adapted for a use that infringes one or more claims of the '461 patent, including at least claim 1, and that there is no substantial non-infringing use for Navinta's ANDA Product.

106. Plaintiff will be substantially and irreparably damaged and harmed if Navinta's infringement of the '461 patent is not enjoined.

107. Plaintiff does not have an adequate remedy at law.

108. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT II – INFRINGEMENT BY AUROBINDO OF THE '461 PATENT

109. Plaintiff incorporates each of the preceding paragraphs 1–108 as if fully set forth herein.

110. Aurobindo's submission of Aurobindo's ANDA with a Paragraph IV Certification for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Aurobindo's ANDA Product in or into the United States before the expiration of the '461 patent is an act of infringement of the '461 patent under 35 U.S.C. § 271(e)(2)(A).

111. There is a justiciable controversy between Plaintiff and Aurobindo as to the infringement of the '461 patent.

112. Unless enjoined by this Court, upon FDA approval of Aurobindo's ANDA, Aurobindo will infringe one or more claims of the '461 patent under 35 U.S.C. § 271(a), including

at least claim 1, by making, using, offering to sell, selling, and/or importing Aurobindo's ANDA Product in the United States.

113. Unless enjoined by this Court, upon FDA approval of Aurobindo's ANDA, Aurobindo will induce infringement of one or more claims of the '461 patent under 35 U.S.C. § 271(b), including at least claim 1, by making, using, offering to sell, selling, and/or importing Aurobindo's ANDA Product in the United States. On information and belief, upon FDA approval of Aurobindo's ANDA, Aurobindo will intentionally encourage acts of direct infringement with knowledge of the '461 patent and knowledge that its acts are encouraging infringement.

114. Unless enjoined by this Court, upon FDA approval of Aurobindo's ANDA, Aurobindo will contributorily infringe one or more claims of the '461 patent under 35 U.S.C. § 271(c), including at least claim 1, by making, using, offering to sell, selling, and/or importing Aurobindo's ANDA Product in the United States. On information and belief, Aurobindo has had and continues to have knowledge that Aurobindo's ANDA Product is especially adapted for a use that infringes one or more claims of the '461 patent, including at least claim 1, and that there is no substantial non-infringing use for Aurobindo's ANDA Product.

115. Plaintiff will be substantially and irreparably damaged and harmed if Aurobindo's infringement of the '461 patent is not enjoined.

116. Plaintiff does not have an adequate remedy at law.

117. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

COUNT III – INFRINGEMENT BY ESJAY OF THE '461 PATENT

118. Plaintiff incorporates each of the preceding paragraphs 1–117 as if fully set forth herein.

119. Esjay's submission of Esjay's ANDA with a Paragraph IV Certification for the purpose of obtaining approval to engage in the commercial manufacture, use, offer for sale, sale, and/or importation of Esjay's ANDA Product in or into the United States before the expiration of the '461 patent is an act of infringement of the '461 patent under 35 U.S.C. § 271(e)(2)(A).

120. There is a justiciable controversy between Plaintiff and Esjay as to the infringement of the '461 patent.

121. Unless enjoined by this Court, upon FDA approval of Esjay's ANDA, Esjay will infringe one or more claims of the '461 patent under 35 U.S.C. § 271(a), including at least claim 1, by making, using, offering to sell, selling, and/or importing Esjay's ANDA Product in the United States.

122. Unless enjoined by this Court, upon FDA approval of Esjay's ANDA, Esjay will induce infringement of one or more claims of the '461 patent under 35 U.S.C. § 271(b), including at least claim 1, by making, using, offering to sell, selling, and/or importing Esjay's ANDA Product in the United States. On information and belief, upon FDA approval of Esjay's ANDA, Esjay will intentionally encourage acts of direct infringement with knowledge of the '461 patent and knowledge that its acts are encouraging infringement.

123. Unless enjoined by this Court, upon FDA approval of Esjay's ANDA, Esjay will contributorily infringe one or more claims of the '461 patent under 35 U.S.C. § 271(c), including at least claim 1, by making, using, offering to sell, selling, and/or importing Esjay's ANDA Product in the United States. On information and belief, Esjay has had and continues to have knowledge that Esjay's ANDA Product is especially adapted for a use that infringes one or more claims of the '461 patent, including at least claim 1, and that there is no substantial non-infringing use for Esjay's ANDA Product.

124. Plaintiff will be substantially and irreparably damaged and harmed if Esjay's infringement of the '461 patent is not enjoined.

125. Plaintiff does not have an adequate remedy at law.

126. This case is an exceptional one, and Plaintiff is entitled to an award of its reasonable attorneys' fees under 35 U.S.C. § 285.

PRAYER FOR RELIEF

WHEREFORE, Plaintiff requests the following relief:

a. Navinta

(a) A judgment that the '461 patent has been infringed under 35 U.S.C. § 271(e)(2) by Navinta's submission to the FDA of Navinta's ANDA;

(b) A judgment, pursuant to 35 U.S.C. § 271(e)(4)(A), ordering that the effective date of any FDA approval of ANDA No. 219480 for the commercial manufacture, use, or sale of Navinta's ANDA Product, or any other drug product that infringes or the use of which infringes the '461 patent, be not earlier than the expiration date of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(c) A preliminary and permanent injunction, pursuant to 35 U.S.C. §§ 271(e)(4)(B) and 283 and Rule 65, Fed. R. Civ. P., enjoining Navinta, and all persons acting in concert with Navinta, from the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Navinta's ANDA Product, or any other drug product covered by or whose use is covered by the '461 patent, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(d) A judgment declaring that the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Navinta's ANDA Product, or any other drug product that is covered by or whose use is covered by the '461 patent, prior to the

expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity, will infringe, induce the infringement of, and contribute to the infringement by others of the '461 patent;

(e) A declaration that Navinta's commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States, of Navinta's ANDA Product, or inducing or contributing to such conduct, would constitute infringement of one or more claims of the '461 patent by Navinta, including at least claim 1, under one or more of 35 U.S.C. § 271(a), (b), and (c);

(f) An award of damages or other relief, pursuant to 35 U.S.C. § 271(e)(4)(C), if Navinta engages in the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Navinta's ANDA Product, or any product that infringes the '461 patent, or induces or contributes to such conduct, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(g) A judgment that Navinta willfully and deliberately infringed the '461 patent;

(h) A declaration that this is an exceptional case and an award of attorneys' fees pursuant to 35 U.S.C. § 285;

(i) Costs and expenses in this action; and

(j) Such further and other relief as this Court may deem just and proper.

b. Aurobindo

(a) A judgment that the '461 patent has been infringed under 35 U.S.C. § 271(e)(2) by Aurobindo's submission to the FDA of Aurobindo's ANDA;

(b) A judgment, pursuant to 35 U.S.C. § 271(e)(4)(A), ordering that the effective date of any FDA approval of ANDA No. 219711 for the commercial manufacture, use,

or sale of Aurobindo's ANDA Product, or any other drug product that infringes or the use of which infringes the '461 patent, be not earlier than the expiration date of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(c) A preliminary and permanent injunction, pursuant to 35 U.S.C. §§ 271(e)(4)(B) and 283 and Rule 65, Fed. R. Civ. P., enjoining Aurobindo, and all persons acting in concert with Aurobindo, from the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Aurobindo's ANDA Product, or any other drug product covered by or whose use is covered by the '461 patent, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(d) A judgment declaring that the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Aurobindo's ANDA Product, or any other drug product that is covered by or whose use is covered by the '461 patent, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity, will infringe, induce the infringement of, and contribute to the infringement by others of the '461 patent;

(e) A declaration that Aurobindo's commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States, of Aurobindo's ANDA Product, or inducing or contributing to such conduct, would constitute infringement of one or more claims of the '461 patent, including at least claim 1, by Aurobindo under one or more of 35 U.S.C. § 271(a), (b), and (c);

(f) An award of damages or other relief, pursuant to 35 U.S.C. § 271(e)(4)(C), if Aurobindo engages in the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Aurobindo's ANDA Product, or any product that

infringes the '461 patent, or induces or contributes to such conduct, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(g) A judgment that Aurobindo willfully and deliberately infringed the '461 patent;

(h) A declaration that this is an exceptional case and an award of attorneys' fees pursuant to 35 U.S.C. § 285;

(i) Costs and expenses in this action; and

(j) Such further and other relief as this Court may deem just and proper.

c. Esjay

(a) A judgment that the '461 patent has been infringed under 35 U.S.C. § 271(e)(2) by Esjay's submission to the FDA of Esjay's ANDA;

(b) A judgment, pursuant to 35 U.S.C. § 271(e)(4)(A), ordering that the effective date of any FDA approval of ANDA No. 219512 for the commercial manufacture, use, or sale of Esjay's ANDA Product, or any other drug product that infringes or the use of which infringes the '461 patent, be not earlier than the later of the expiration date of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(c) A preliminary and permanent injunction, pursuant to 35 U.S.C. §§ 271(e)(4)(B) and 283 and Rule 65, Fed. R. Civ. P., enjoining Esjay, and all persons acting in concert with Esjay, from the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Esjay's ANDA Product, or any other drug product covered by or whose use is covered by the '461 patent, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(d) A judgment declaring that the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Esjay's ANDA Product, or any

other drug product that is covered by or whose use is covered by the '461 patent, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity, will infringe, induce the infringement of, and contribute to the infringement by others of the '461 patent;

(e) A declaration that Esjay's commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States, of Esjay's ANDA Product, or inducing or contributing to such conduct, would constitute infringement of one or more claims of the '461 patent, including at least claim 1, by Esjay under one or more of 35 U.S.C. § 271(a), (b), and (c);

(f) An award of damages or other relief, pursuant to 35 U.S.C. § 271(e)(4)(C), if Esjay engages in the commercial manufacture, use, offer for sale, or sale in the United States, or importation into the United States of Esjay's ANDA Product, or any product that infringes the '461 patent, or induces or contributes to such conduct, prior to the expiration of the '461 patent, inclusive of any extension(s) and additional period(s) of exclusivity;

(g) A judgment that Esjay willfully and deliberately infringed the '461 patent;

(h) A declaration that this is an exceptional case and an award of attorneys' fees pursuant to 35 U.S.C. § 285;

(i) Costs and expenses in this action; and

(j) Such further and other relief as this Court may deem just and proper.

Dated: April 3, 2026

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LOCAL CIVIL RULES 11.2 AND 40.1 CERTIFICATION

Pursuant to Local Civil Rules 11.2 and 40.1, I hereby certify that the matter in controversy involves the same defendants, one of the same plaintiffs, the same ANDAs, and the same drug product that are at issue in the matter captioned *Ultragenyx Pharmaceutical Inc., et al. v. Navinta LLC, et al.*, No. 2:24-cv-09483 (ES) (MAH) (D.N.J.), currently pending in this Judicial District.

I further certify that the matter in controversy involves one of the same plaintiffs, one of the same patents, and the same drug product that are at issue in the matter captioned *Ultragenyx Pharmaceutical Inc., et al. v. Somerset Therapeutics, LLC, et al.*, No. 2:26-cv-03509 (ES) (MAH) (D.N.J.), currently pending in this Judicial District.

I further certify that, to the best of my knowledge, the matter in controversy is not the subject of any other action pending in any court or of any pending arbitration or administrative proceeding.

Dated: April 3, 2026

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EXHIBIT A

(12) **United States Patent**
Klopp et al.

(10) **Patent No.:** **US 12,551,461 B2**
(45) **Date of Patent:** ***Feb. 17, 2026**

(54) **COMPOSITIONS OF TRIGLYCERIDES AND USES THEREOF**

(71) Applicant: **Ultragenyx Pharmaceutical Inc.,**
Novato, CA (US)

(72) Inventors: **John Klopp,** Novato, CA (US);
Gabrielle Morris, Novato, CA (US);
Emil Kakkis, Novato, CA (US); **Steven Jungles,** Novato, CA (US)

(73) Assignee: **Ultragenyx Pharmaceutical Inc.,**
Novato, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/167,551**

(22) Filed: **Feb. 10, 2023**

(65) **Prior Publication Data**

US 2023/0181513 A1 Jun. 15, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/445,557, filed on Jun. 19, 2019, now abandoned, which is a continuation of application No. 14/419,601, filed as application No. PCT/US2014/065693 on Nov. 14, 2014, now abandoned.

(Continued)

(51) **Int. Cl.**

A61K 31/23 (2006.01)
A61K 9/14 (2006.01)
A61K 9/16 (2006.01)
A61K 31/20 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 31/23** (2013.01); **A61K 9/143** (2013.01); **A61K 9/146** (2013.01); **A61K 9/1611** (2013.01); **A61K 9/1652** (2013.01); **A61K 31/20** (2013.01)

(58) **Field of Classification Search**

None
See application file for complete search history.

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(Continued)

Primary Examiner — Isis A Ghali

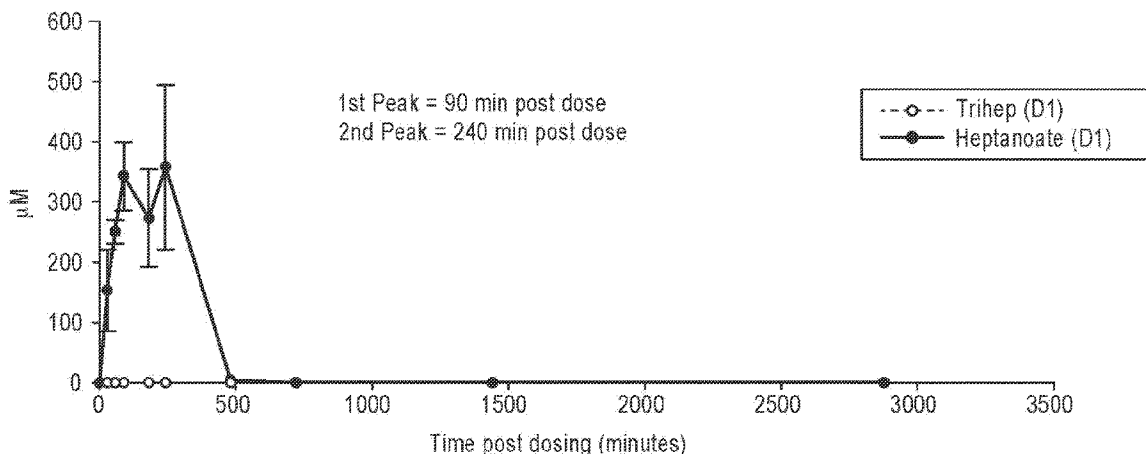
(74) *Attorney, Agent, or Firm* — COOLEY LLP

(57) **ABSTRACT**

The present invention includes solid compositions of triglycerides with one or more fatty acids, such as triheptanoin and glycerol phenylbutyrate, and therapeutic use thereof. The solid compositions can be prepared by spray-drying or other processes.

5 Claims, 40 Drawing Sheets

Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Oil Dosing on Day 1 in Male and Female Mini-pigs Combined



US 12,551,461 B2

Page 2

Related U.S. Application Data

- (60) Provisional application No. 61/904,369, filed on Nov. 14, 2013.

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Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Oil
Dosing on Day 1 in Male and Female Mini-pigs Combined

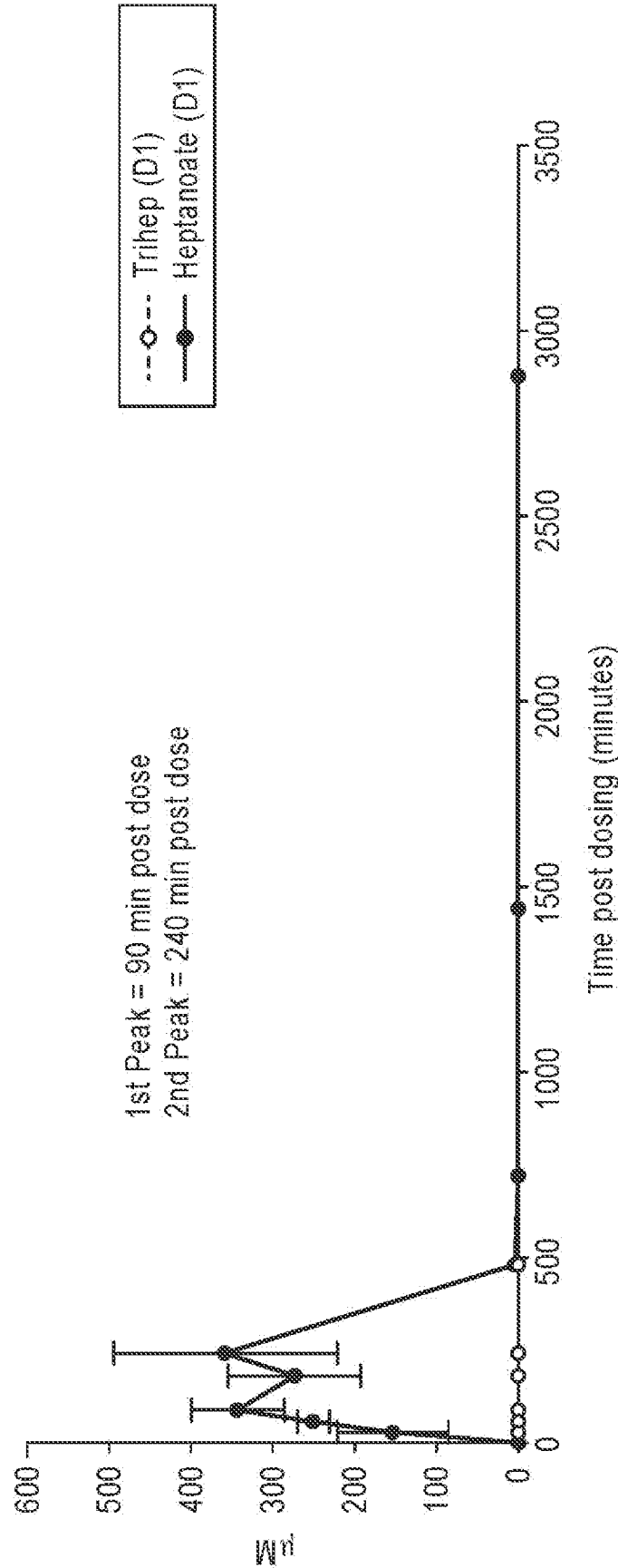


Figure 1

Triheptanoic acid and Heptanoate (C7) Metabolite PK Following Triheptanoic acid Powder-ER
Dosing on Day 1 in Male and Female Mini-pigs Combined

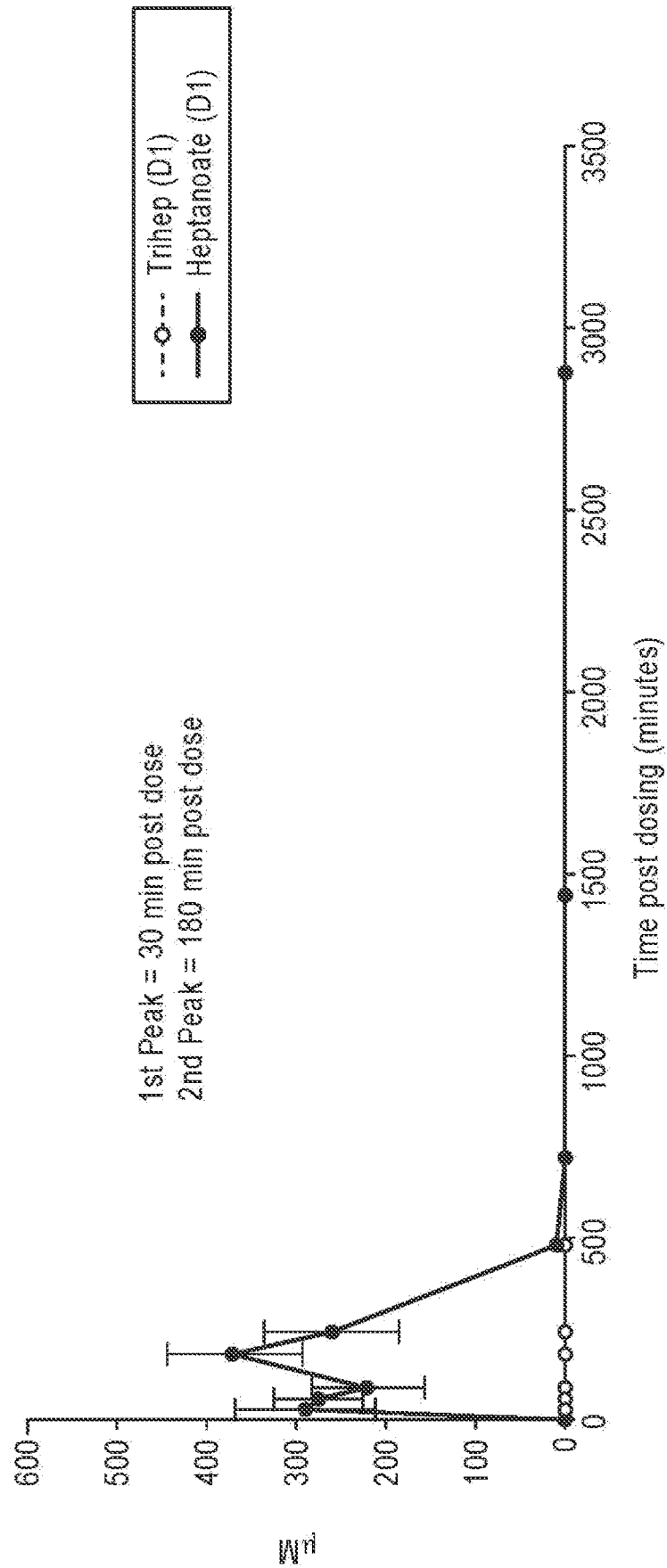


Figure 2

Triheptanoin and Heptanoate (C7) Metabolite PK Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

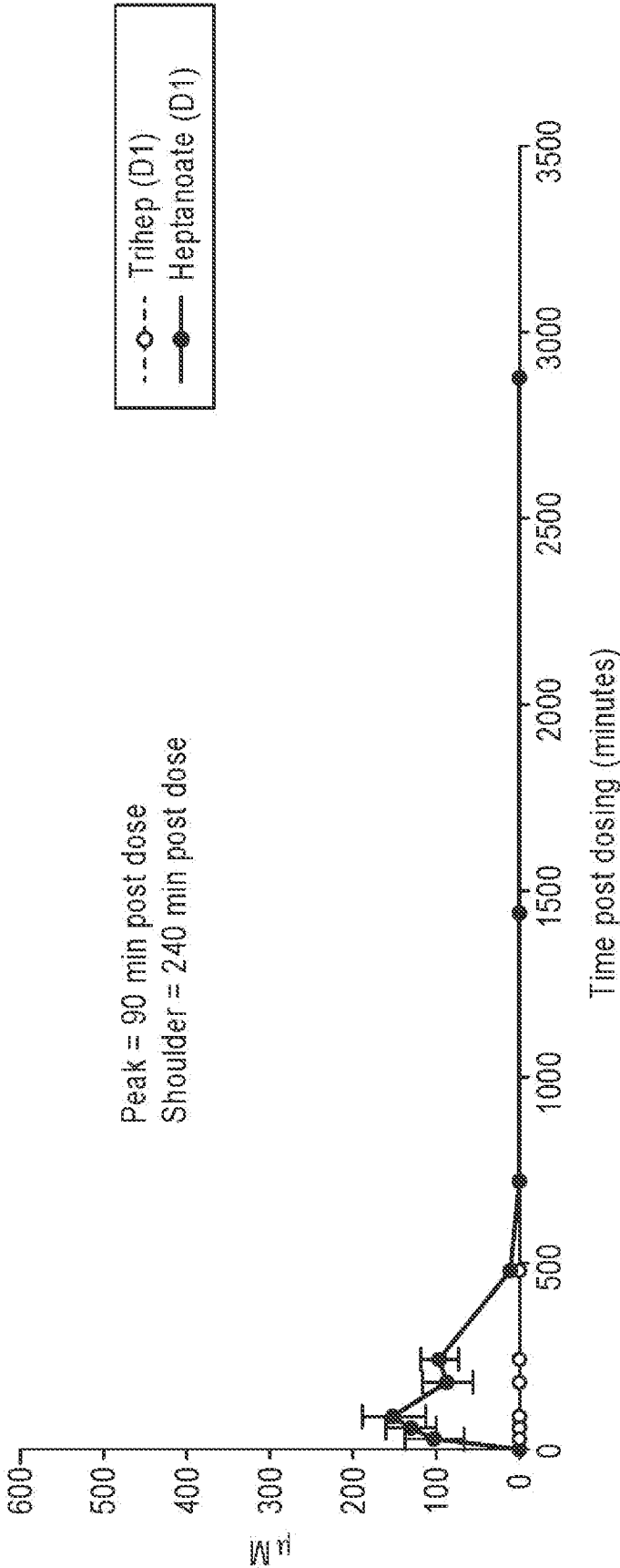


Figure 3

PK up to 48hr timepoint

PK up to 8hr timepoint

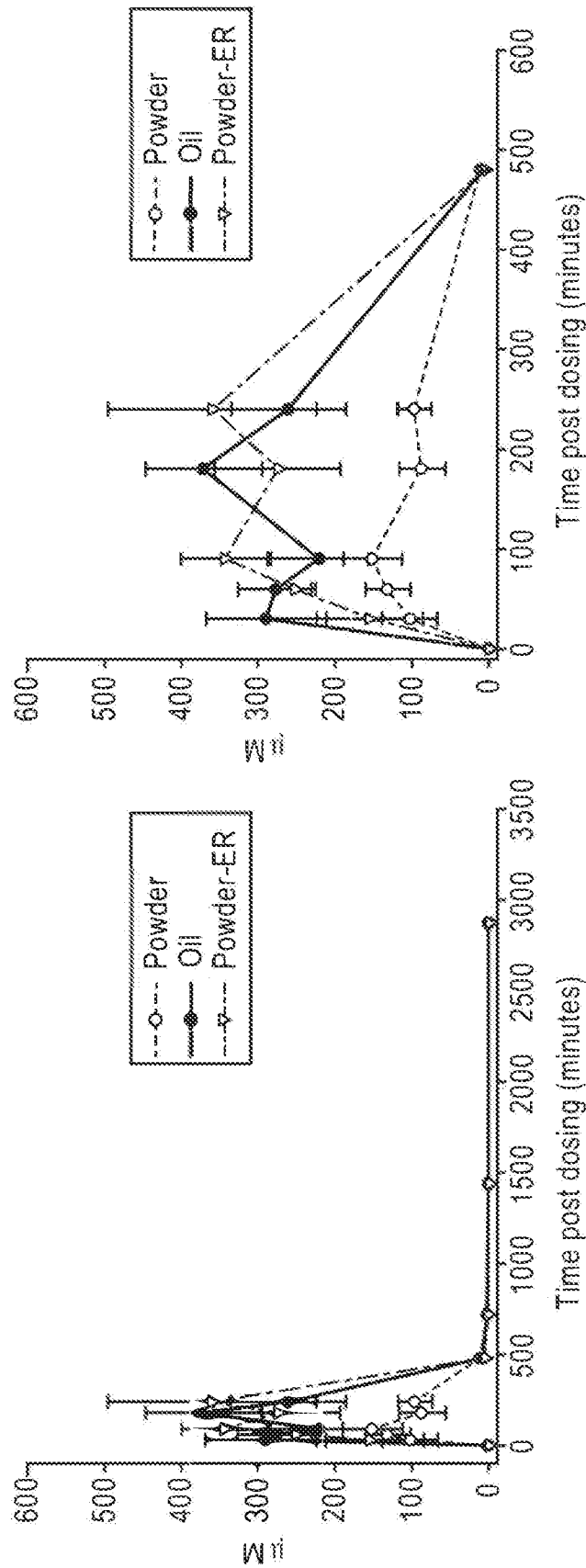


Figure 4

B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK
Following Triheptanoin Powder-ER Dosing on Day 1 in Male and Female Mini-pigs
Combined

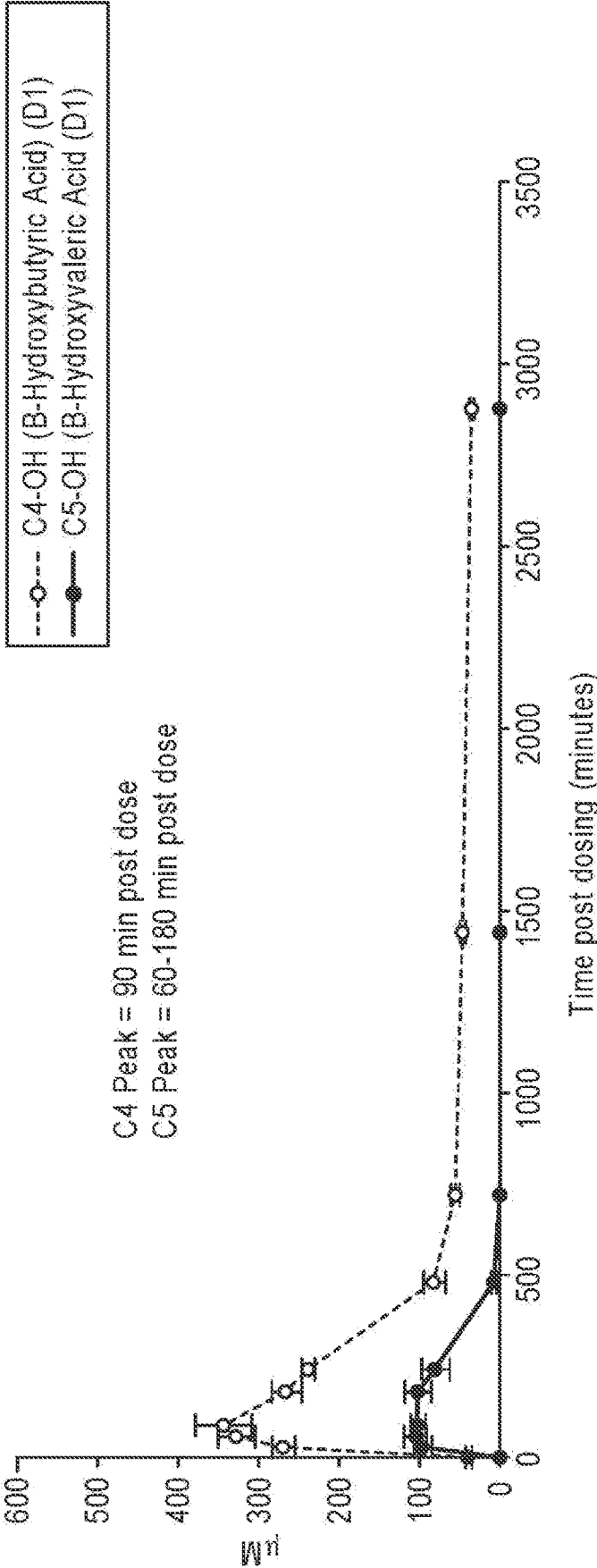


Figure 5

B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK Following Triheptanoin Powder Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

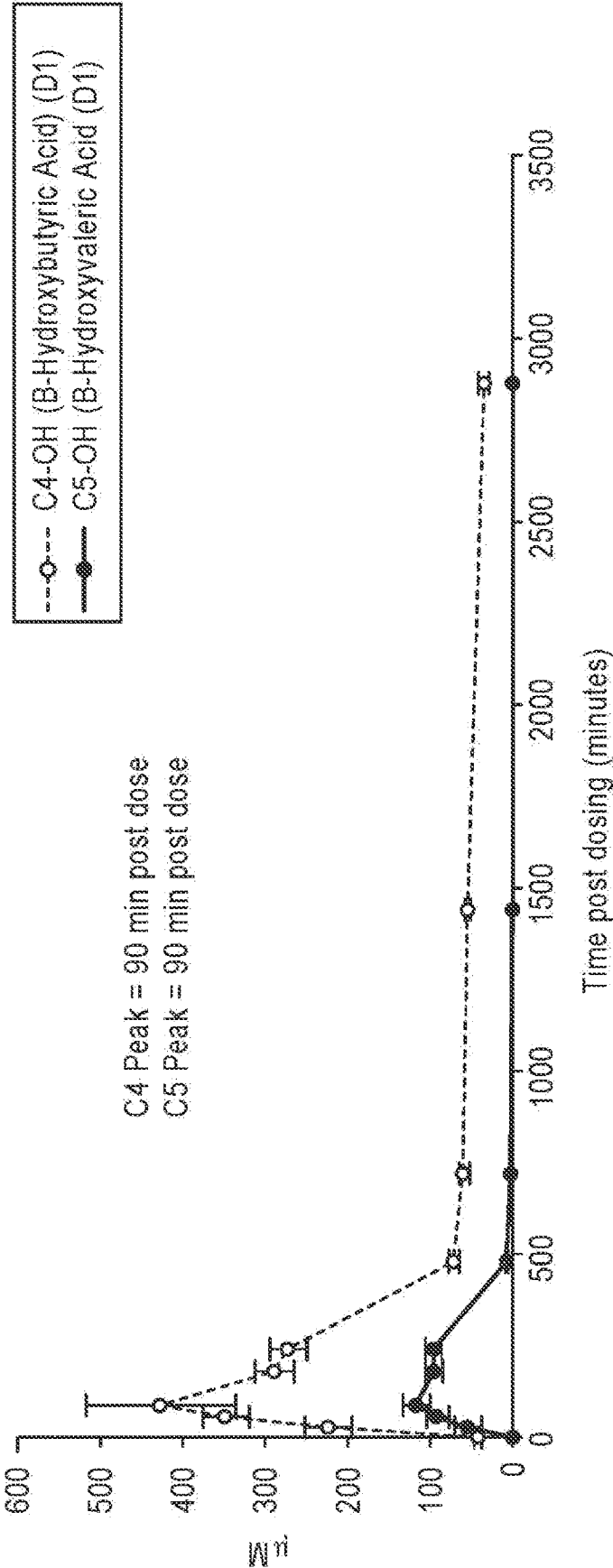


Figure 6

B-Hydroxybutyric Acid (C4-OH) and B-Hydroxyvaleric Acid (C5-OH) Metabolite PK Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

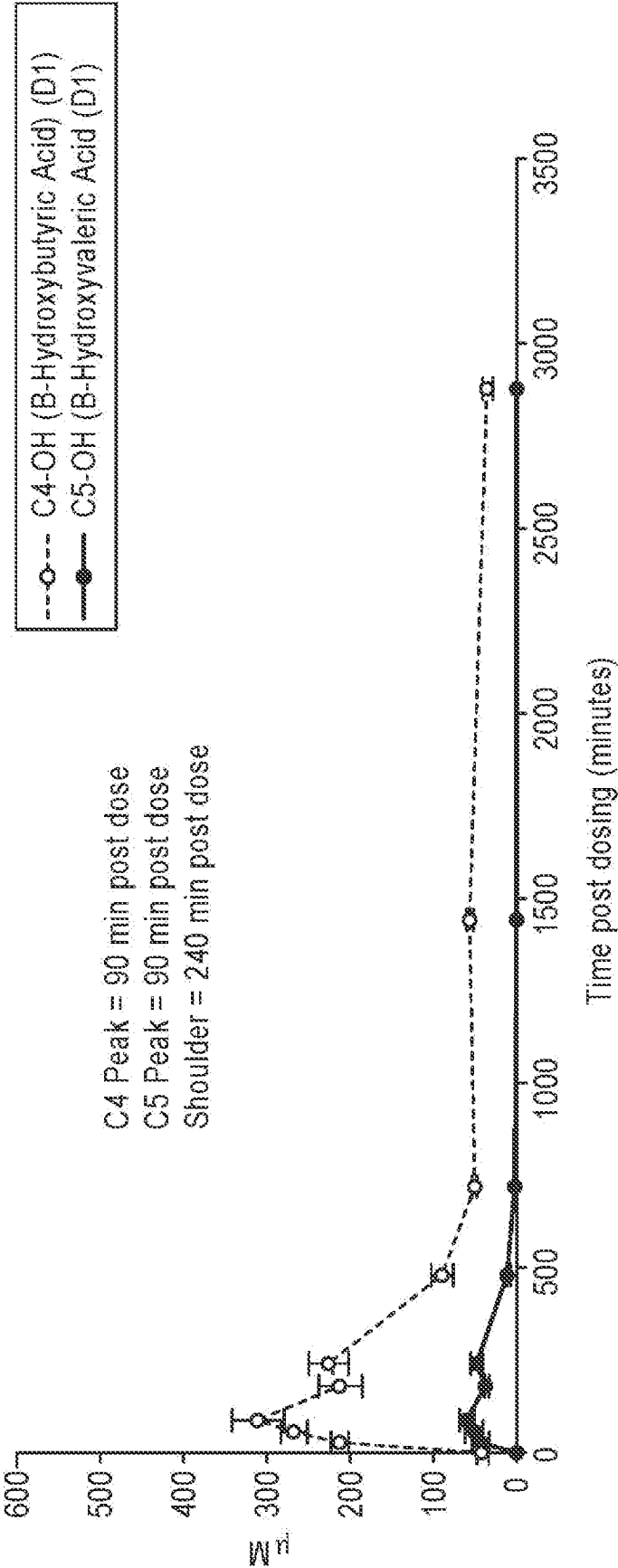
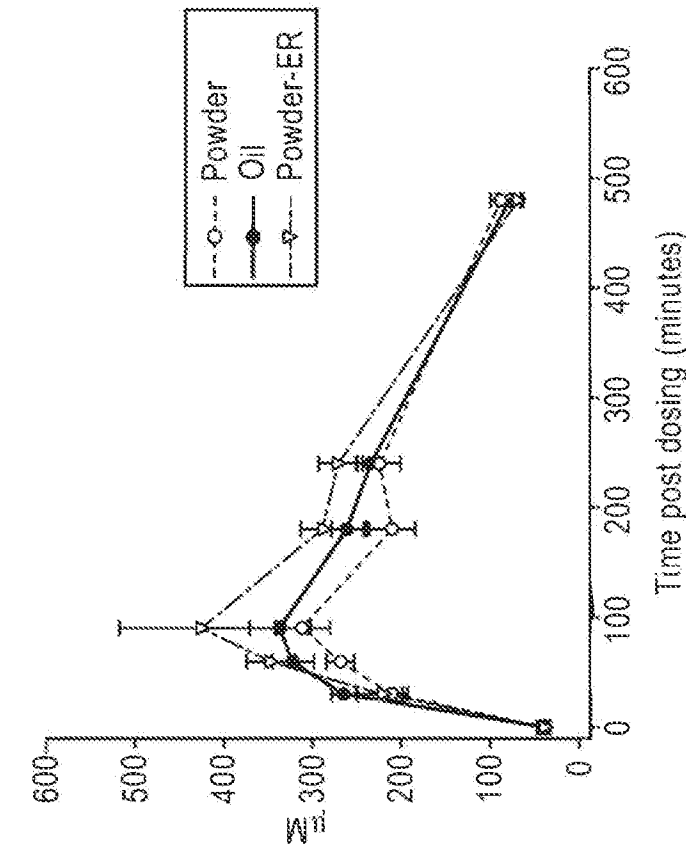


Figure 7

PK up to 8hr timepoint



PK up to 48hr timepoint

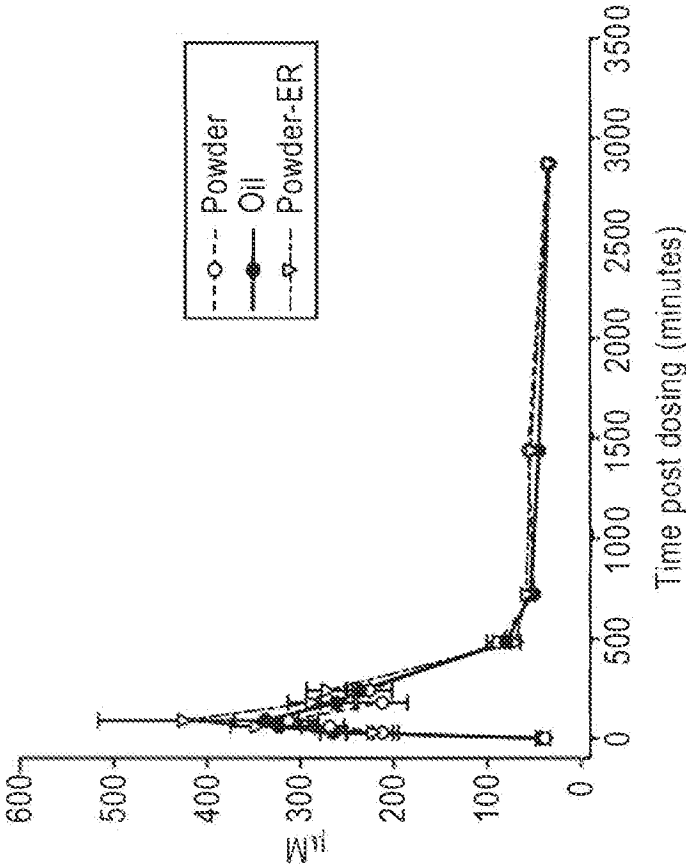


Figure 8

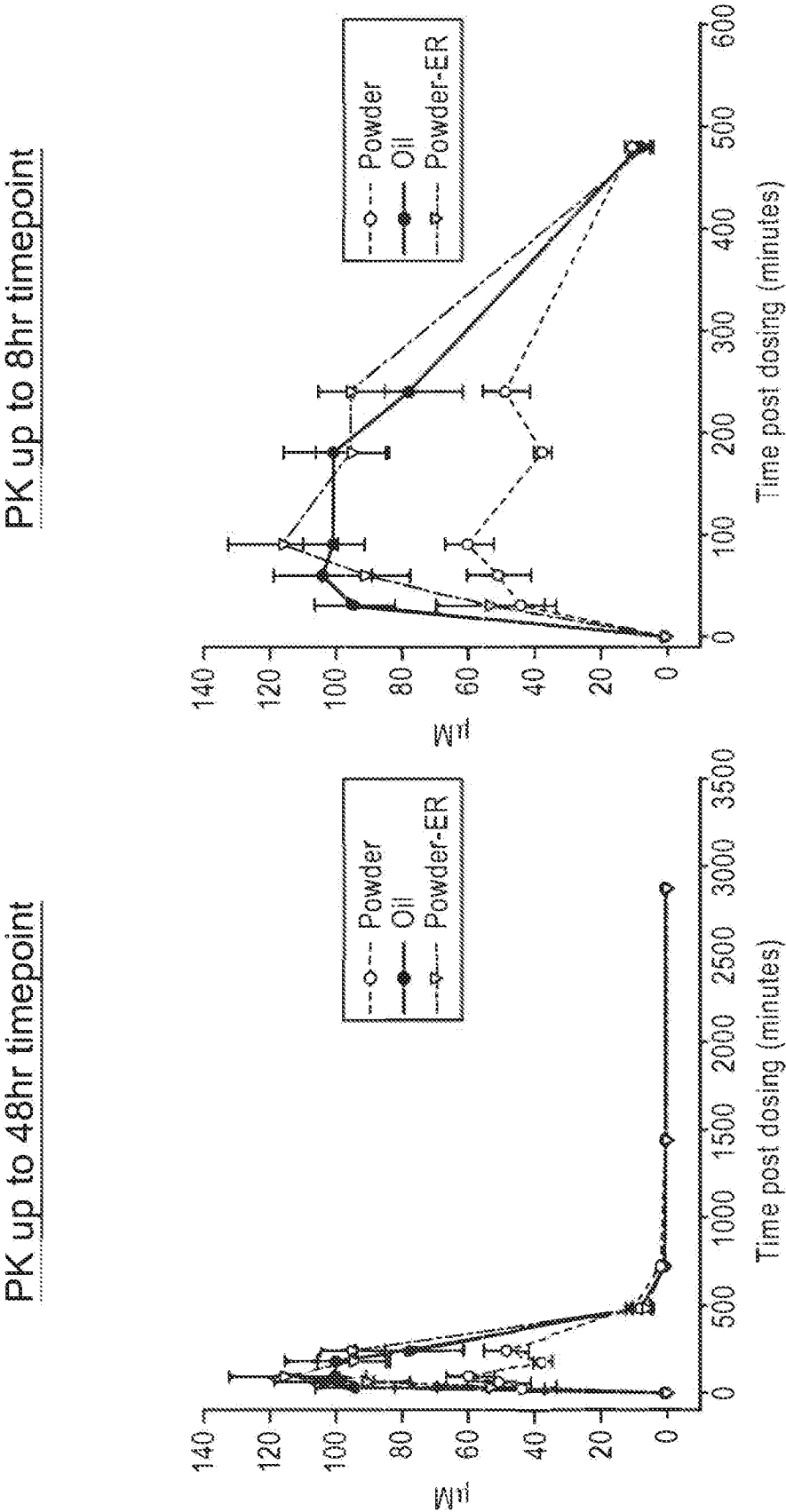


Figure 9

Levels of Aspartate Following Triheptanoin Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

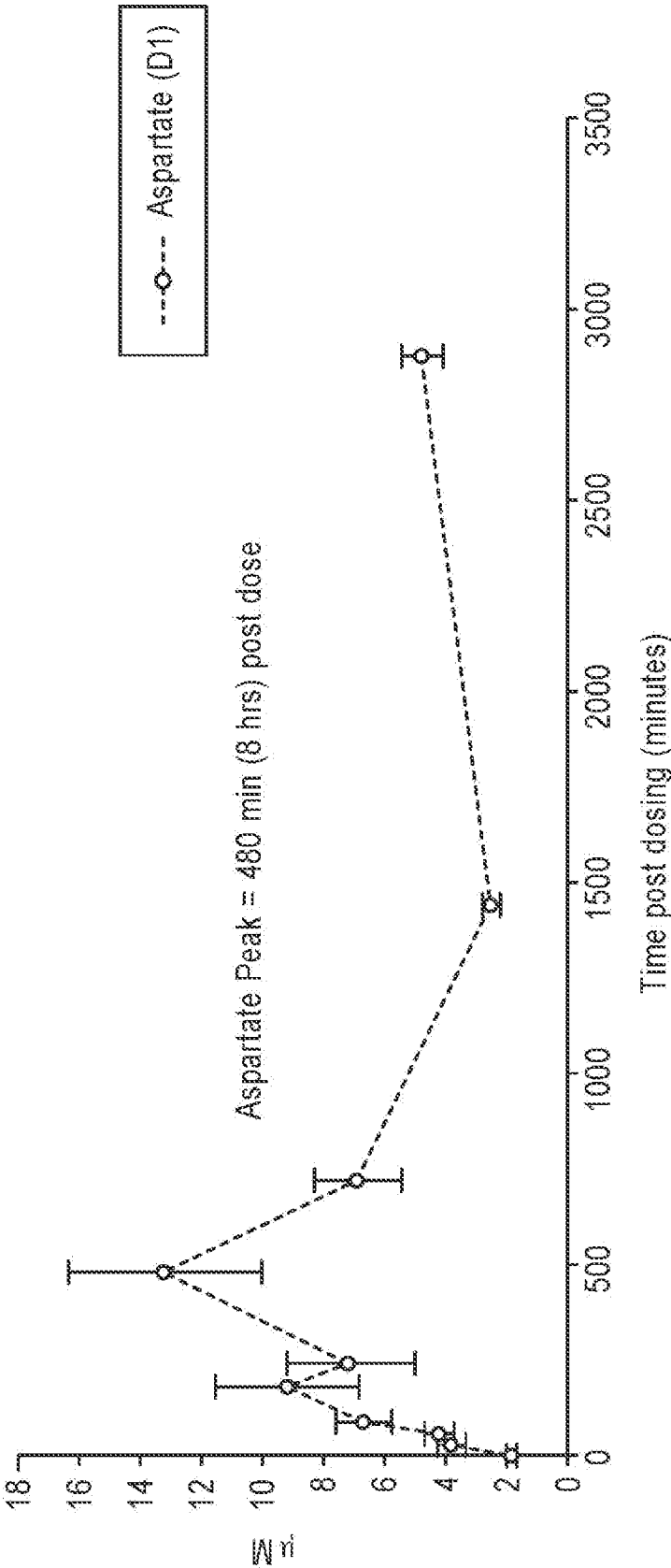


Figure 10

Levels of Aspartate Following Triheptanoic Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

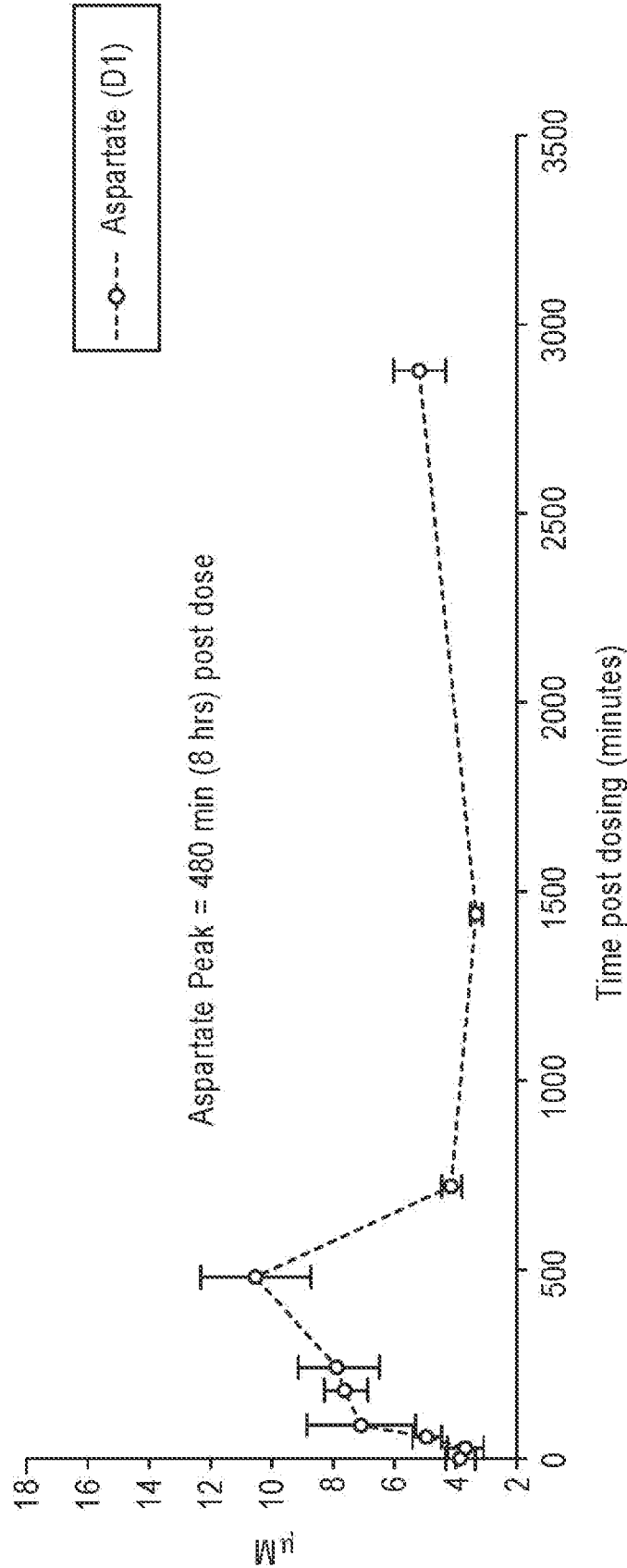


Figure 11

Levels of Aspartate Following Triheptanoic Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

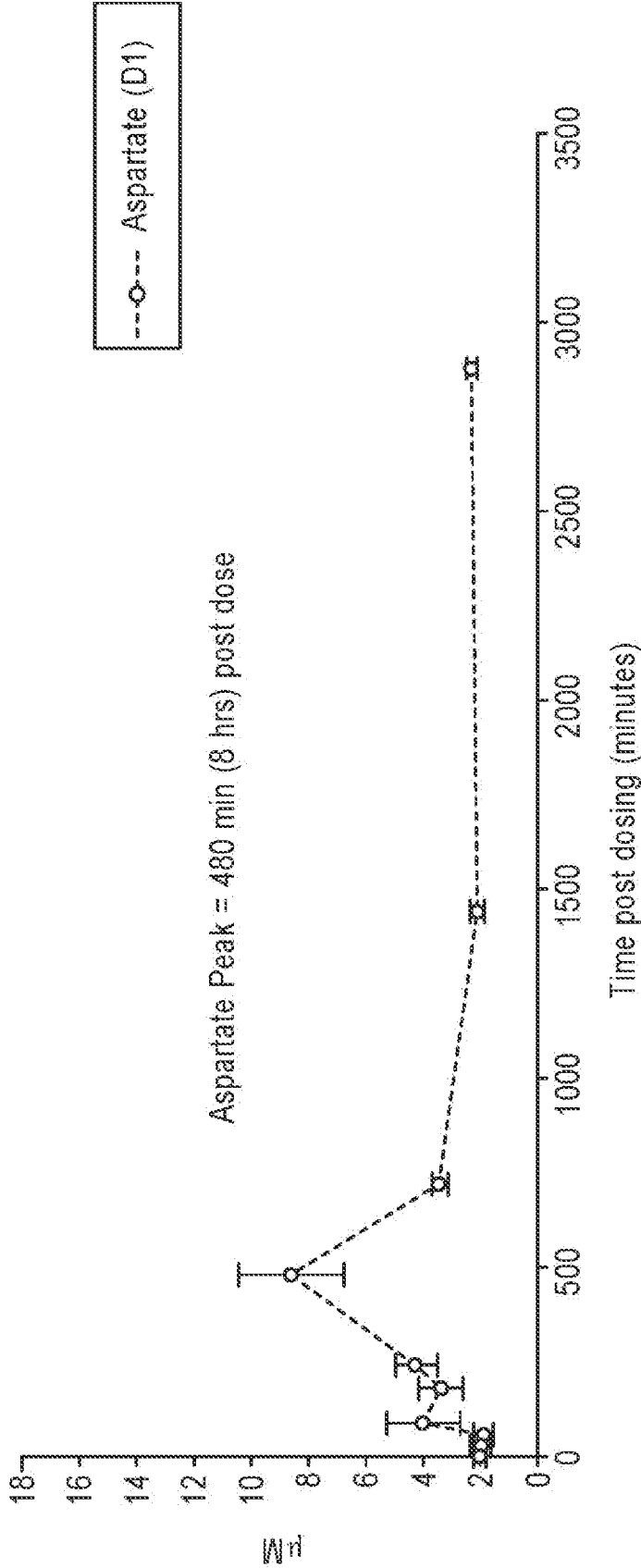


Figure 12

Levels of Glutamate Following Triheptanoin Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

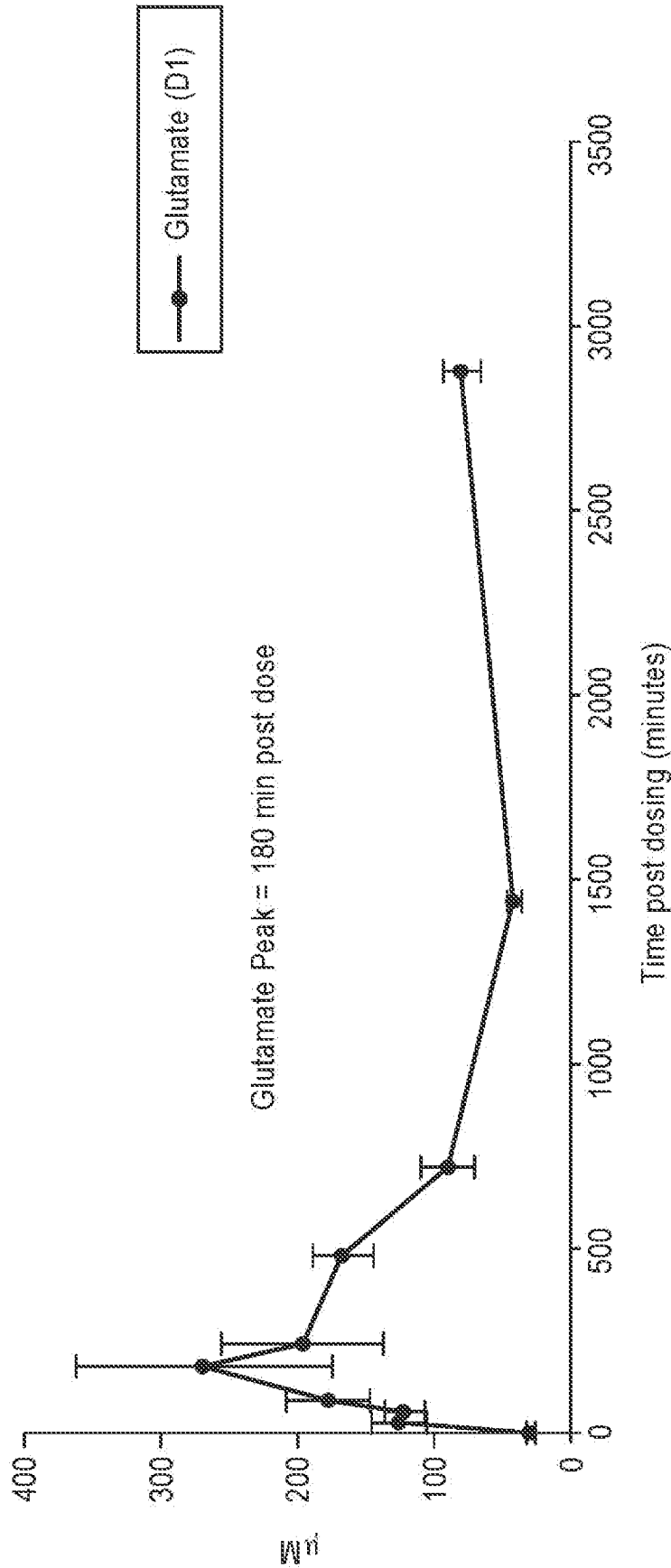


Figure 13

Levels of Glutamate Following Triheptanoin Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

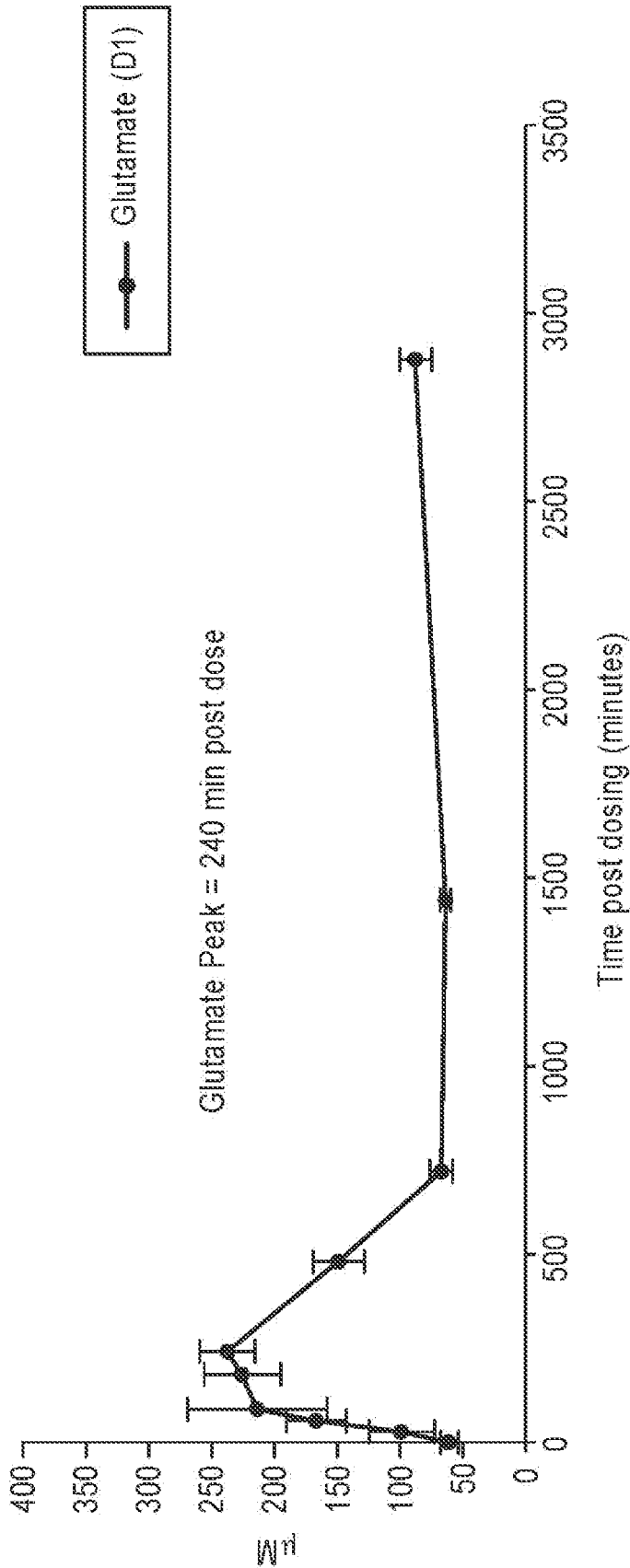


Figure 14

Levels of Glutamate Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

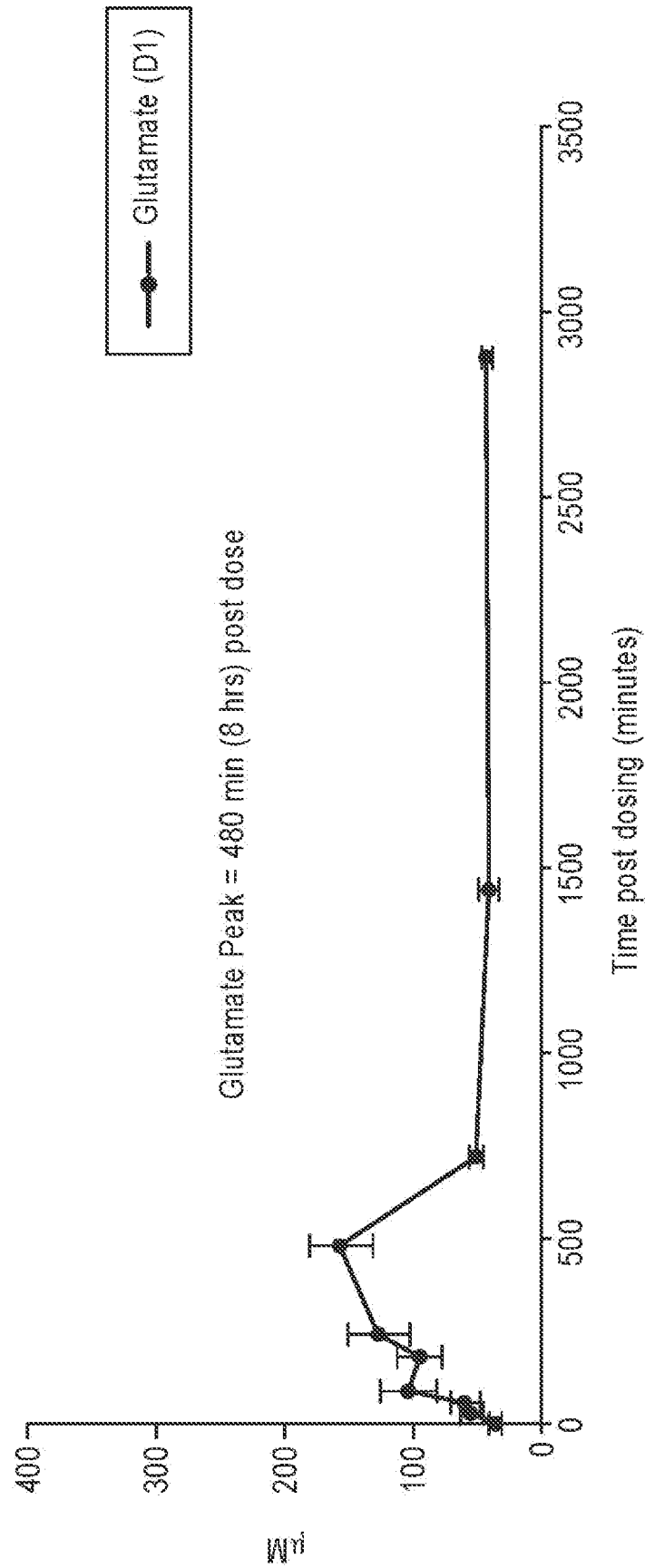


Figure 15

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following Triheptanoic Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

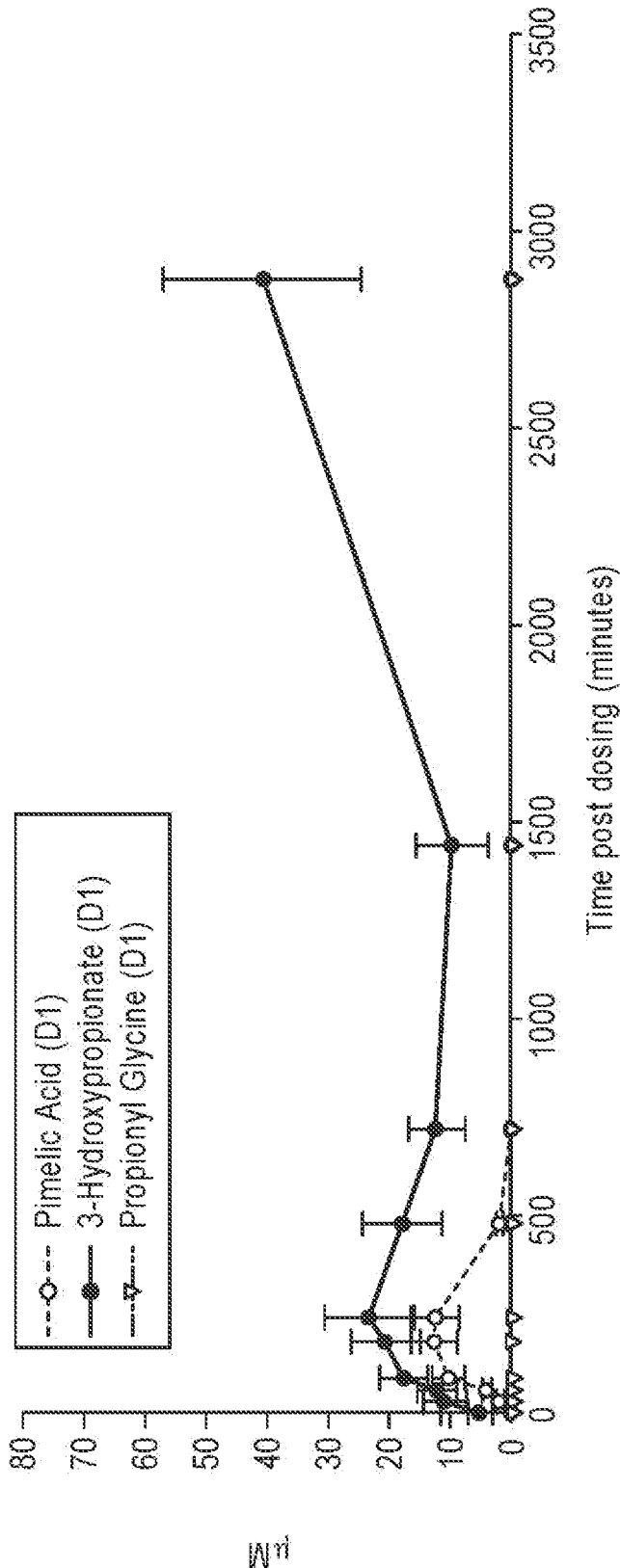


Figure 16

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following Triheptanoil Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

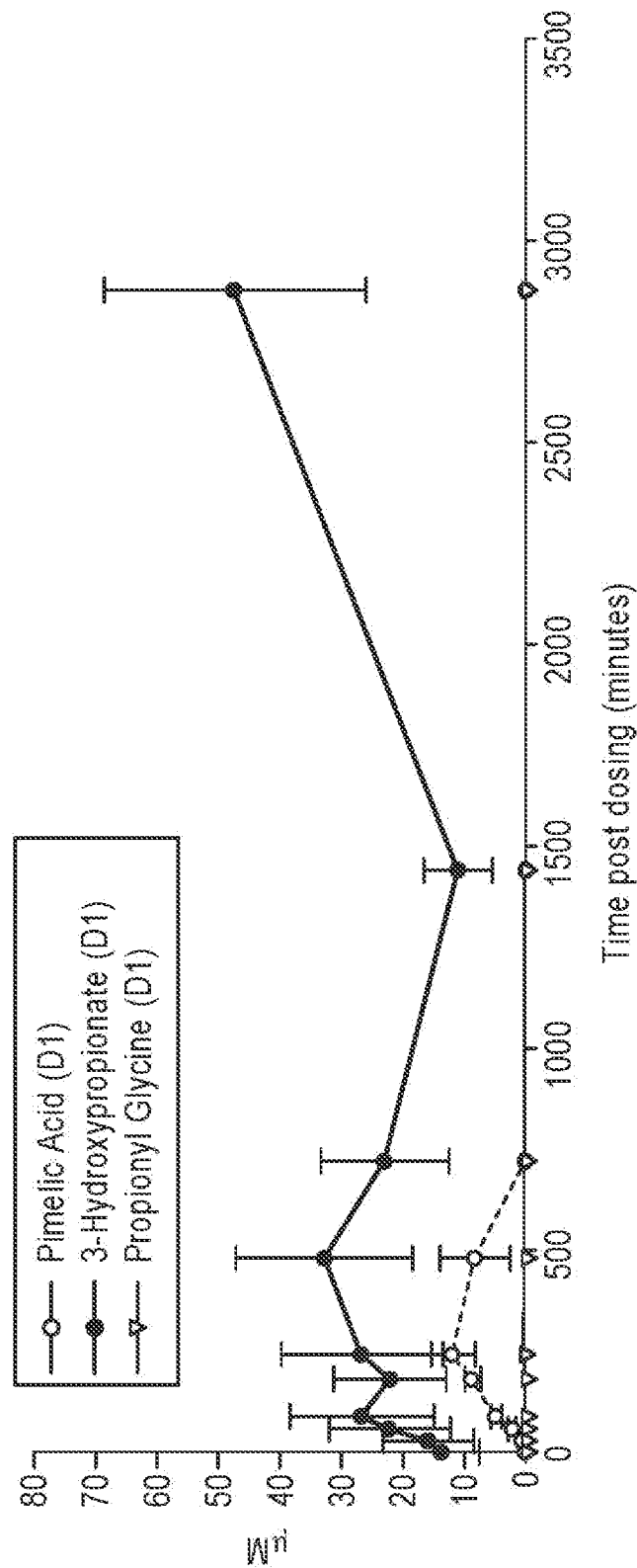


Figure 17

Pimelic Acid, 3-Hydroxypropionate and Propionyl Glycine Metabolite PK Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

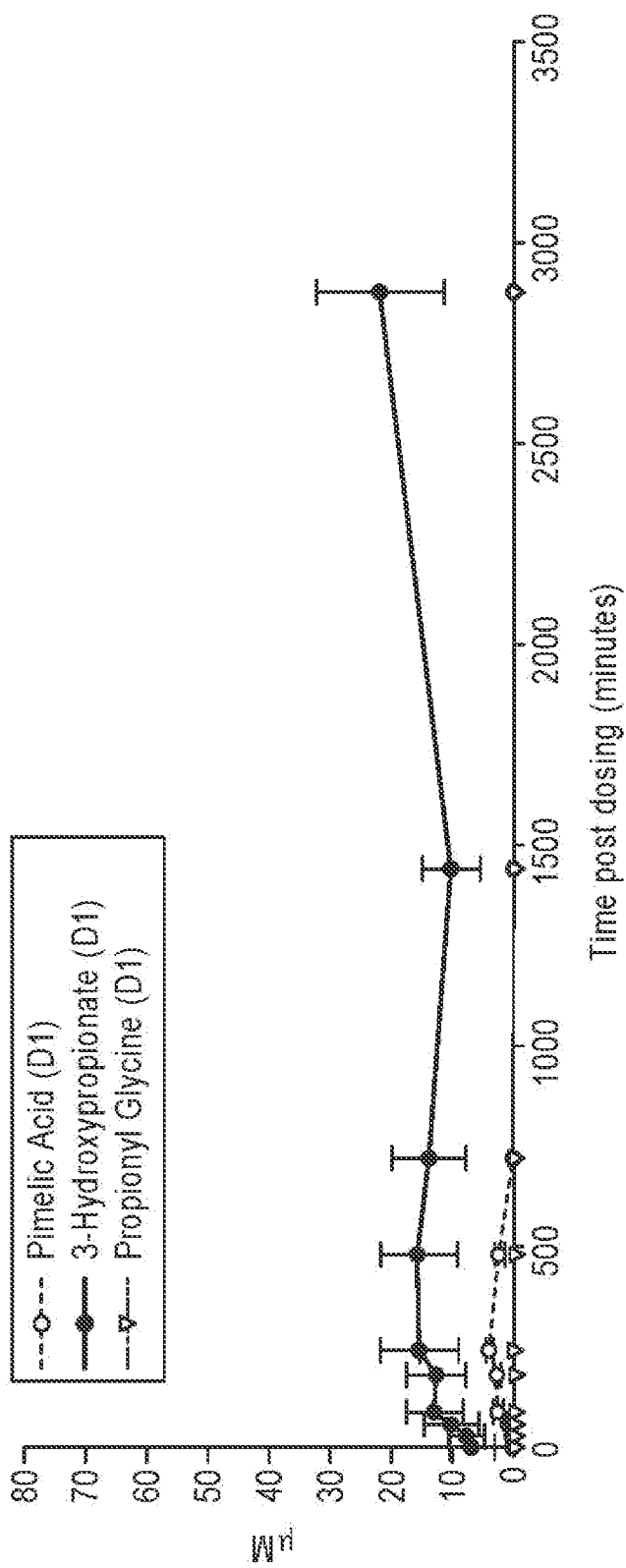


Figure 18

Average Energy and Alternative Metabolites at Each Timepoint Following Triheptanoic Powder-ER Dosing on Day 1 in Male and Female Mini-pigs Combined

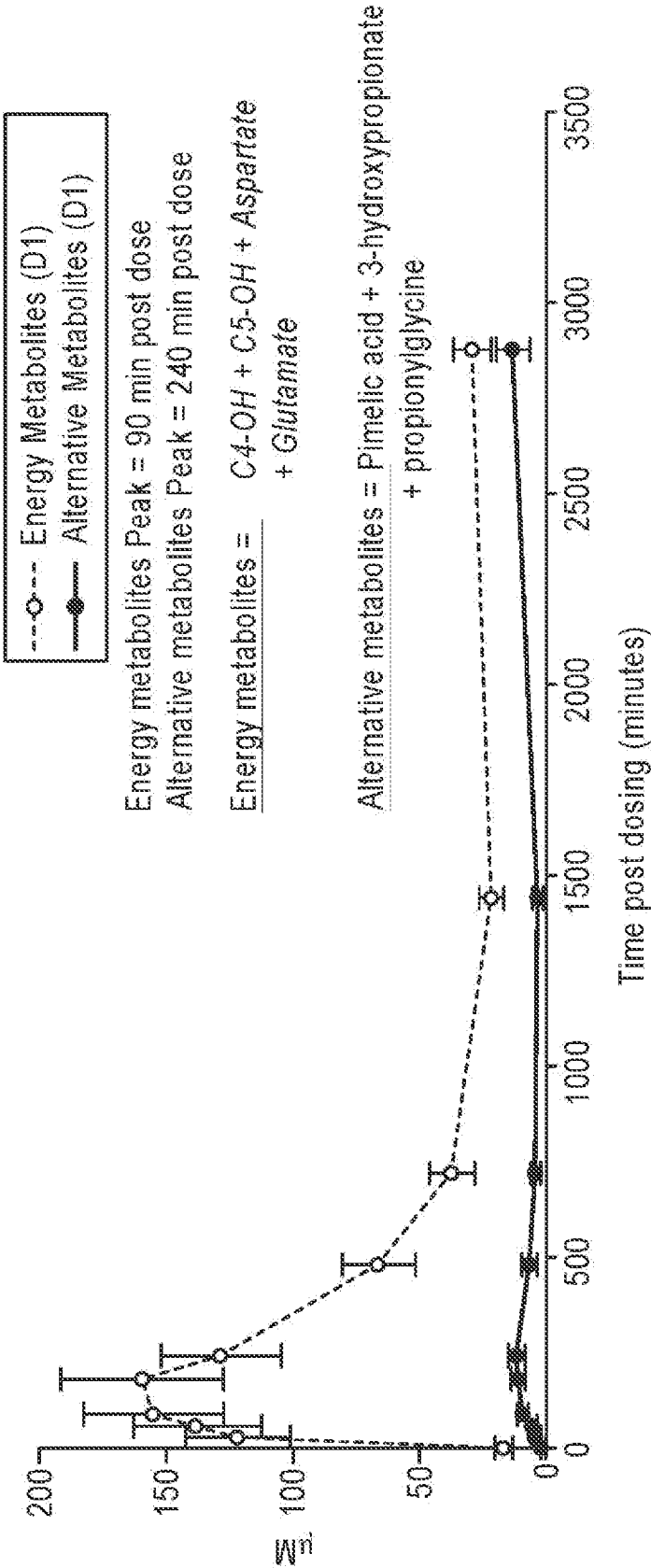


Figure 19

Average Energy and Alternative Metabolites at Each Timepoint Following Triheptanoil Oil Dosing on Day 1 in Male and Female Mini-pigs Combined

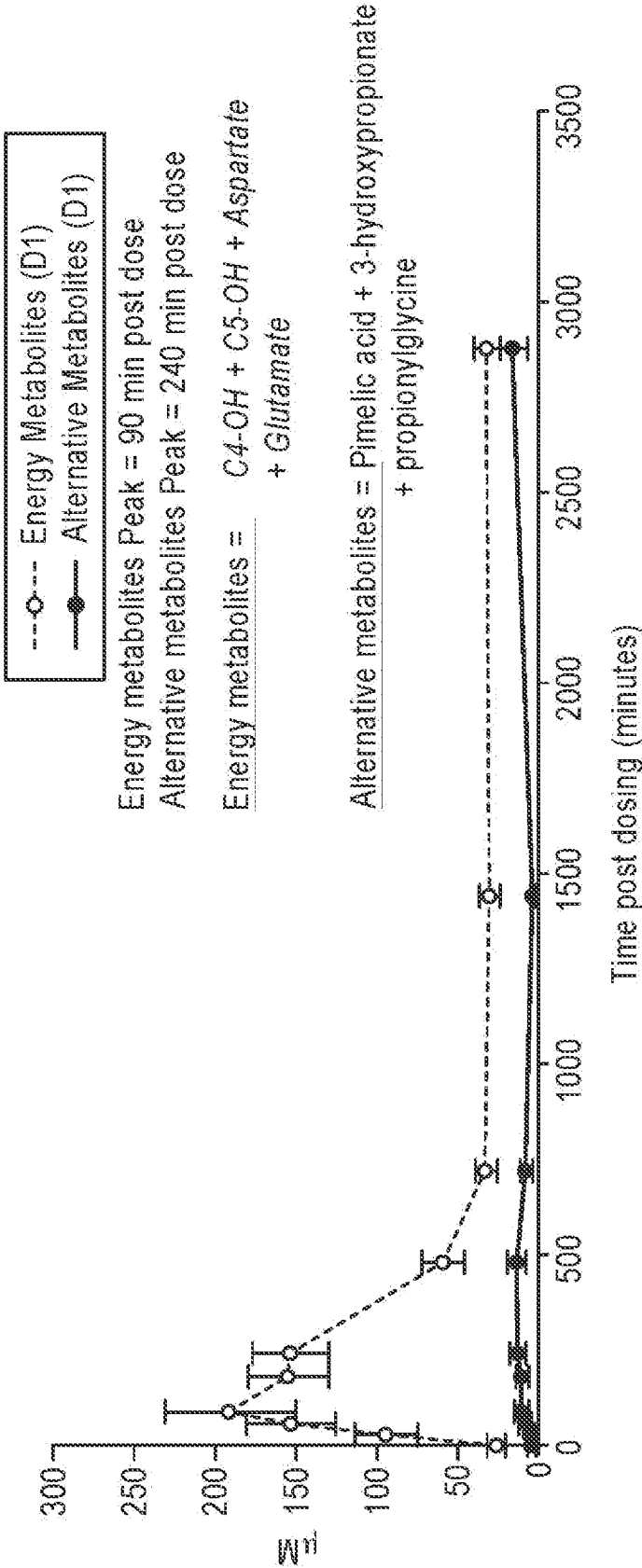


Figure 20

Average Energy and Alternative Metabolites at Each Timepoint Following Triheptanoin Powder Dosing on Day 1 in Male and Female Mini-pigs Combined

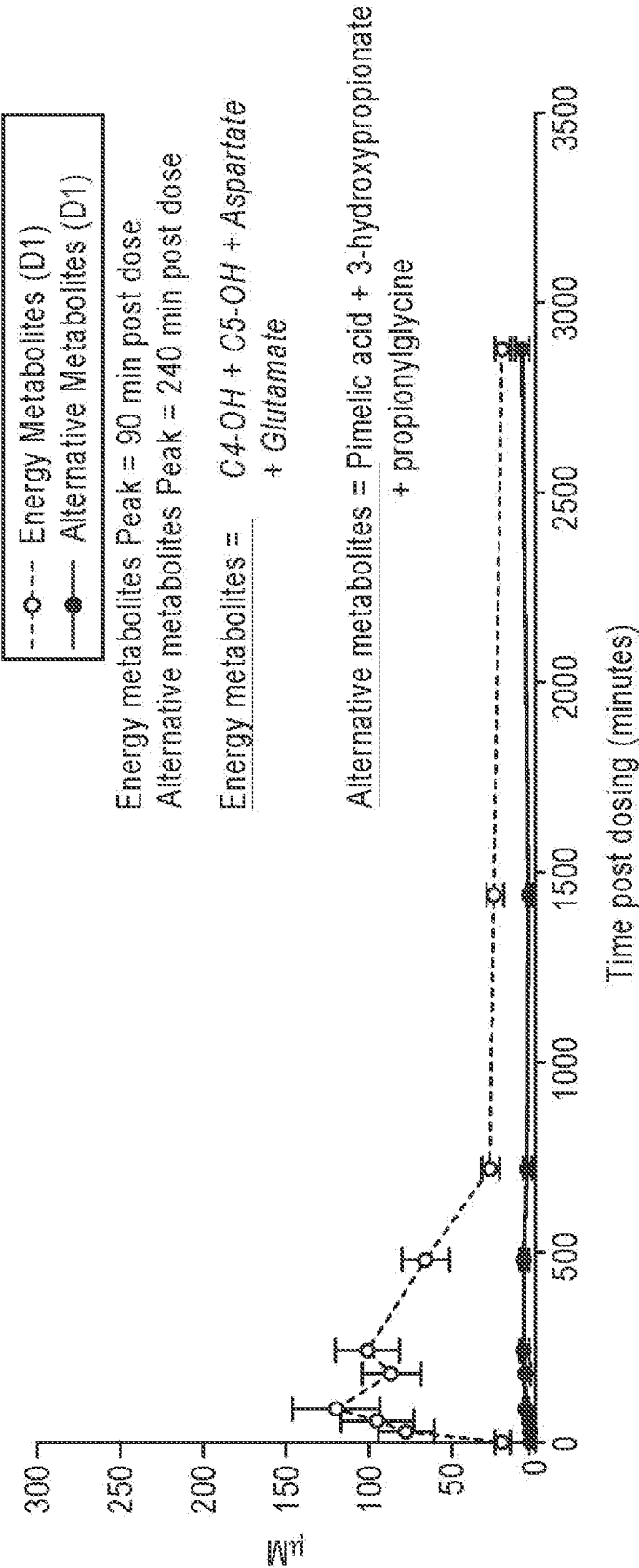


Figure 21

Heptanoic Acid (C7) Metabolite PK Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

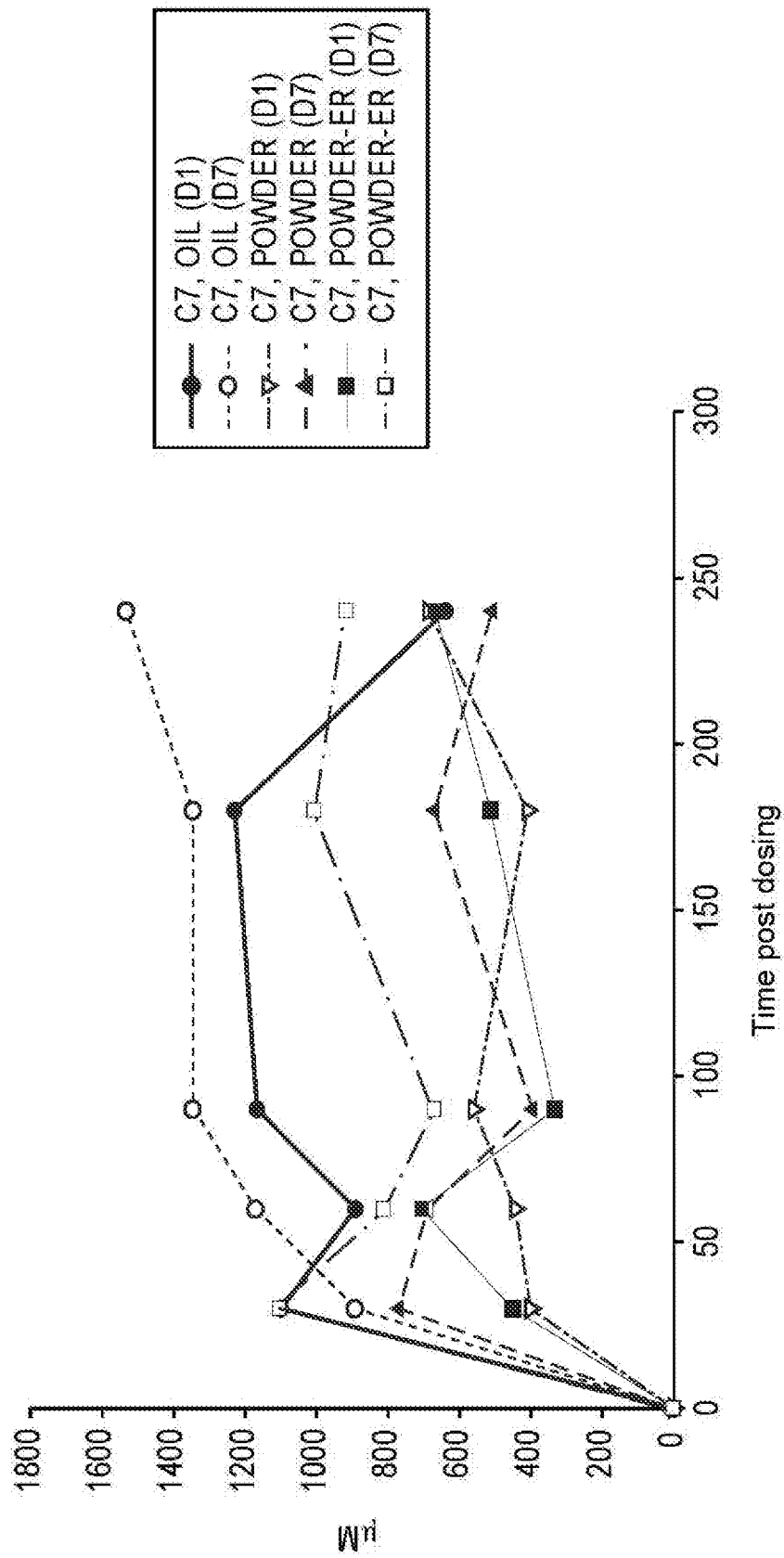


Figure 22

C4 Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

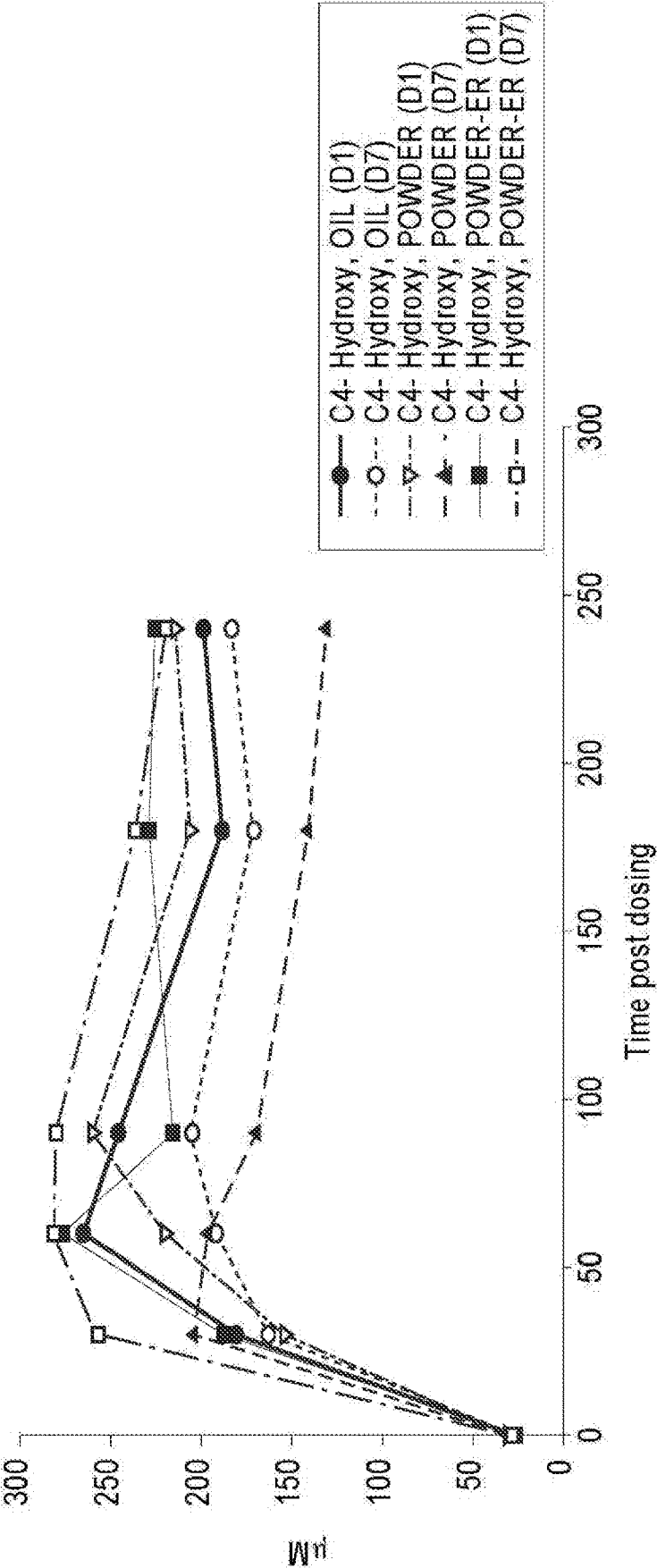


Figure 23

C5 Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

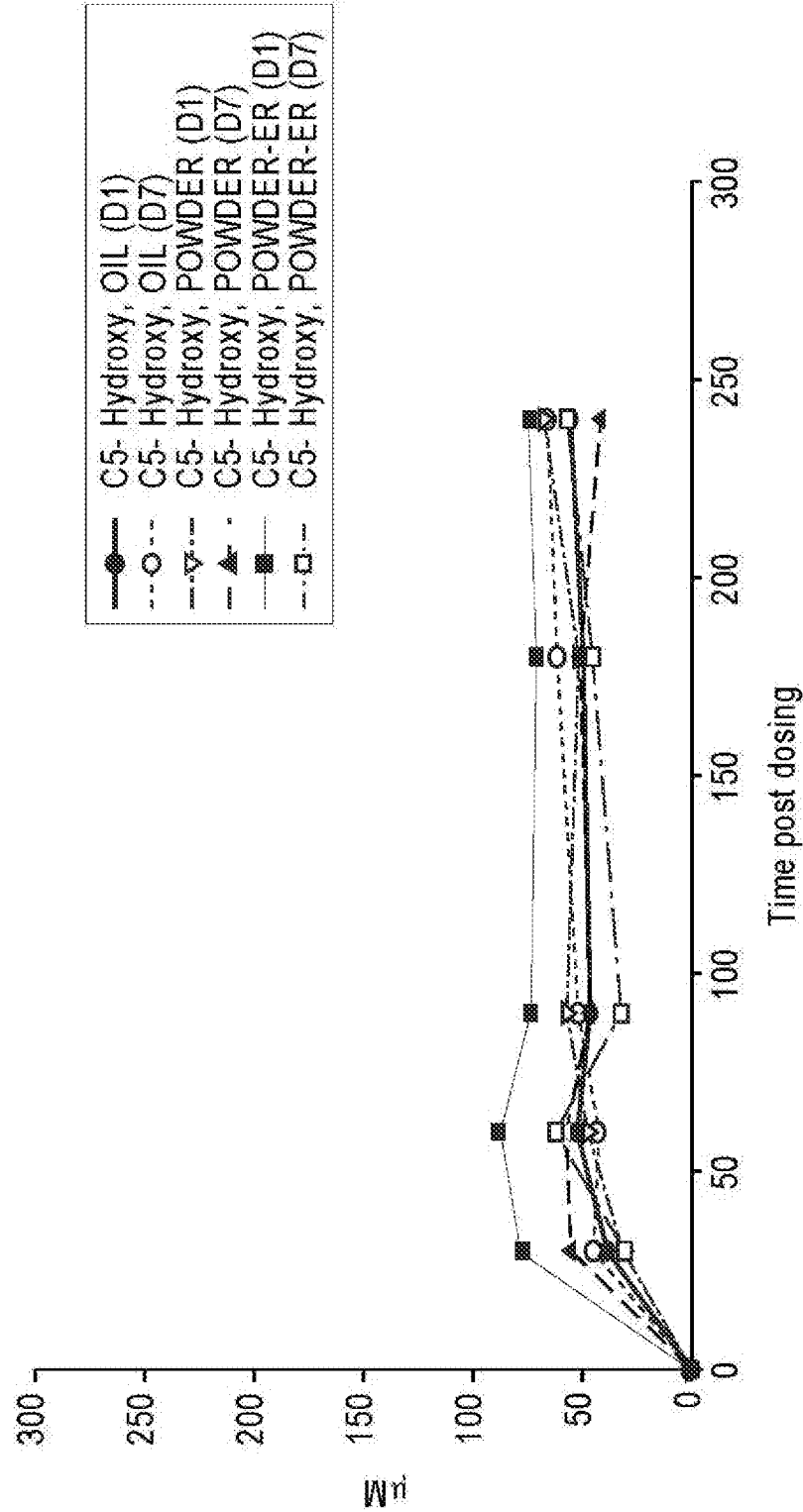


Figure 24

Aspartic Acid Metabolite PK (Normalized to Alanine) Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

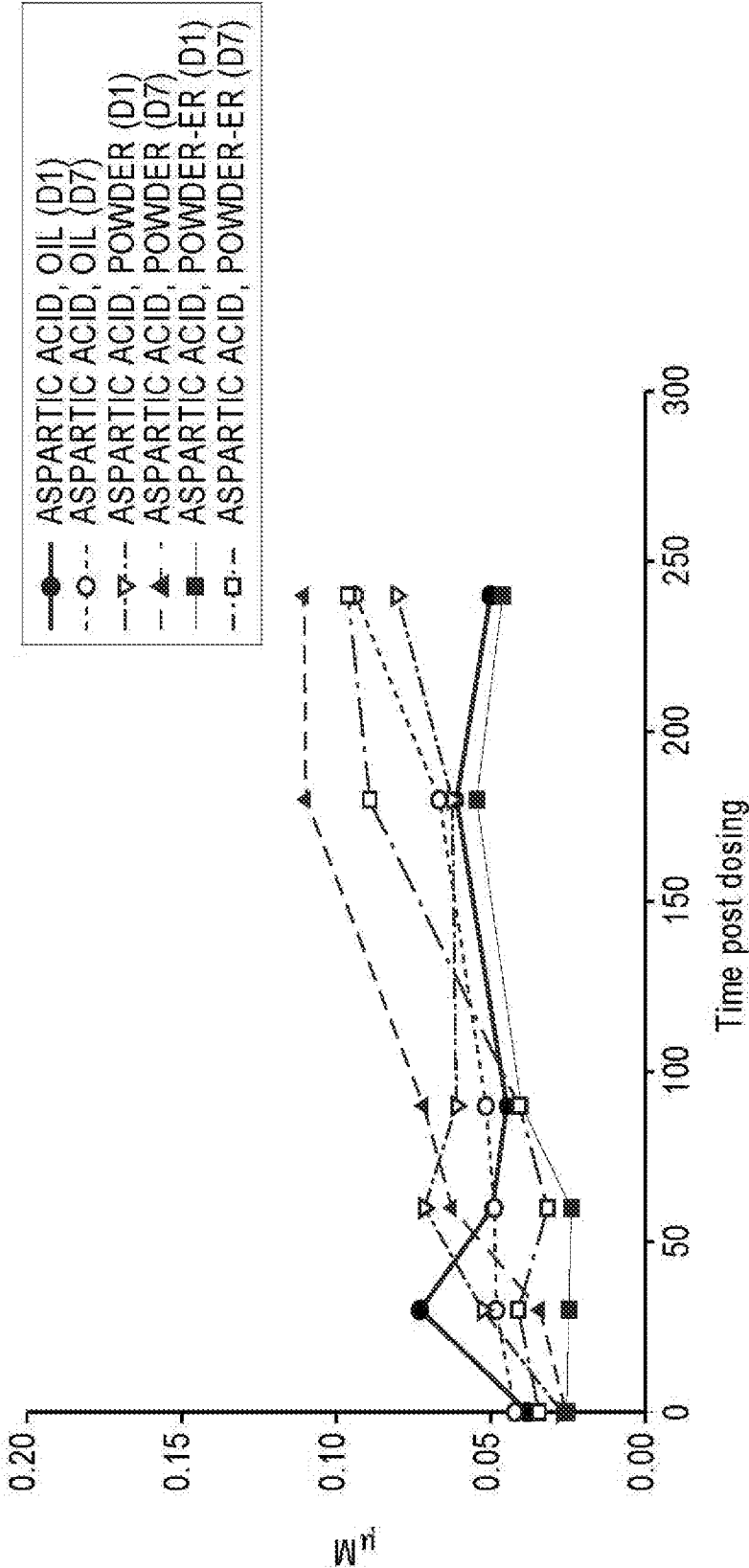


Figure 25

Glutamic Acid Metabolite PK (Relative to Alanine) Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

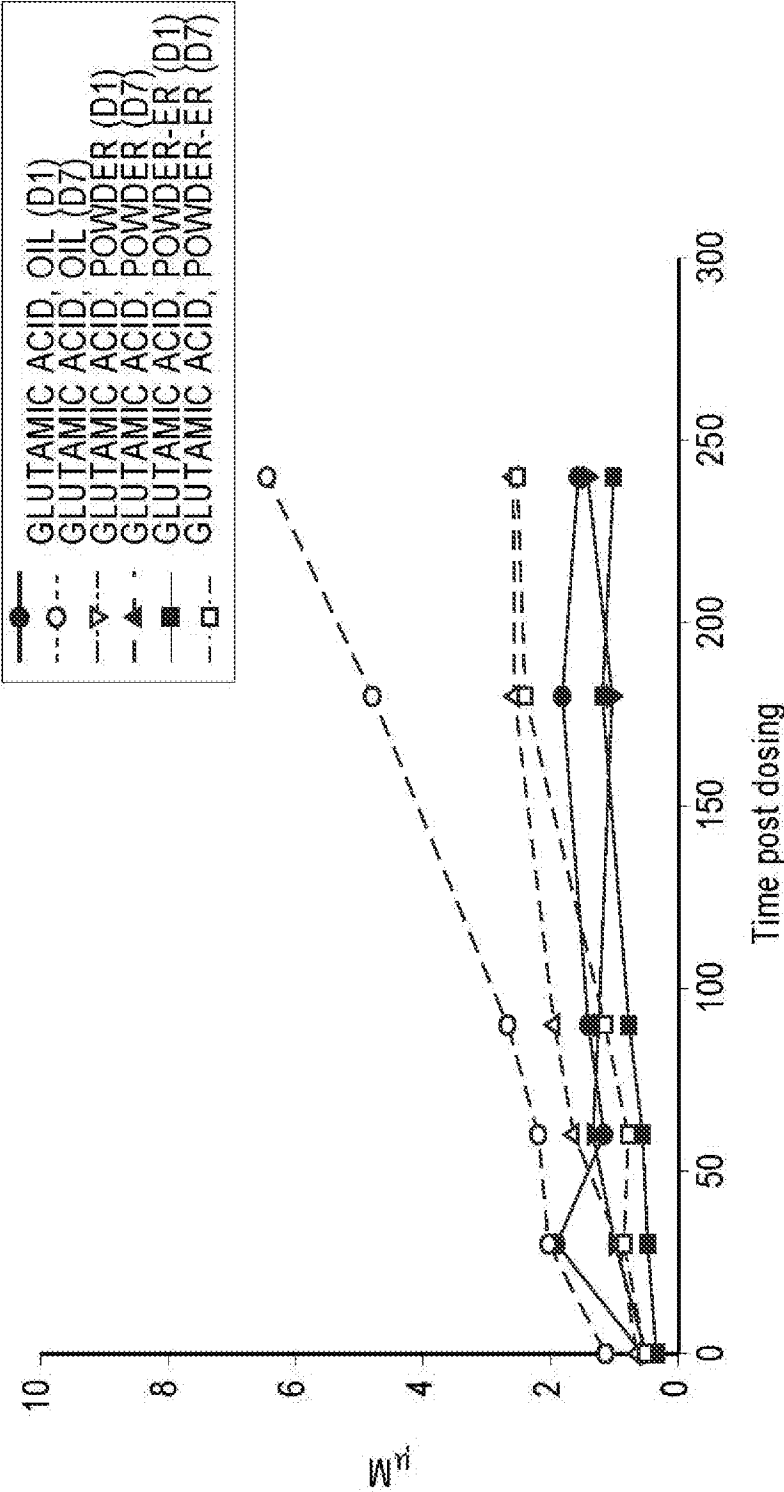


Figure 26

Pimelic Acid Metabolite PK Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

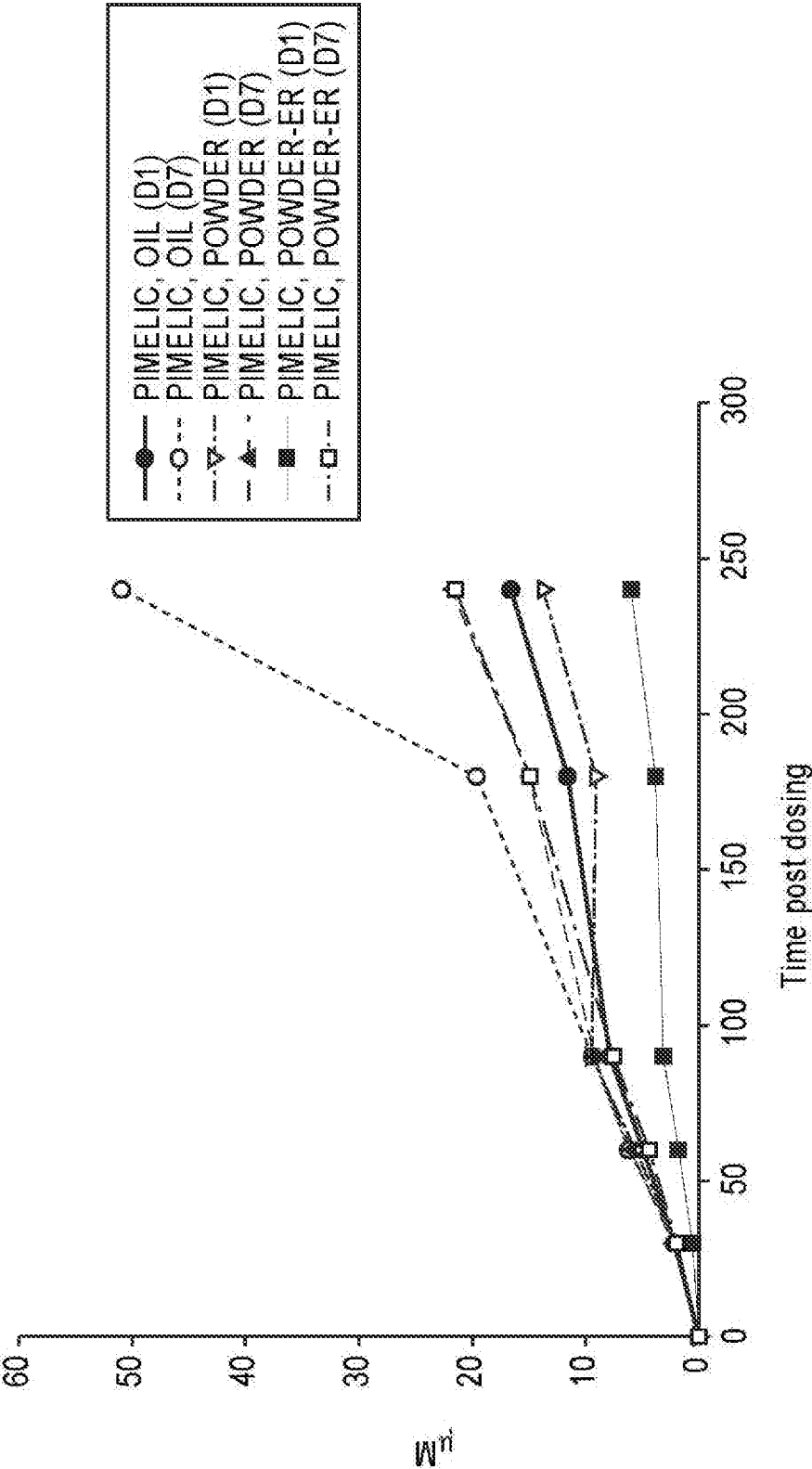


Figure 27

3-Hydroxypropionate Metabolite PK Following Triheptanoil Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

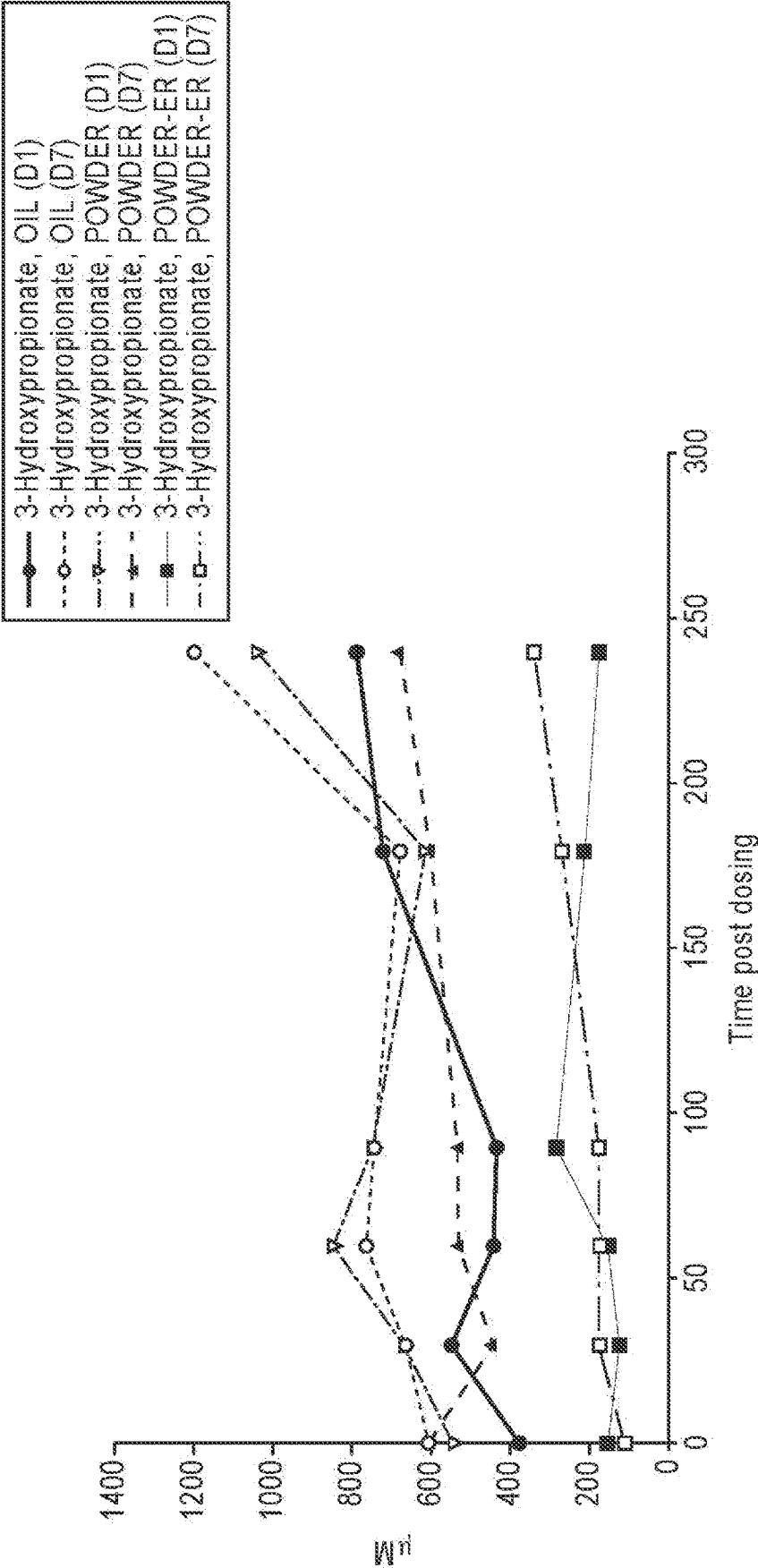


Figure 28

Propionyl Glycine Metabolite PK Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

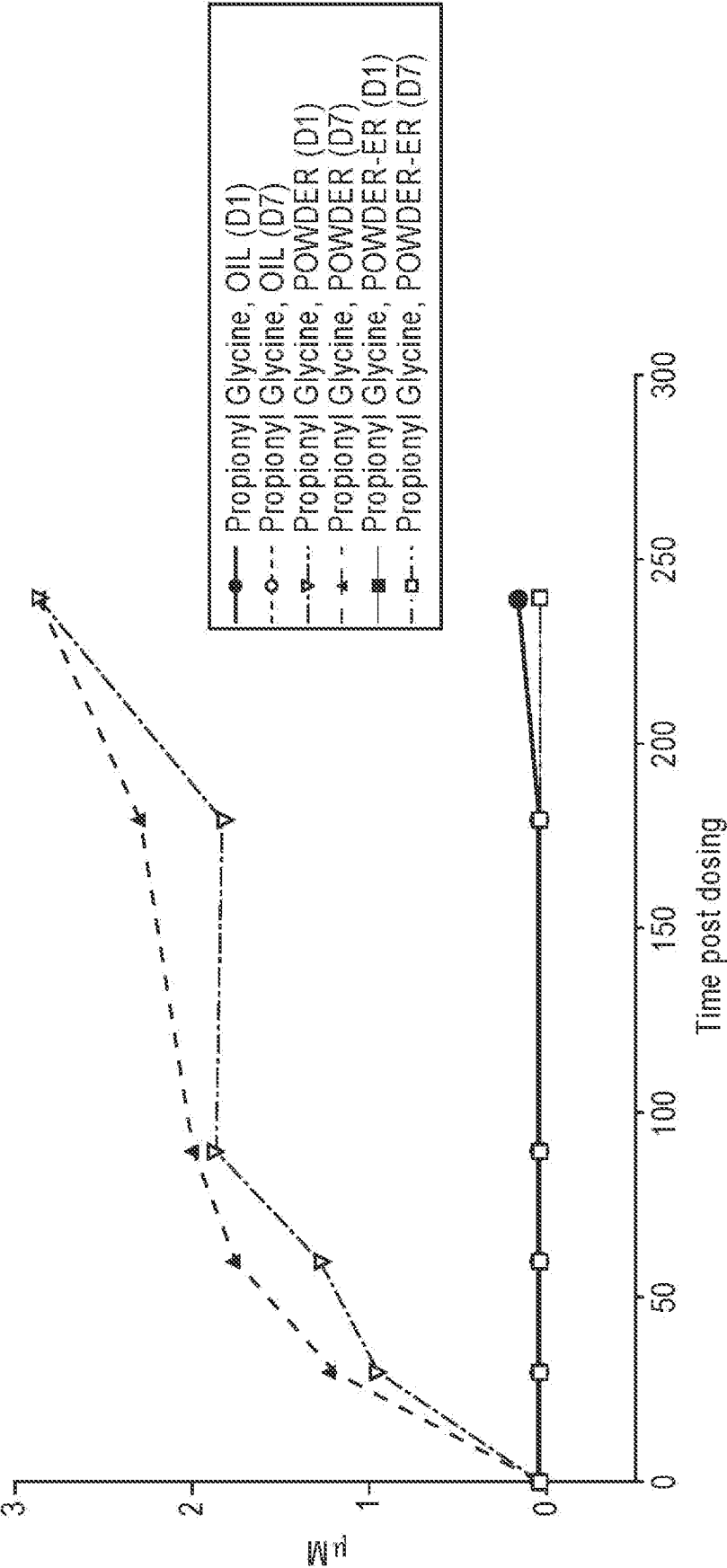


Figure 29

Sum of all Metabolites During PK Timepoints Following Triheptanoin Oil, Powder and Powder-ER dosing on Day 1 and Day 7 in Male/Female Combined Mini-pigs

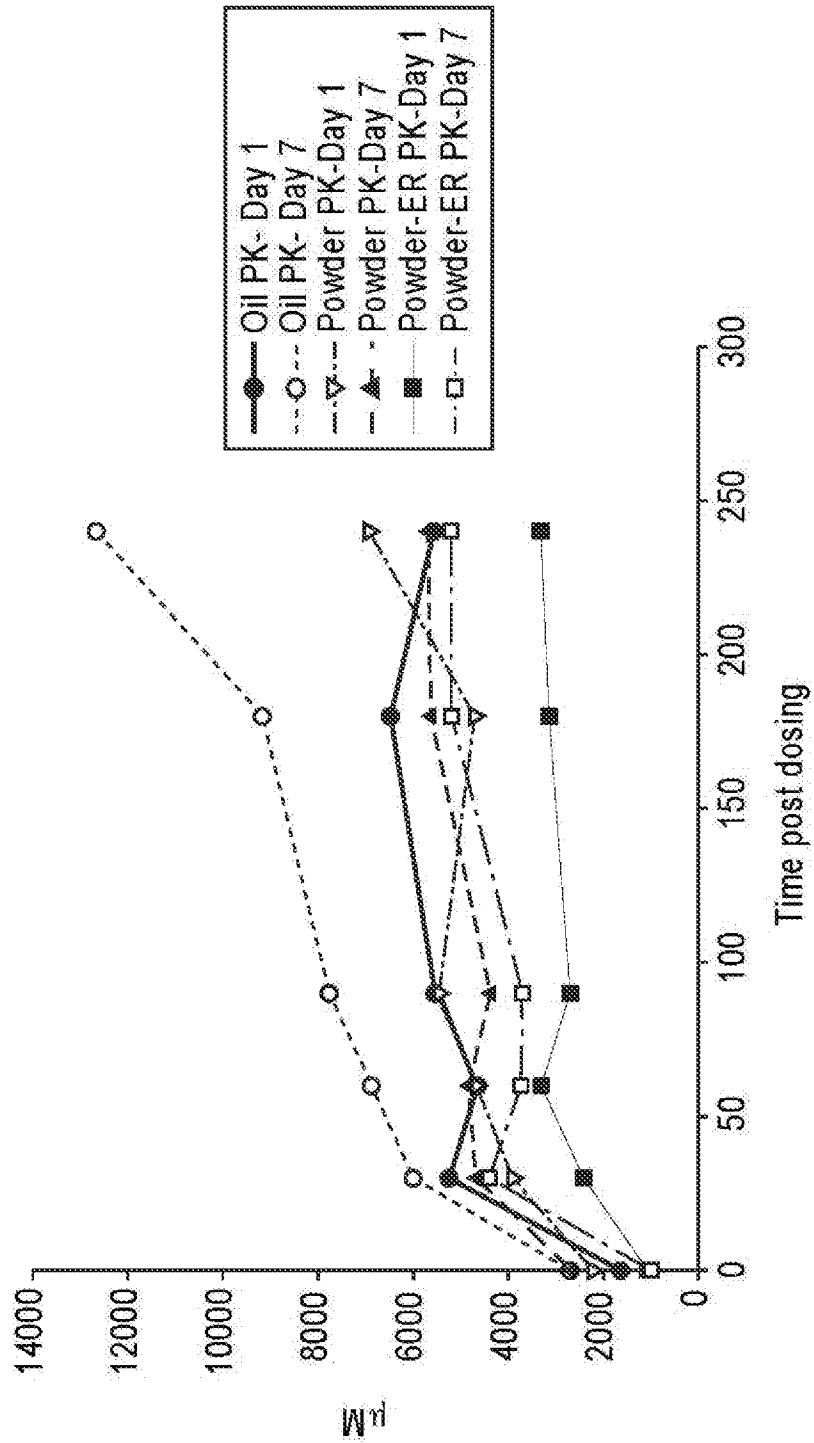


Figure 30

C4-Hydroxy Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

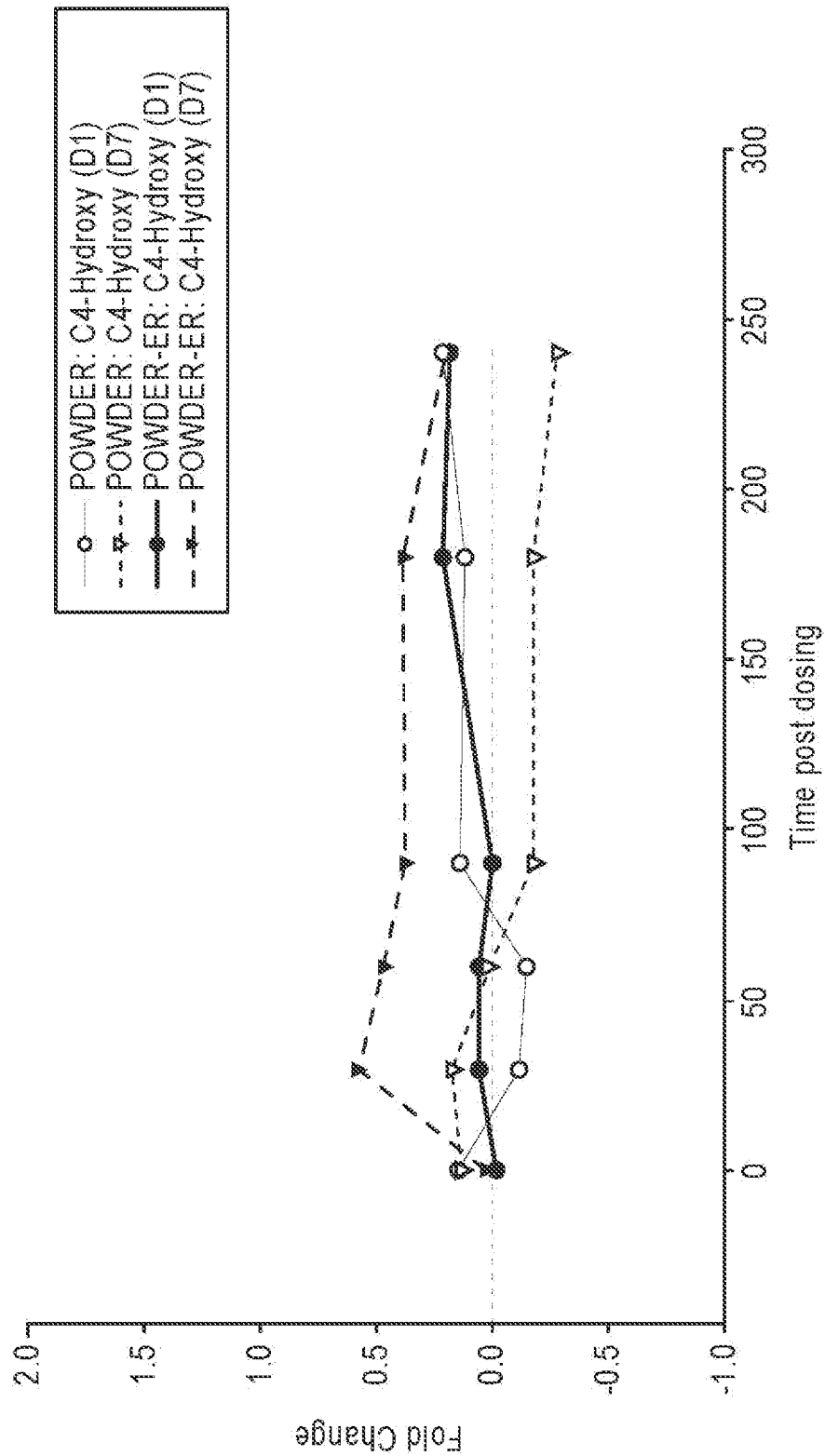


Figure 31

C5-Hydroxy Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

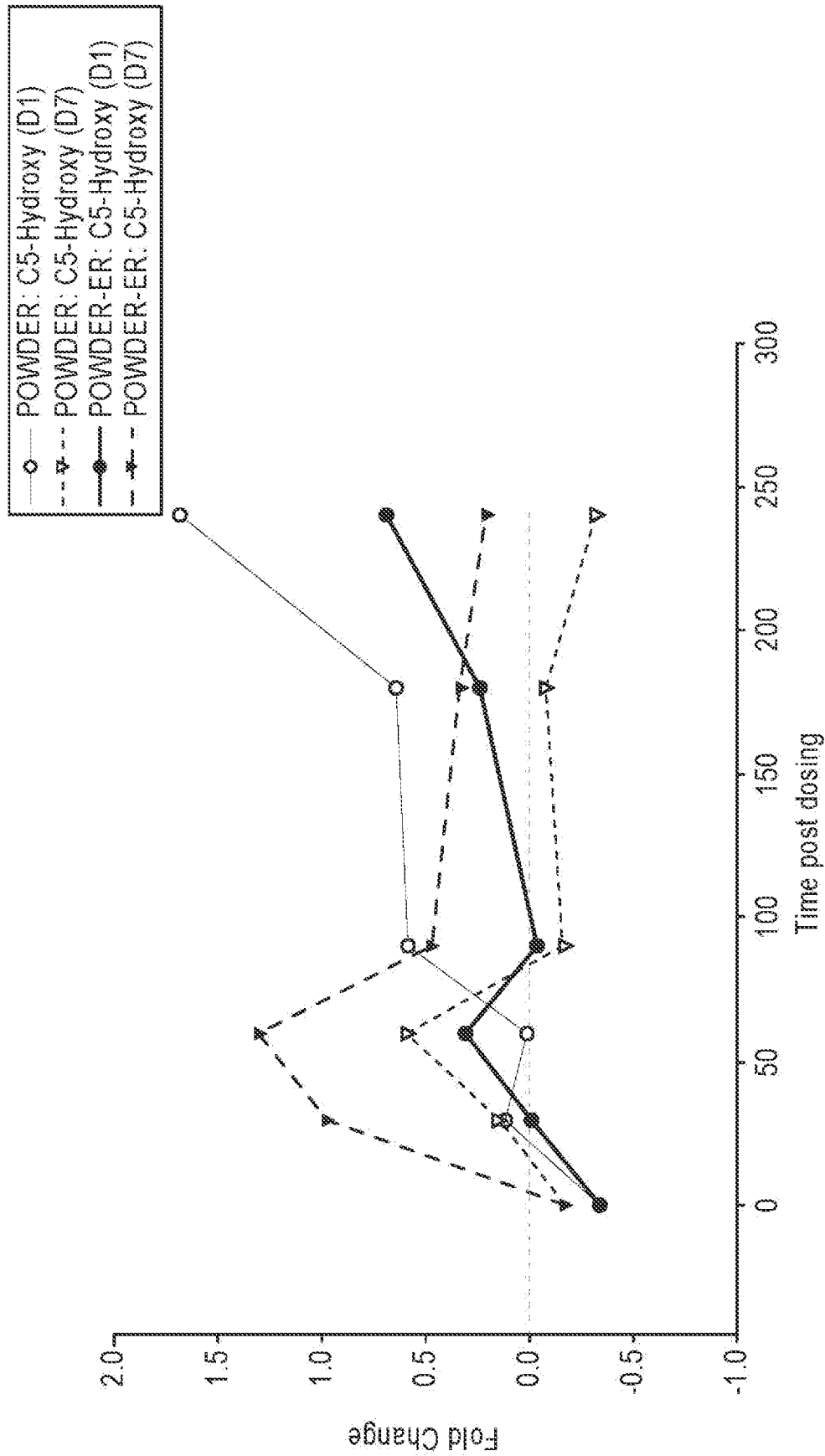


Figure 32

Heptanoic Acid Metabolite PK Following Triheptanoic Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

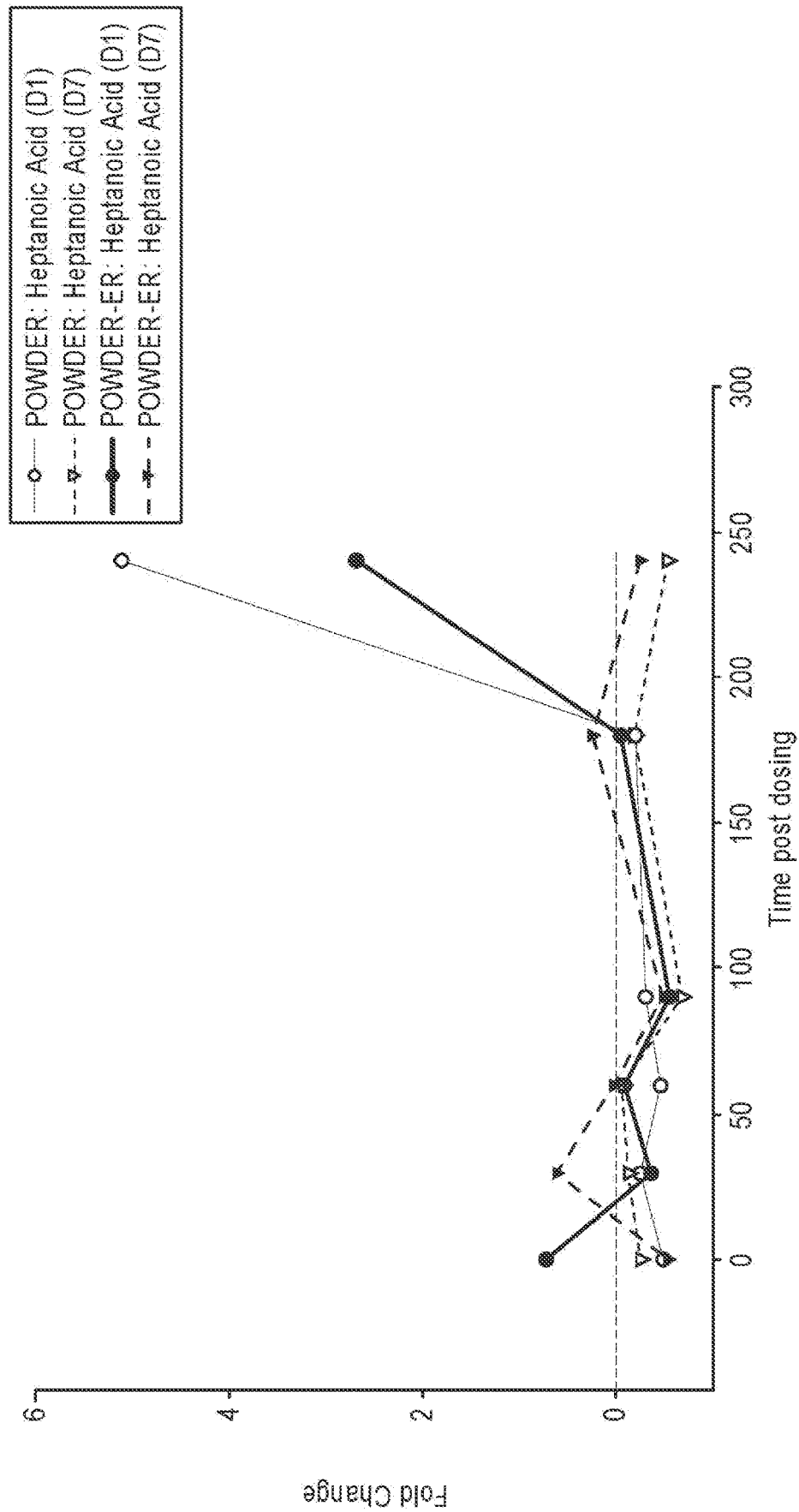


Figure 33

Pimelic Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

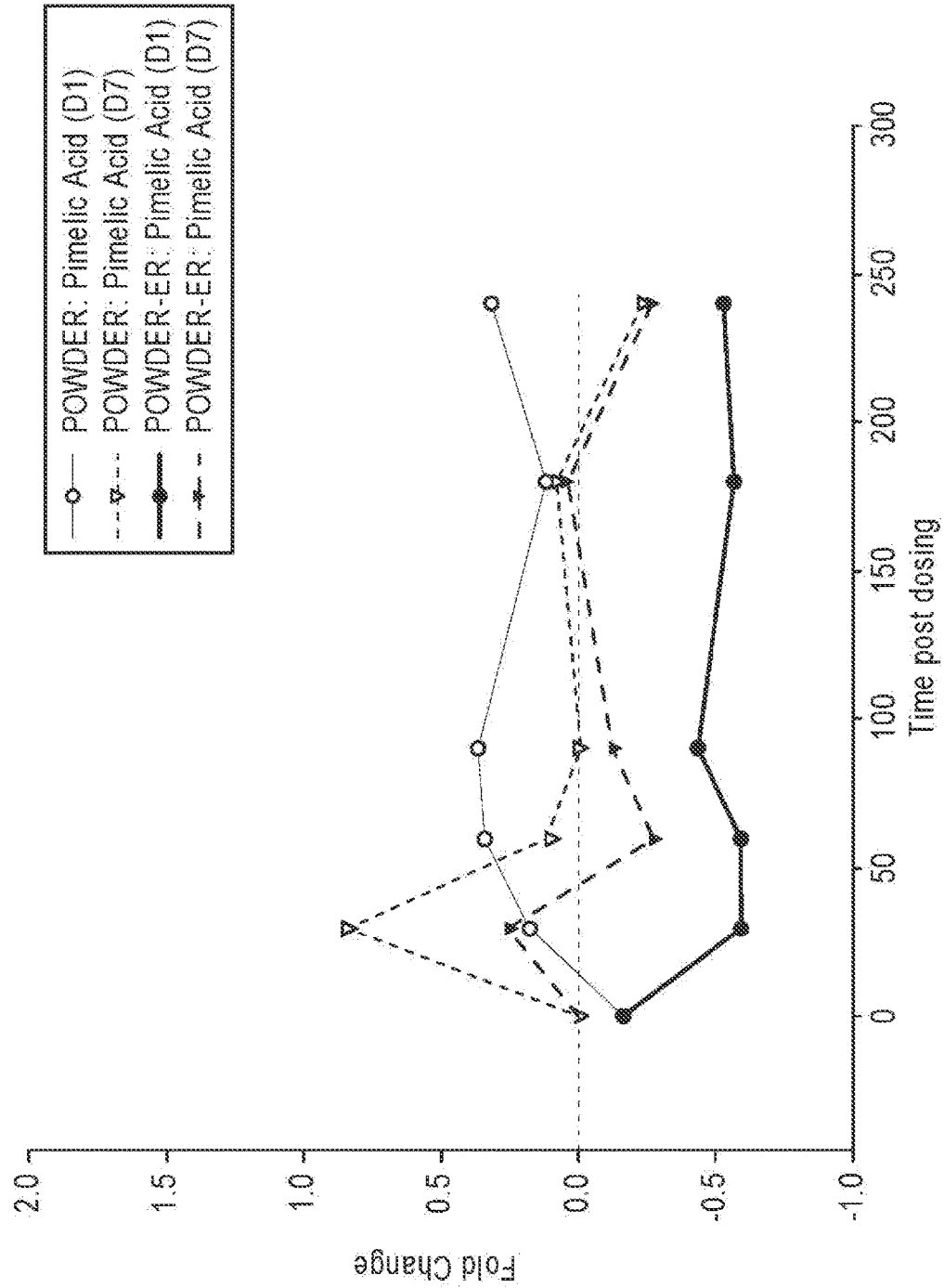


Figure 34

Glutaric Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

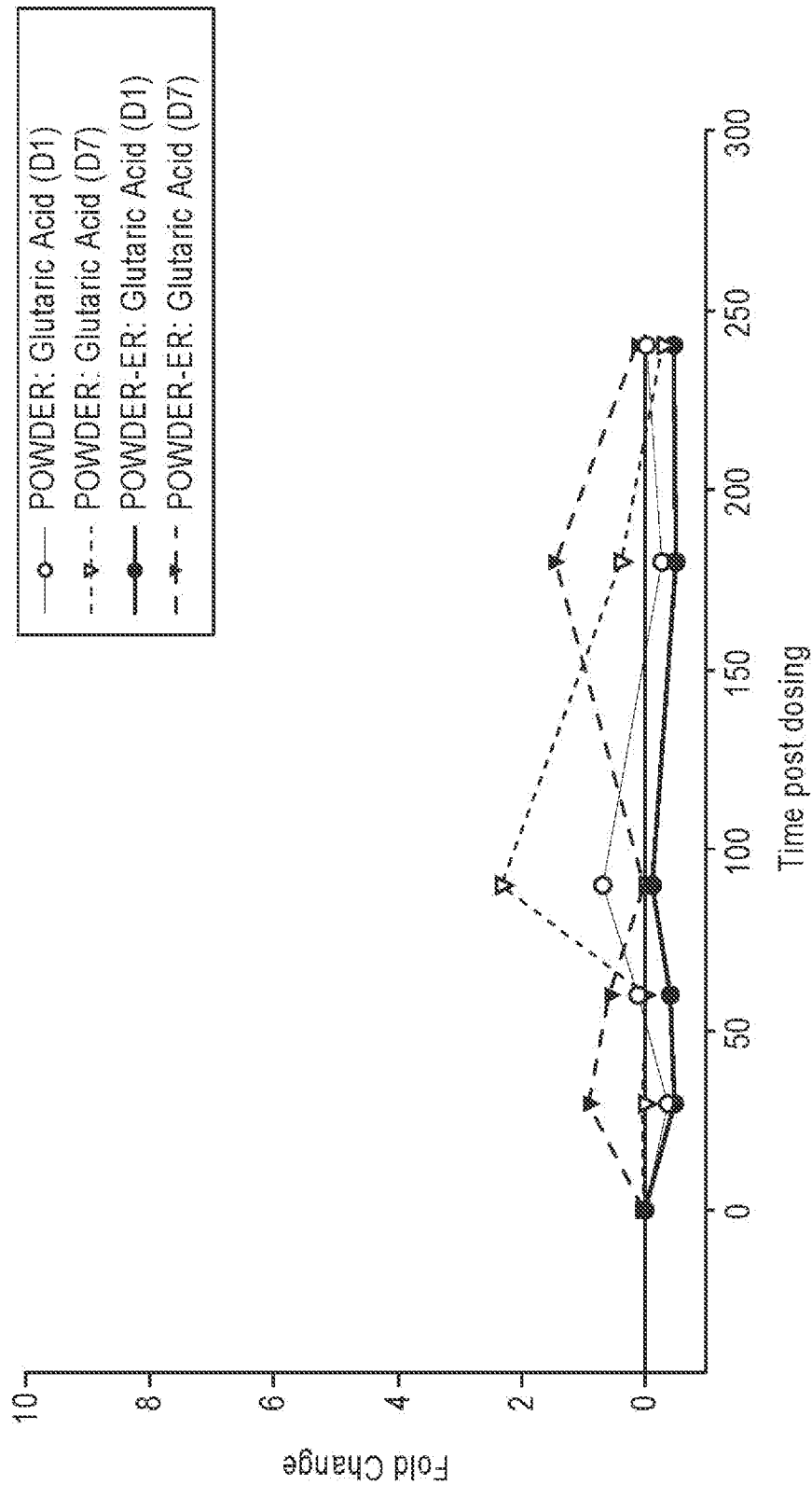


Figure 35

Aspartic Acid Metabolite PK Following Triheptanoil Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

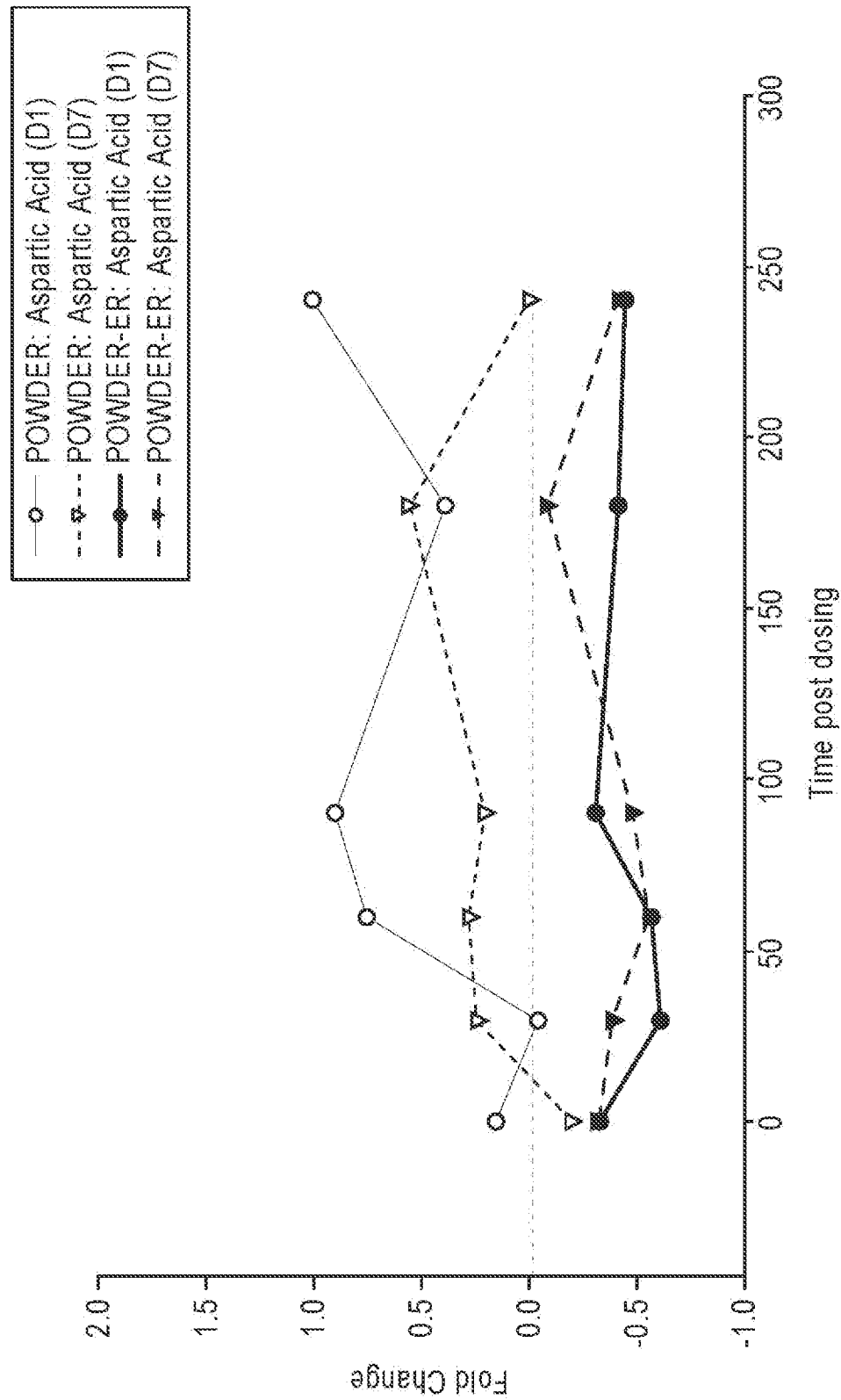


Figure 36

Glutamic Acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

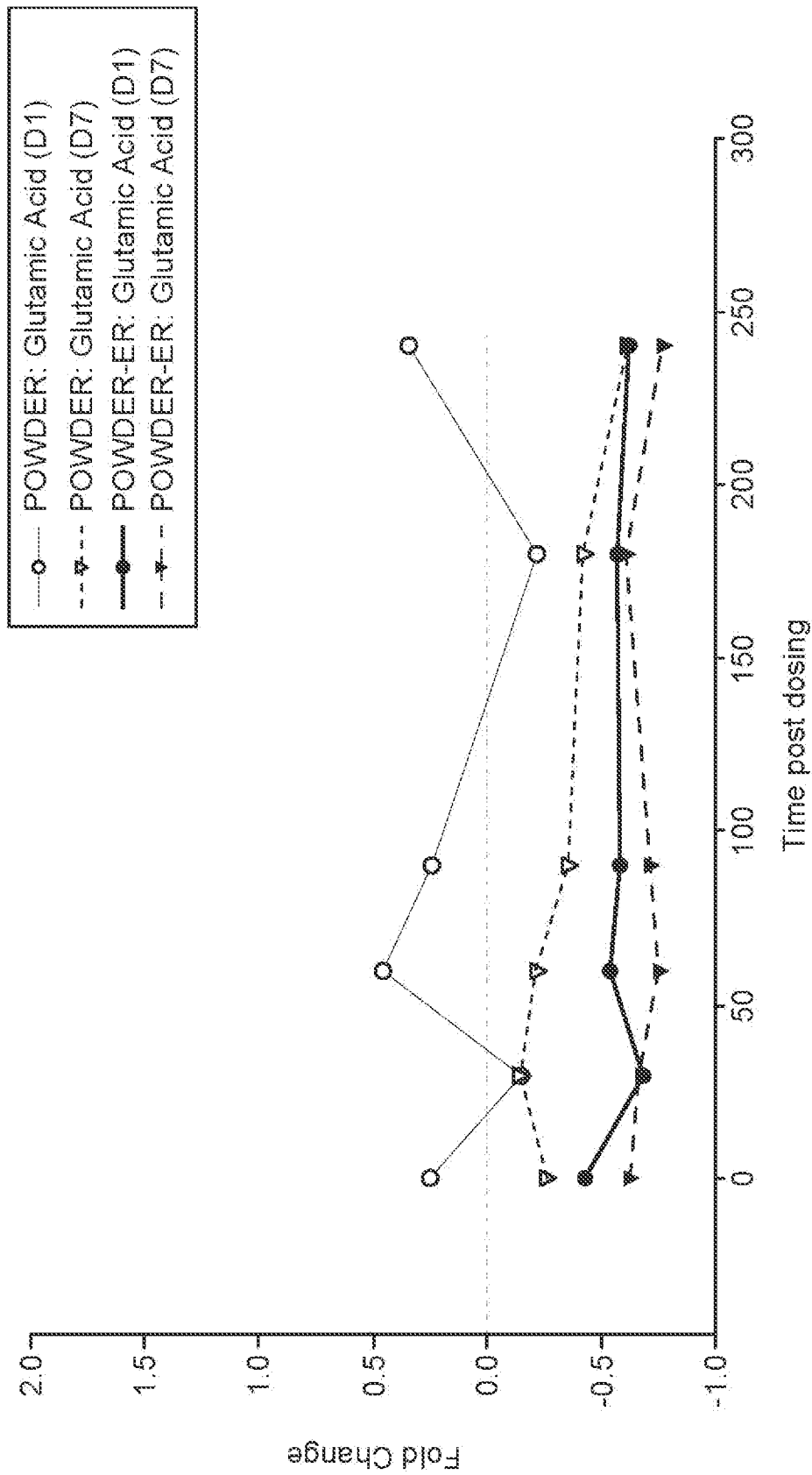


Figure 37

Alanine Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing
(Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

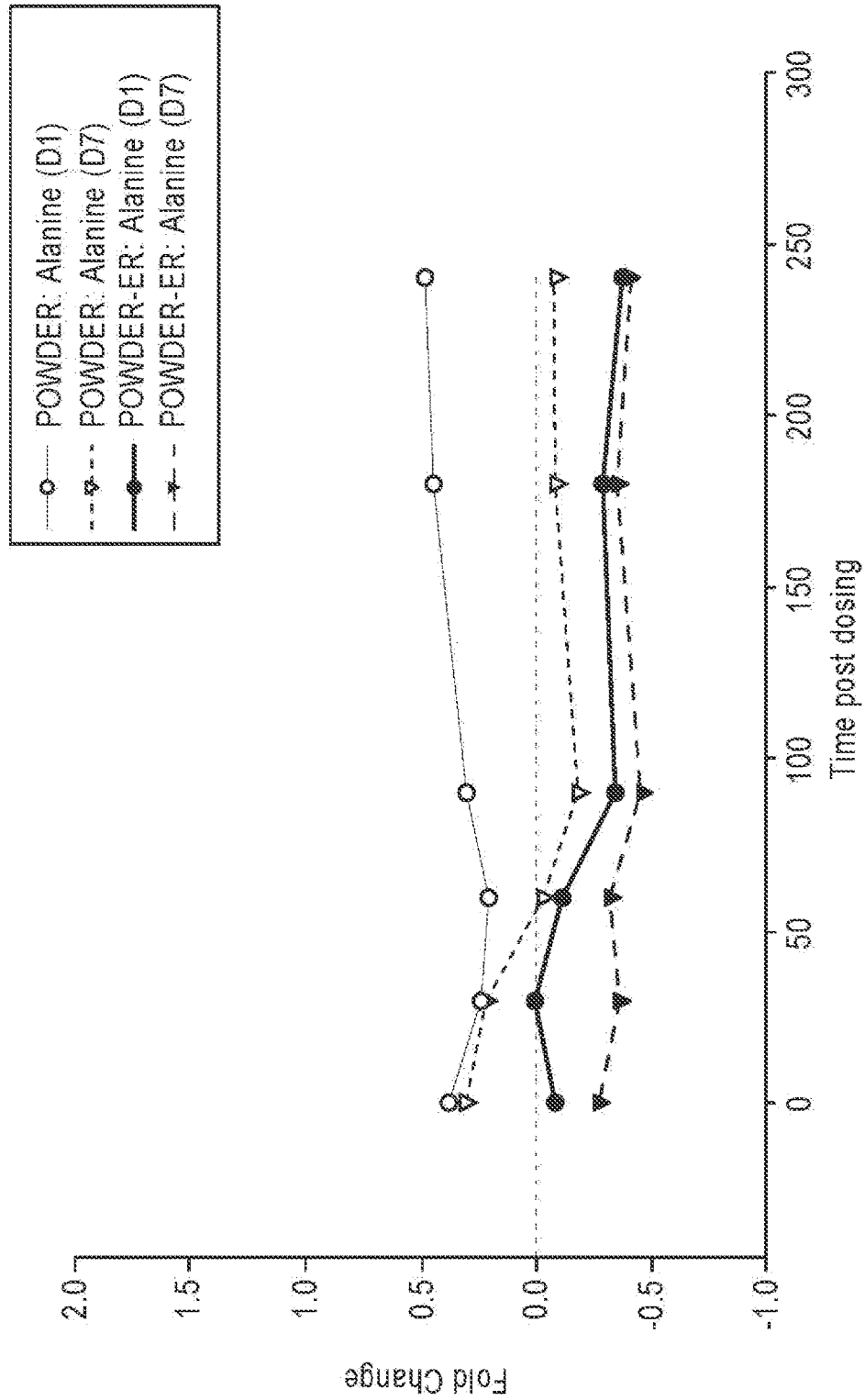


Figure 38

3-Hydroxypropionic acid Metabolite PK Following Triheptanoin Powder and Powder-ER Dosing (Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

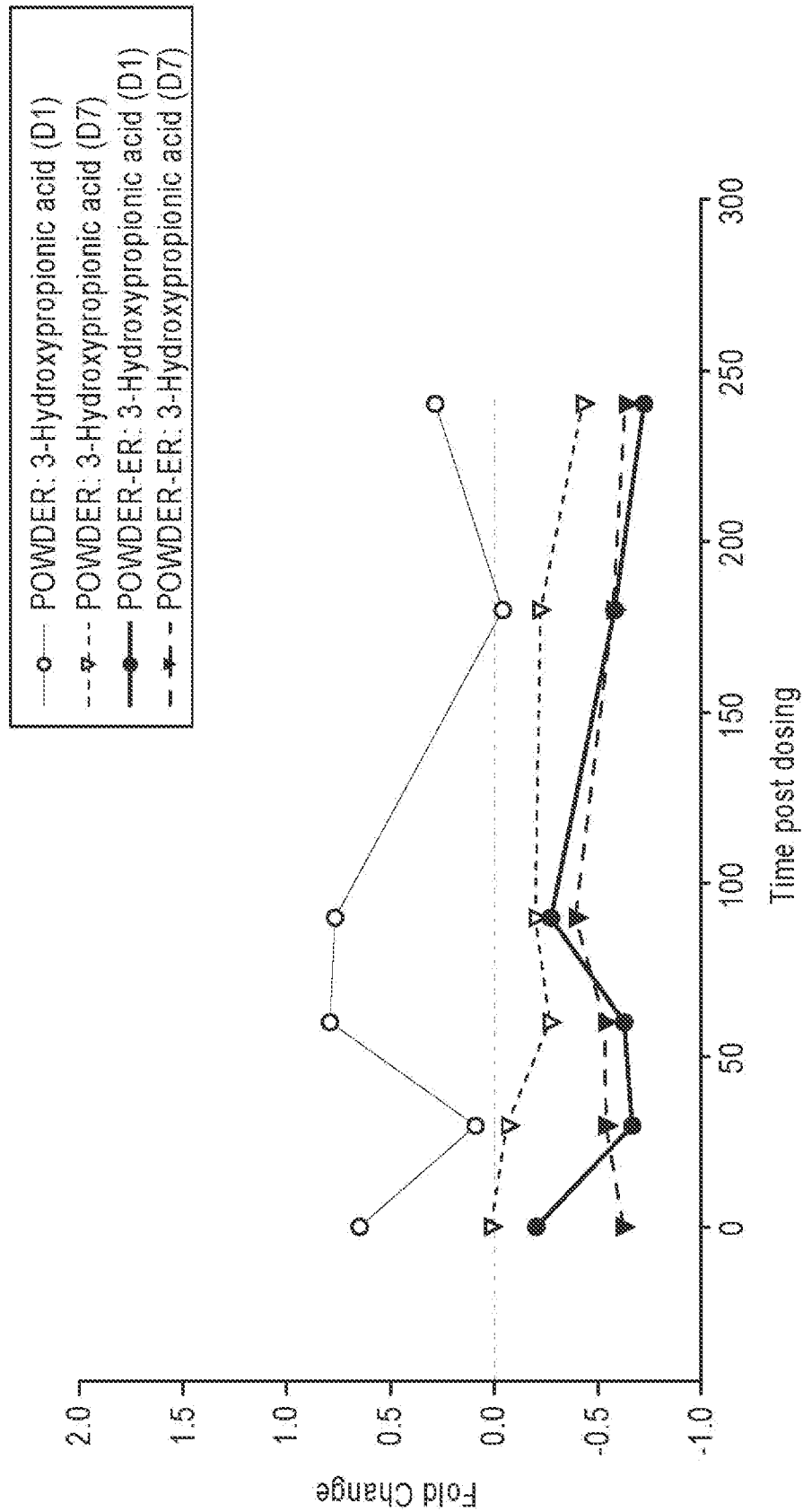


Figure 39

Propionyl Glycine Metabolite PK Following Triheptanoic Powder and Powder-ER
Dosing (Fold Change Relative to Oil Dosing) on Days 1 and 7 in Male and Female Mini-pigs

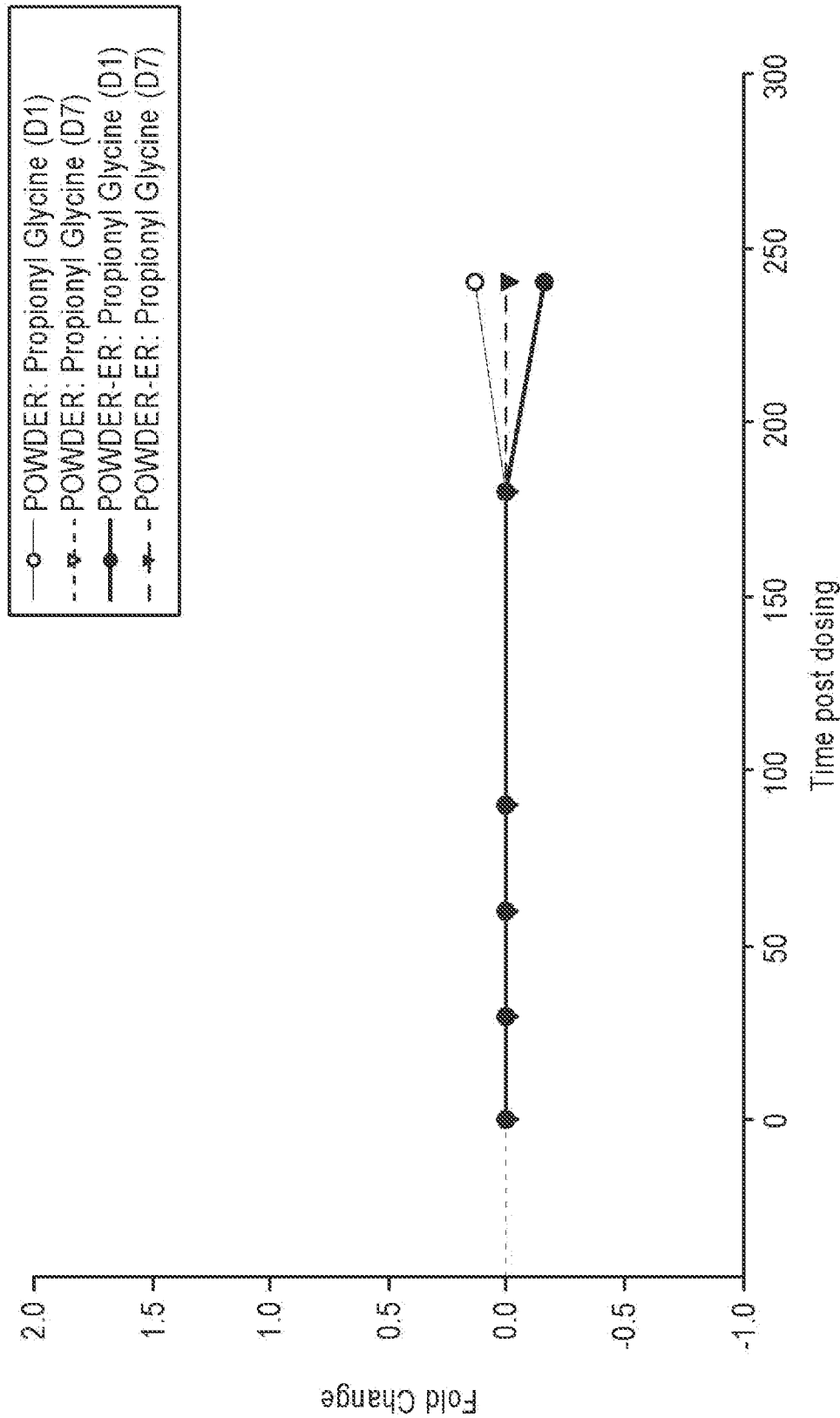


Figure 40

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**COMPOSITIONS OF TRIGLYCERIDES AND
USES THEREOF****CROSS-REFERENCE TO RELATED
APPLICATION**

This application is a Continuation of U.S. application Ser. No. 16/445,557, filed on Jun. 19, 2019, which is a Continuation of U.S. application Ser. No. 14/419,601, filed on Feb. 4, 2015, which is a U.S. National Stage Application of International Application No. PCT/US2014/065693, filed on Nov. 14, 2014, which claims the benefit of U.S. Provisional Patent Application No. 61/904,369 filed Nov. 14, 2013, the contents of each of which are herein incorporated by reference in their entirety for all purposes.

FIELD OF THE INVENTION

The present invention relates to solid compositions of triglycerides with one or more fatty acids, such as triheptanoin and glycerol phenylbutyrate, and the therapeutic use thereof.

BACKGROUND OF THE INVENTION

Current dosage form of the triglycerides drug, e.g., triheptanoin, is a liquid with the consistency of oil. Several problems are associated with the liquid dosage form due to its physical and physiological properties. First, the liquid dosage forms, such as oil, are difficult to be administered due to low miscibility with food or drinks. Second, the oil formulations are difficult to be handled, carried, and dispensed. Furthermore, upon administration, the oil composition is hydrolyzed and released rapidly in the stomach leading to gastric upset, gastric retention, and likely gastrin-related stomach spasm and emesis. In addition, the oil formulations can cause diarrhea by reforming oil droplets and causing a mineral-oil like excess lubrication. Upon repetitive administration of the oil, some patients experience gastric stress which causes vomiting and/or diarrhea. Thus, the tolerability in the oil form can be a dose-limiting toxicity or lead to adverse effects, and the reduced administrable dose would negatively impact the treatment effect for a fatty acid oxidation disorder or deficiency (FAOD); adult polyglucosan body disease; a mitochondrial fat oxidation defect; a glycogen storage disease; a mitochondrial myopathy; glucose transporter type 1 (GLUT1) deficiency syndrome, or other related diseases.

SUMMARY OF THE INVENTION

Among other things, the present invention is directed to a solid composition comprising an ester derived from a polyol and one or more fatty acids as an active ingredient, and a solid substance. To give an example of the ester, triglycerides with one or more fatty acids are suitable for use in accordance with the present invention.

In one aspect, the present invention provides a solid composition comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides. The present invention also provides a solid composition comprising a plurality of solid particles, each

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particle comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient adsorbed onto a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides.

In another aspect, the present invention provides a solid composition comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides. The present invention also provides a solid composition comprising a plurality of solid particles, each particle comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient adsorbed onto a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides.

In some embodiments, the solid composition comprises at least about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90% of the active ingredient by weight of the composition.

In some embodiments, the solid substance includes a solid carrier. The solid carrier can be a fumed silica. Additionally or alternatively, the solid carrier can be selected from the group consisting of SiO₂, TiO₂, Al₂O₃, zeolites, Cab-O-Sil, and combinations thereof.

The solid substance can further comprise one or more sustained release polymers.

In some embodiments, the sustained release polymer is a film-forming, water insoluble polymer. The film-forming, water insoluble polymer can be selected from the group consisting of ethylcellulose, cellulose acetate, cellulose propionate (lower, medium or higher molecular weight), cellulose acetate propionate, cellulose acetate butyrate, cellulose acetate phthalate, cellulose triacetate, poly(methyl methacrylate), poly(ethyl methacrylate), poly(butyl methacrylate), poly(isobutyl methacrylate), poly(hexyl methacrylate), poly(isodecyl methacrylate), poly(lauryl methacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate), poly(ethylene), poly(ethylene) low density, poly(ethylene) high density, poly(propylene), poly(ethylene oxide), poly(ethylene terephthalate), poly(vinyl isobutyl ether), poly(vinyl acetate), poly(vinyl chloride) or polyurethane, or mixtures thereof. In one embodiment, the film-forming, water insoluble polymer is cellulose acetate.

In some embodiments, the sustained release polymer comprises one or more pH dependent polymers. The pH dependent polymer can be selected from a group consisting of a methyl acrylate-methacrylic acid copolymer, a cellulose acetate succinate, a hydroxy propyl methyl cellulose phthalate, a hydroxy propyl methyl cellulose acetate succinate (hypromellose acetate succinate), a polyvinyl acetate phthalate (PVAP), a methyl methacrylate-methacrylic acid copolymer, alginate and stearic acid, and any combinations thereof.

In some embodiments, the solid composition comprises, by weight of the composition, about 50% to about 80% of the active ingredient; about 10% to about 30% of the solid carrier; and about 10% to about 30% of the sustained release polymer. In some embodiments, the active ingredient, the solid carrier, and the sustained release polymer are in a weight ratio of about 2:1:1.

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The solid composition in accordance with the present invention can be in form of a powder. In some embodiments, the powder comprises particles having an average diameter of less than about 10 micron. The particles can have an average diameter of less than about 9 microns, less than about 8 microns, less than about 7 microns, less than about 6 microns, less than about 5 microns, less than about 4 microns, less than about 3 microns, less than about 2 microns, less than about 1000 nm, less than about 900 nm, less than about 800 nm, less than about 700 nm, less than about 600 nm, less than about 500 nm, less than about 400 nm, less than about 300 nm, less than about 290 nm, less than about 280 nm, less than about 270 nm, less than about 260 nm, less than about 250 nm, less than about 240 nm, less than about 230 nm, less than about 220 nm, less than about 210 nm, less than about 200 nm, less than about 190 nm, less than about 180 nm, less than about 170 nm, less than about 160 nm, less than about 150 nm, less than about 140 nm, less than about 130 nm, less than about 120 nm, less than about 110 nm, less than about 100 nm, less than about 90 nm, less than about 80 nm, less than about 70 nm, less than about 60 nm, less than about 50 nm, less than about 40 nm, less than about 30 nm, less than about 20 nm, less than about 10 nm, less than about 9 nm, less than about 8 nm, less than about 7 nm, less than about 6 nm, or less than about 5 nm.

In some embodiments, the solid substance has a surface area of at least 20 m²/g.

In some embodiments, the solid particles are formed by spray drying a spray suspension comprising the triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and the solid substance.

The solid composition in accordance with the present invention can include triglycerides with seven carbon fatty acids. In some embodiments, the solid composition further comprises triglycerides with one or more odd-numbered carbon fatty acids selected from the group consisting of C5, C9, C11, C13, C15, and any combinations thereof.

In some embodiments, the triglycerides is a triheptanoin oil comprising greater than about 98% pure glycerol triheptanoate as an active ingredient. The triheptanoin oil can comprise less than 1% w/w of a non-C7 triglyceride or a combination of non-C7 triglycerides.

The solid composition in accordance with the present invention can further include a pharmaceutically acceptable excipient.

In some embodiments, the solid composition includes an active ingredient that has purity greater than 97% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 1.0% by weight after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 98% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.5% by weight after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 99% after about four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.35% by weight after about

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four weeks of exposure to about 25° C. at about 60% relative humidity when package in a sealed container.

In some embodiments, the solid composition includes an active ingredient that has purity greater than 97% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 1.0% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 98% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.5% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition includes an active ingredient that has purity greater than 99% after about four weeks of exposure to about 40° C. at about 75% relative humidity when package in a sealed container. In some embodiments, the solid composition has a water content of no more than 0.45% by weight after about four weeks of exposure to 40° C. at about 75% relative humidity when package in a sealed container.

The present invention provides a method of treating a disease, disorder, or condition in a subject comprising orally administering to the subject a therapeutically effective amount of a solid composition in accordance with the present invention, wherein the disease, disorder, or condition is selected from any one or more of the following: a fatty acid oxidation disorder or deficiency; adult polyglucosan body disease; a mitochondrial fat oxidation defect (e.g., relating to carnitine palmitoyl transferase I, carnitine palmitoyl transferase II, carnitine acylcarnitine translocase, very long chain acyl-CoA dehydrogenase, trifunctional protein, long chain hydroxyacyl-CoA dehydrogenase, multiple acyl-CoA dehydrogenase, short chain acyl CoA dehydrogenase, alpha glucosidase, brancher enzyme, debrancher enzyme, myophosphorylase, or phosphofructokinase); a glycogen storage disease (e.g., glycogen storage disease Type II); glucose transporter type 1 (GLUT1) deficiency syndrome; and a mitochondrial myopathy.

Also provided is a method of treating a disease, disorder, or condition in a subject comprising orally administering to the subject a therapeutically effective amount of a solid composition in accordance with the present invention, wherein the disease, disorder, or condition is selected from any one or more of the following: urea cycle disorders (UCD) and hepatic encephalopathy (HE).

In some embodiments, the solid composition is co-administered with a food, drink, or comestible composition.

BRIEF DESCRIPTION OF FIGURES

FIG. 1 is the metabolite pharmacokinetics (PK) profile of triheptanoin and heptanoate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 2 is the metabolite PK profile of triheptanoin and heptanoate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 3 is the metabolite PK profile of triheptanoin and heptanoate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 4 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of heptanoate following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female

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mini-pigs combined. Heptanoate (C7) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 5 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 6 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 7 is the metabolite PK profile of B-hydroxybutyric acid (C4-OH) and B-hydroxyvaleric acid (C5-OH) following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 8 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of B-hydroxybutyric acid (C4-OH) following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined. B-Hydroxybutyric acid (C4-OH) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 9 is the metabolite PK profile up to 48 hours (left) or 8 hours (right) of B-hydroxyvaleric acid (C5-OH) following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined. B-Hydroxyvaleric acid (C5-OH) metabolite PK following triheptanoin oil, powder and powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 10 is the metabolite PK profile of aspartate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 11 is the metabolite PK profile of aspartate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 12 is the metabolite PK profile of aspartate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 13 is the metabolite PK profile of glutamate following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 14 is the metabolite PK profile of glutamate following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 15 is the metabolite PK profile of glutamate following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 16 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 17 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

FIG. 18 is the metabolite PK profile of pimelic acid, 3-hydroxypropionate and propionyl glycine following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 19 is the profile of average energy and alternative metabolites following triheptanoin powder-ER dosing on Day 1 in male and female mini-pigs combined.

FIG. 20 is the profile of average energy and alternative metabolites following triheptanoin oil dosing on Day 1 in male and female mini-pigs combined.

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FIG. 21 is the profile of average energy and alternative metabolites following triheptanoin powder dosing on Day 1 in male and female mini-pigs combined.

FIG. 22 is the metabolite PK profile of heptanoic acid following triheptanoin oil dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 23 is the metabolite PK profile of C4-hydroxy following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 24 is the metabolite PK profile of C5-hydroxy following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 25 is the metabolite PK profile of aspartic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 26 is the metabolite PK profile of glutamic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 27 is the metabolite PK profile of pimelic acid following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 28 is the metabolite PK profile of 3-hydroxypropionate following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 29 is the metabolite PK profile of propionyl glycine following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 30 is the sum of all metabolite PK profiles following triheptanoin oil, powder and powder-ER dosing on Day 1 and Day 7 in male and female mini-pigs combined.

FIG. 31 is the metabolite PK profile of C4-hydroxy following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 32 is the metabolite PK profile of C5-hydroxy following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 33 is the metabolite PK profile of heptanoic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 34 is the metabolite PK profile of pimelic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 35 is the metabolite PK profile of glutaric acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 36 is the metabolite PK profile of aspartic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 37 is the metabolite PK profile of glutamic acid following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 38 is the metabolite PK profile of alanine following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

FIG. 39 is the metabolite PK profile of 3-hydroxypropionate following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

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FIG. 40 is the metabolite PK profile of propionyl glycine following triheptanoin powder and powder-ER dosing (fold change relative to the oil dosing) on Day 1 and Day 7 in male and female mini-pigs.

DETAILED DESCRIPTION OF THE INVENTION

A solid composition in accordance with the present invention comprises an ester derived from a polyol and one or more fatty acids as an active ingredient, and a solid substance. The term “polyol” denotes an alcohol containing two or more hydroxyl groups. Examples of polyols include, but are not limited to, diol (e.g., ethylene glycol, propylene glycol, and resorcinol), triol (e.g., glycerol and ethane-1,1,2-triol), tetraol (e.g., pentaerythritol), and sugar alcohols (e.g., maltitol, sorbitol, xylitol, and erythritol). In one embodiment, the polyol is glycerol.

In one embodiment, the present solid compositions are particularly useful as pharmaceutical formulation with both improved physical properties and physiological properties. In one embodiment, the solid compositions are easier to be administered because they are miscible with food, drink, or other comestible compositions with ease. In another embodiment, the solid compositions, such as powder or granule dosage forms, are more portable and easier to handle and dispense during regular daily use or while traveling. In another embodiment, the solid compositions are more stable during storage and easier to be handled and transported during manufacture and commercialization. In another embodiment, the solid compositions may have masked taste and improve dose tolerability and reduce side effects. In another embodiment, the solid compositions, e.g., the powder dosage forms, have delayed release characteristics in the stomach, with limited gastric upset, distress, or spasms. In another embodiment, the solid compositions do not cause diarrhea from the oil leading to excess lubrication of the GI tract through its delayed release and stabilized physical presence allowing better and more complete digestion during passage in the gastrointestinal tract. In another embodiment, the solid compositions comprising a plurality of particles wherein the active ingredient oil is adsorbed on the surface of solid substance thereby enhancing the surface area of the oil on substrate particles. Such enhanced surface area of the oil can enhance absorption efficiency for each dose thereby improving efficacy and reducing diarrhea at the same time. In one embodiment, the solid composition can reduce the gastric stress and other side effects, enhance the therapeutic effect, and improve patient compliance. In another embodiment, the solid composition can reduce gastric and diarrheal tolerability issues, allow higher daily doses of triheptanoin to be achieved and allow for better, and more complete digestion and absorption via the suspended stabilized particles in the GI tract with high surface area. Triglycerides Solid Compositions

In the present invention, “triglyceride” refers to an ester derived from glycerol and one or more fatty acids. The fatty acids can have a carbon chain that is optionally substituted alkyl, optionally substituted alkenyl, or optionally substituted aryl.

“Alkyl,” by itself or as part of another substituent, refers to a saturated branched, straight-chain or cyclic monovalent hydrocarbon radical derived by the removal of one hydrogen atom from a single carbon atom of a parent alkane. The term “alkyl” includes “cycloalkyl” as defined herein below. Typical alkyl groups include, but are not limited to, methyl; ethyl; propyls such as propan-1-yl, propan-2-yl (isopropyl),

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cyclopropan-1-yl, etc.; butan-1-yl, butan-2-yl (sec-butyl), 2-methyl-propan-1-yl (isobutyl), 2-methyl-propan-2-yl (t-butyl), cyclobutan-1-yl, etc.; and the like. In some embodiments, an alkyl group comprises from 1 to 20 carbon atoms (C_1 - C_{20} alkyl). In other embodiments, an alkyl group comprises from 1 to 10 carbon atoms (C_1 - C_{10} alkyl). In still other embodiments, an alkyl group comprises from 1 to 6 carbon atoms (C_1 - C_6 alkyl) or 1 to 4 carbon atoms (C_1 - C_4 alkyl). C_1 - C_6 alkyl is also known as “lower alkyl”.

“Alkenyl,” by itself or as part of another substituent, refers to an unsaturated branched, straight-chain or cyclic monovalent hydrocarbon radical having at least one carbon-carbon double bond derived by the removal of one hydrogen atom from a single carbon atom of a parent alkene. The term “alkenyl” includes “cycloalkenyl” as defined herein below. The group may be in either the cis or trans conformation about the double bond(s). Typical alkenyl groups include, but are not limited to, ethenyl; propenyls such as prop-1-en-1-yl, prop-1-en-2-yl, prop-2-en-1-yl (allyl), prop-2-en-2-yl, cycloprop-1-en-1-yl; cycloprop-2-en-1-yl; butenyls such as but-1-en-1-yl, but-1-en-2-yl, 2-methyl-prop-1-en-1-yl, but-2-en-1-yl, but-2-en-2-yl, buta-1,3-dien-1-yl, buta-1,3-dien-2-yl, cyclobut-1-en-1-yl, cyclobut-1-en-3-yl, cyclobuta-1,3-dien-1-yl, etc.; and the like. In some embodiments, an alkenyl group comprises from 2 to 20 carbon atoms (C_2 - C_{20} alkenyl). In other embodiments, an alkenyl group comprises from 2 to 10 carbon atoms (C_2 - C_{10} alkenyl). In still other embodiments, an alkenyl group comprises from 2 to 6 carbon atoms (C_2 - C_6 alkenyl) or 2 to 4 carbon atoms (C_2 - C_4 alkenyl). C_2 - C_6 alkenyl is also known as “lower alkenyl”.

“Aryl,” by itself or as part of another substituent, refers to a monovalent aromatic hydrocarbon group derived by the removal of one hydrogen atom from a single carbon atom of a parent aromatic ring system, as defined herein. Typical aryl groups include, but are not limited to, groups derived from aceanthrylene, acenaphthylene, acephenanthrylene, anthracene, azulene, benzene, chrysene, coronene, fluoranthene, fluorene, hexacene, hexaphene, hexalene, as-indacene, s-indacene, indane, indene, naphthalene, octacene, octaphene, octalene, ovalene, penta-2,4-diene, pentacene, pentalene, pentaphene, perylene, phenalene, phenanthrene, picene, pleiadene, pyrene, pyranthrene, rubicene, triphenylene, trinaphthalene and the like. In some embodiments, an aryl group comprises from 6 to 20 carbon atoms (C_6 - C_{20} aryl). In other embodiments, an aryl group comprises from 6 to 15 carbon atoms (C_6 - C_{15} aryl). In still other embodiments, an aryl group comprises from 6 to 10 carbon atoms (C_6 - C_{10} aryl). In a specific embodiment, an aryl group comprises a phenyl group.

The term “substituted” specifically envisions and allows for one or more substitutions that are common in the art. However, it is generally understood by those skilled in the art that the substituents should be selected so as to not adversely affect the useful characteristics of the compound or adversely interfere with its function. Suitable substituents may include, for example, halogen groups, perfluoroalkyl groups, perfluoroalkoxy groups, alkyl groups, alkenyl groups, alkynyl groups, hydroxy groups, oxo groups, mercapto groups, alkylthio groups, alkoxy groups, aryl or heteroaryl groups, aryloxy or heteroaryloxy groups, arylalkyl or heteroarylalkyl groups, arylalkoxy or heteroarylalkoxy groups, amino groups, alkyl- and dialkylamino groups, carbamoyl groups, alkylcarbonyl groups, carboxyl groups, alkoxy carbonyl groups, alkylaminocarbonyl groups, dialkylamino carbonyl groups, arylcarbonyl groups, aryloxy carbonyl groups, alkylsulfonyl groups, arylsulfonyl groups,

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cycloalkyl groups, cyano groups, C₁-C₆ alkylthio groups, arylthio groups, nitro groups, keto groups, acyl groups, boronate or boronyl groups, phosphate or phosphonyl groups, sulfamyl groups, sulfonyl groups, sulfinyl groups, and combinations thereof. In the case of substituted combinations, such as “substituted arylalkyl,” either the aryl or the alkyl group may be substituted, or both the aryl and the alkyl groups may be substituted with one or more substituents. Additionally, in some cases, suitable substituents may combine to form one or more rings as known to those of skill in the art.

The term “optionally substituted” denotes the presence or absence of the substituent group(s). That is, it means “substituted or unsubstituted”. For example, optionally substituted alkyl includes both unsubstituted alkyl and substituted alkyl. The substituents used to substitute a specified group can be further substituted, typically with one or more of the same or different groups selected from the various groups specified above.

In some embodiments, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from the odd-numbered carbon fatty acids of C5, C7, C9, C11, C13, and C15. In some embodiments, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from phenylalkanoic acids and phenylalkenoic acids.

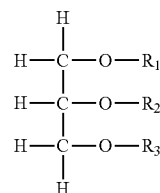
The present invention is based, in part, on the discovery that triglycerides with one or more odd-numbered carbon fatty acids, such as C5, C7, C9, C11, C13, C15, and any combinations thereof, having a purity greater than food grade can be formulated with a solid substance to form a solid composition. In one embodiment, the solid composition contains at least about 50% by weight the triglycerides, which can be in either solid or oil form prior to being formulated with the solid substance. In one embodiment, the triglycerides are an oil prior to being formulated with the solid substance to form the solid composition.

In one embodiment, the present invention provides a solid composition comprising triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 98% and the one or more odd-numbered carbon fatty acids are selected from the group consisting of C5, C7, C9, C11, C13, C15, and any combinations thereof and the solid composition comprises at least about 50% by weight the triglycerides. In one embodiment, the triglyceride is an ester derived from glycerol and three fatty acids independently selected from the odd-numbered carbon fatty acids of C5, C7, C9, C11, C13, and C15. In one specific embodiment, the triglyceride is an ester derived from glycerol and three C7 fatty acids, i.e., triheptanoin. In one embodiment, the solid composition comprises triglycerides with seven carbon fatty acids. In another embodiment, the solid composition further comprises triglycerides with one or more odd-numbered carbon fatty acids selected from the group consisting of C5, C9, C11, C13, C15, and any combinations thereof. In another embodiment, the triglycerides with seven carbon fatty acids is a triheptanoin oil comprising greater than about 98% pure glycerol triheptanoate as an active ingredient.

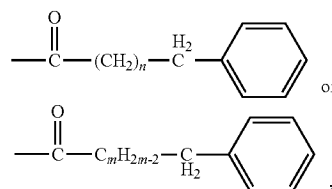
In some embodiment, the present invention provides a solid composition comprising triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids as an active ingredient and a solid substance; wherein the active ingredient has purity greater than 95% and the solid composition comprises at least about 50% by weight the triglycerides. In one embodiment, the triglyceride is an ester derived from glycerol and three fatty acids independently

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selected from phenylalkanoic acids and phenylalkenoic acids. In one embodiment, the triglyceride is a compound of formula (I):



wherein R₁, R₂, and R₃ are independently H,



and n is zero or an even number from 2-24 and m is an even number from 2-24, provided that at least one of R₁, R₂, and R₃ is other than H. In one embodiment, n and m are an even number from 2 to 24. In one embodiment, n and m are 0, 2, 4, or 6. In one specific embodiment, the triglyceride is an ester derived from glycerol and three phenylbutyrate, i.e., glycerol phenylbutyrate (e.g., RAVICTI®). Examples of triglycerides with one or more phenylalkanoic acids and/or phenylalkenoic acids are further described in U.S. Pat. No. 5,968,979, the contents of which are hereby incorporated by reference in entirety for all purpose.

In one embodiment, the solid composition comprises at least about 50%, about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, about 85%, or about 90% of the triglycerides. In another embodiment, the solid composition comprises at least about 55%, about 60%, about 65%, about 70%, about 75%, about 80%, or about 85% of the triglycerides. In one specific embodiment, the solid composition comprises about 60% of the triglycerides. In one specific embodiment, the solid composition comprises about 75%, about 80%, or about 85% of the triglycerides.

In some embodiments, the active ingredient in the solid composition has a purity of about or at least about 95, 95.5, 96, 96.5, 97, 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99.0, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, or 99.9% or more, or any range derivable therein.

In one embodiment, the solid composition is in a powder form which comprises a plurality of particles, and each particle comprises triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient adsorbed onto a solid substance. In one embodiment, the solid substance has a surface area of at least 20 m²/g.

In one embodiment, the solid substance comprises a solid carrier. The term “solid carrier”, as used herein, denotes any pharmaceutically acceptable solid material, which has a high surface area and does not adversely interact with the present triglycerides. By “high surface area”, it is meant that the solid carrier has a surface area of at least 20 m²/g, or at least 30 m²/g, or at least 50 m²/g, or at least 70 m²/g, or at least 100 m²/g, or at least 150 m²/g, or at least 180 m²/g. In one

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embodiment, the solid carrier can have surface areas of up to 200 m²/g, up to 400 m²/g, or up to 600 m²/g, or more. The surface area of the substrate can be measured using standard procedures, such as low-temperature nitrogen adsorption, based on the Brunauer, Emmett, and Teller (BET) method, which is well known in the art.

In one embodiment, the particles have an average diameter of less than about 100 micron. In one embodiment, the particles have an average diameter of less than about 90 micron, less than about 80 microns, less than about 70 microns, less than about 60 microns, less than about 50 microns, less than about 40 microns, less than about 30 microns, less than about 20 microns, or less than about 10 microns. In another embodiment, the particles have an average diameter of less than about 9 microns, less than about 8 microns, less than about 7 microns, less than about 6 microns, less than about 5 microns, less than about 4 microns, less than about 3 microns, less than about 2 microns, less than about 1000 nm, less than about 900 nm, less than about 800 nm, less than about 700 nm, less than about 600 nm, less than about 500 nm, less than about 400 nm, less than about 300 nm, less than about 290 nm, less than about 280 nm, less than about 270 nm, less than about 260 nm, less than about 250 nm, less than about 240 nm, less than about 230 nm, less than about 220 nm, less than about 210 nm, less than about 200 nm, less than about 190 nm, less than about 180 nm, less than about 170 nm, less than about 160 nm, less than about 150 nm, less than about 140 nm, less than about 130 nm, less than about 120 nm, less than about 110 nm, less than about 100 nm, less than about 90 nm, less than about 80 nm, less than about 70 nm, less than about 60 nm, less than about 50 nm, less than about 40 nm, less than about 30 nm, less than about 20 nm, less than about 10 nm, less than about 9 nm, less than about 8 nm, less than about 7 nm, less than about 6 nm, or less than about 5 nm. The powder may first be formed in the form of small particles ranging in size of from about 5 nm to about 1 micron, or from about 10 nm to about 50 nm, or from about 20 nm to about 100 nm. These particles may in turn form agglomerates ranging in size from about 10 nm to about 100 micron, or 20 nm to about 100 micron, or about 50 nm to about 100 micron.

Examples of the solid carrier include fumed silica and inorganic oxides, such as SiO₂, TiO₂, ZnO₂, ZnO, Al₂O₃, MgAlSilicate, CaSilicate, Al(OH)₃, zeolites, and other inorganic molecular sieves; water insoluble polymers, such as cross-linked cellulose acetate phthalate, cross-linked hydroxypropyl methyl cellulose acetate succinate, cross-linked polyvinyl pyrrolidinone, (also known as cross povidone) microcrystalline cellulose, polyethylene/polyvinyl alcohol copolymer, polyethylene polyvinyl pyrrolidinone copolymer, cross-linked carboxymethyl cellulose, sodium starch glycolate, cross-linked polystyrene divinyl benzene; and activated carbons, including those made by carbonization of polymers such as polyimides, polyacrylonitrile, phenolic resins, cellulose acetate, regenerated cellulose, and rayon. In some specific embodiments, the solid carrier can be Cab-O-Sil, Neusilin, hypromellose acetate succinate (HPMCAS), Casein, or combinations thereof. In one embodiment, the solid carrier is fumed silica. Suitable fumed silica are available from any major producers under various brand names, such as are Evonik (who sells it under the name Aerosil), Cabot Corporation (Cab-O-Sil), Wacker Chemie (HDK), Dow Corning, and OCI (Konasil).

In one embodiment, the solid substance further comprises one or more sustained release polymer. In one embodiment, the one or more sustained release polymer comprises one or

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more pH dependent polymers. In one embodiment, the pH dependent polymer is selected from a group consisting of a methyl acrylate-methacrylic acid copolymer, a cellulose acetate succinate, a hydroxy propyl methyl cellulose phthalate, a hydroxy propyl methyl cellulose acetate succinate (hypromellose acetate succinate), a polyvinyl acetate phthalate (PVAP), a methyl methacrylate-methacrylic acid copolymer, alginate and stearic acid, and any combinations thereof.

In one embodiment, the solid composition may further comprise one or more additional pharmaceutically acceptable excipient, such as fillers, surfactants, and flavorants. Those excipients can be added to improve the powder flowability, dissolvability, and taste.

“Pharmaceutically acceptable” refers to being suitable for use in contact with the tissues of humans and animals without undue toxicity, irritation, allergic response, and the like, commensurate with a reasonable benefit/risk ratio, and effective for their intended use within the scope of sound medical judgment. “Excipient” denotes a diluent, adjuvant, vehicle or carrier with which the active ingredient is administered. Examples of the pharmaceutically acceptable excipient include, are not limited to a filler (or diluent), a binder, a disintegrant, a lubricant, or a combination thereof.

Fillers may be one or more compounds which are capable of providing compactability and good flow. Examples of fillers include microcrystalline cellulose, starch, lactose, sucrose, glucose, mannitol, maltodextrin, sorbitol, dextrose, silicic acid, dibasic calcium phosphate, or a combination comprising at least one of the foregoing fillers. Exemplary lactose forms include lactose monohydrate, NF (Fast Flo), lactose spray-dried monohydrate, and lactose anhydrous. Exemplary microcrystalline celluloses (MCC) include, for example, AVICEL® PH101 and AVICEL® PH102, which are commercially available from FMC Biopolymer, Philadelphia, PA.

Binders may be used to impart cohesive qualities to a formulation, for example, a tablet formulation, and thus ensure that the tablet remains intact after compaction. Examples of binders include starches (for example, Starch 1500® or pregelatinized starch), alginates, gelatin, carboxymethylcellulose, sugars (for example, sucrose, glucose, dextrose, and maltodextrin), polyethylene glycol, waxes, natural and synthetic gums, polyvinylpyrrolidone, and celulosic polymers (for example, microcrystalline cellulose, hydroxypropyl cellulose, hydroxypropyl methylcellulose, methyl cellulose, and hydroxyethyl cellulose) and combinations comprising one or more of the foregoing binders.

Disintegrants are used to facilitate disintegration or “breakup” of a composition, for example, a tablet, after administration. Examples of disintegrants include sodium starch glycolate, sodium croscarmellose (cross-linked carboxy methyl cellulose), crosslinked polyvinylpyrrolidone (PVP-XL), anhydrous calcium hydrogen phosphate, agar-agar, potato or tapioca starch, alginic acid, or a combination comprising one or more of the foregoing disintegrants.

A lubricant may be added to the composition for a minimum period of time to obtain good dispersal. Examples of lubricants include magnesium stearate, calcium stearate, zinc stearate, stearic acid, talc, glyceryl behenate, polyethylene glycol, polyethylene glycol, polyethylene oxide, sodium lauryl sulfate, magnesium lauryl sulfate, sodium oleate, sodium stearyl fumarate, DL-leucine, colloidal silica, or a combination comprising one or more of the foregoing lubricants.

If desired, the solid composition may optionally comprise small amounts of nontoxic auxiliary substances such as

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wetting or emulsifying agents, or pH buffering agents, for example, sodium acetate, sorbitan monolaurate, triethanolamine sodium acetate, triethanolamine oleate, sodium lauryl sulfate, dioctyl sodium sulfosuccinate, and polyoxyethylene sorbitan fatty acid esters.

In another embodiment of the present invention, the solid composition comprises one or more sustained release polymer. In one embodiment, the solid substance comprises one or more sustained release polymer. By “sustained release polymer”, it is meant any polymer which can control the release of the active ingredient from the composition in such a way to obtain the desired release profile. In one embodiment, the sustained release polymer is a hydrophilic polymer. The term “hydrophilic polymer” refers to a polymer having a strong affinity for water and tending to dissolve in, mix with, or be wetted by water. Examples of the hydrophilic polymer include, but are not limited to polyethylene oxide, hydroxypropyl cellulose, hydroxypropyl methyl cellulose, hydroxyethyl cellulose, sodium carboxymethylcellulose, calcium carboxymethyl cellulose, methyl cellulose, polyacrylic acid, maltodextrin, pre-gelatinized starch, guar gum, sodium alginate, polyvinyl alcohol, chitosan, locust bean gum, amylase, any other water-swelling polymer, and a combination thereof. In another embodiment, the sustained release polymer is a film-forming, water insoluble polymer. Examples of the film-forming, water insoluble polymer include, but are not limited to ethylcellulose, cellulose acetate, cellulose propionate (lower, medium or higher molecular weight), cellulose acetate propionate, cellulose acetate butyrate, cellulose acetate phthalate, cellulose triacetate, poly(methyl methacrylate), poly(ethyl methacrylate), poly(butyl methacrylate), poly(isobutyl methacrylate), poly(hexyl methacrylate), poly(isodecyl methacrylate), poly(lauryl methacrylate), poly(phenyl methacrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate), poly(ethylene), poly(ethylene) low density, poly(ethylene) high density, poly(propylene), poly(ethylene oxide), poly(ethylene terephthalate), poly(vinyl isobutyl ether), poly(vinyl acetate), poly(vinyl chloride) or polyurethane, or any other water insoluble polymer, or mixtures thereof. In another embodiment, the sustained release polymer is a film-forming, water soluble polymer. Examples of the film-forming, water soluble polymer include, but are not limited to polyvinyl alcohol, polyvinylpyrrolidone, methyl cellulose, hydroxypropyl cellulose, hydroxypropylmethyl cellulose and polyethylene glycol, Pluronic F108, Pluronic F127, Pluronic F68 or mixtures thereof. In one embodiment, the present solid composition comprises one or more the film-forming, water insoluble polymers. In one specific embodiment, the present solid composition comprises cellulose acetate.

In another embodiment, the sustained release polymer comprises one or more pH dependent polymers. By “pH dependent polymers”, it is meant polymers which are not soluble at the highly acidic pH but soluble in neutral to basic pH environment. Examples of pH dependent polymers include, but are not limited to, methyl acrylate-methacrylic acid copolymers, cellulose acetate succinates, hydroxy propyl methyl cellulose phthalates, hydroxy propyl methyl cellulose acetate succinates (hypromellose acetate succinates), polyvinyl acetate phthalates (PVAP), methyl methacrylate-methacrylic acid copolymers, alginate and stearic acid, and any combinations thereof. Furthermore, materials, such as fatty acids, waxes, shellac, plastics, and plant fibers, may also be suitable for pH dependent polymers.

The solid compositions as described herein can be used alone for various purposes, such as in treatment methods as

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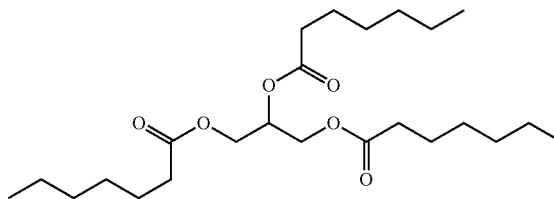
described herein. In this regard, the solid compositions can be pharmaceutically acceptable.

The present solid compositions can be formed by a spray-drying process. In one embodiment, the solid particles are formed by spray drying a spray suspension comprising the triglycerides with one or more odd-numbered carbon fatty acids as an active ingredient and the solid substance. The spray suspension contains the triglycerides dissolved in a solvent and the solid substance suspended therein. Such spray suspension is spray-dried to form a solid powder by rapidly removing the solvent. The formed powder particles may have the triglycerides adsorbed on the surface of the solid substance or the triglycerides and the solid substance interspersed with one another to form a matrix. In another embodiment, the solid composition further comprises a sustained release polymer. For example, the solid particles may be formed by spray drying a spray suspension comprising the triglycerides and the sustained release polymer dissolved in a solvent with the solid substance suspended therein. The spray suspension is spray-dried to form a solid powder by rapidly removing the solvent.

In one embodiment, some powder formulations (with 75%, 80%, 85%, and 90% loading of triheptanoin on solid carrier) were prepared. The triheptanoin oil as described herein was first dissolved in acetone and then mixed with the solid carrier, such as Cab-O-Sil, to form a suspension. The sustained release polymer, e.g., Eudragit, was optionally added. The resulting suspension was spray dried to obtain a powder. The spray dried powder is then dried further to reduce the amount of solvent (acetone) to a desirable level.

Triheptanoin Oil

In one embodiment, the triglyceride in the present solid composition is a triheptanoin oil. Triheptanoin oil as described herein comprises triglycerides with seven carbon fatty acids. Triheptanoin, also known as glycerol triheptanoate, glycerol triheptanoate, glyceryl triheptanoate, 1,3-di(heptanoyloxy)propan-2-yl heptanoate, triheptanthin, 1, 2, 3-triheptanoylglycerol, and propane-1,2,3-triyl triheptanoate, is a triglyceride of the seven carbon (C7) fatty acid heptanoate (CAS Registry No. 620-67-7) and has the following structure:



Triheptanoin oil as described herein is of a purity or grade greater than food grade triheptanoin oil. In some embodiments, the triheptanoin oil is considered ultrapure pharmaceutical grade triheptanoin oil.

In one embodiment, the present triheptanoin oil has the general properties listed in Table 1:

Physical and chemical property	Triheptanoin with Pharmaceutical Grade Purity
Form	Liquid
Color	Light yellow
Cloud point	<0° C.
Flash point	ca. 220° C.

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Physical and chemical property	Triheptanoin with Pharmaceutical Grade Purity
Vapor pressure	<0.01 hPa; 20° C.
Relative Density	ca. 0.96 g/cm ³ ; 20° C.
Water solubility	<0.01 g/l; 20° C.
Partition coefficient (n-octanol/water)	Log Pow: >3.0; glycerides of saturated C ₇ fatty acid
Viscosity, dynamic	ca. 20 mPa · s at 20° C.
Refractive Index	1.4440 to 1.4465

Typically, the total concentration of impurities in triheptanoin oil is less than the total concentration of impurities in food grade triheptanoin oil. In some embodiments, food grade triheptanoin oil can have a purity of about or at most about 95, 95.5, 96, 96.5, 97, or 97.5%, or any range derivable therein.

In some embodiments, total impurities in triheptanoin oil amount to less than 5% w/w. In some embodiments, total impurities in triheptanoin oil amount to less than 5, 4.5, 4, 3.5, 3, 2.5, 2, 1.5, 1, or 0.5% w/w or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2% w/w of triglycerides esterified with acids other than C₇ acids (e.g., C₂, C₃, C₄, C₅, C₆, C₈, or C₉ acids, or others, or combinations thereof (“non-C₇ triglycerides”)). In some embodiments, triheptanoin oil comprises less than 2, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2 or 0.1% w/w or less of non-C₇ triglycerides, or any range derivable therein.

In one embodiment, triheptanoin oil comprises glycerol triheptanoate of a purity of greater than 97.5%. In some embodiments, triheptanoin oil comprises glycerol triheptanoate of a purity of about or at least about 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99.0, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8, or 99.9% or more, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 0.01% w/w ash. In some embodiments, triheptanoin oil comprises less than about 0.01, 0.009, 0.008, 0.007, 0.006, 0.005, 0.004, 0.003, 0.002, or 0.001% w/w ash or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 0.04% w/w water. In some embodiments, triheptanoin oil comprises less than about 0.04, 0.035, 0.03, 0.025, 0.02, 0.015, 0.01, 0.005, or 0.001% w/w water or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w glycerol. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w glycerol or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 1.5% w/w monoheptanoate. In some embodiments, triheptanoin oil comprises less than 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6% w/w monoheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 3.0% w/w diheptanoate. In some embodiments, triheptanoin oil comprises less than 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 1.9, 1.8, or 1.7% w/w diheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w Hexano-Diheptanoate. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1,

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2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w Hexano-Diheptanoate or less, or any range derivable therein.

In one embodiment, triheptanoin oil comprises less than 2.5% w/w Hexanoic acid. In some embodiments, triheptanoin oil comprises less than 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.3, or 1.2% w/w glycerol or less, or any range derivable therein.

In some embodiments, the triheptanoin oil as described herein is a colorless oil of low viscosity and is odorless and tasteless. Triheptanoin oil as described herein is typically stored at room temperature and protected from light, and its stability can exceed 24 months under such conditions.

In one embodiment, the present invention provides a solid composition comprises triheptanoin as the active ingredient; fumed silica; and cellulose acetate. In a further embodiment, the solid composition a plurality of solid particles, each particle comprising triheptanoin adsorbed onto a solid substance comprising fumed silica and cellulose acetate. In a further embodiment, the solid composition comprises, by weight of the composition, about 50% to about 80% of triheptanoin; about 10% to about 30% of fumed silica; and about 10% to about 30% of cellulose acetate. In another embodiment, the solid composition comprises, by weight of the composition, about 60% of triheptanoin; about 20% of fumed silica; and about 20% of cellulose acetate.

In one specific embodiment, the present triheptanoin oil has the characteristics listed in Table 2:

Parameter	Specification	Method
IDENTITY		
Triheptanoate	Comparable to reference	Ph. Eur. 2.2.28
QUALITY		
Appearance	Clear substance and substance is not more intensely colored than reference solution Y ₃	Ph. Eur. 2.2.1, 2.2.2. method I
Relative density	0.95-0.98	Ph. Eur. 2.2.5
Refractive index	1.4440-1.4465	Ph. Eur. 2.2.6
Viscosity	15 mPa · s-23 mPa · s	Ph. Eur. 2.2.9
PURITY		
Composition of fatty acids		
Heptanoic acid	>99%	Ph. Eur. 2.4.22 method C
Hexanoic acid	Max. 0.8%	
All other, each	<0.03%	
Saponification value	360 to 410 mg KOH/g	Ph. Eur. 2.5.6
Acid value	Max. 0.2 mg KOH/g	Ph. Eur. 2.5.1
Hydroxyl value	Max. 10 mg KOH/g	Ph. Eur. 2.5.3 method A
Peroxide value	Max. 1.0 mg KOH/g	Ph. Eur. 2.5.5 method A
Water	Max. 0.2%	Ph. Eur. 2.5.12
Total ash	Max. 0.1%	Ph. Eur. 2.4.16
Heavy metals	Max. 10 ppm	Ph. Eur. 2.4.8
POTENCY/STRENGTH		
Triheptanoate	90% to 110%	Ph. Eur. 2.2.28

In one specific embodiment, the present triheptanoin oil has the characteristics listed in Table 3:

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Parameter	Specification	Method
IDENTITY		
Triheptanoate	Comparable to reference	Ph. Eur. 2.2.28
IR (infrared)	Comparable to reference	Ph. Eur. 2.2.24
QUALITY		
Appearance		
Clarity and Opalescence	Clear substance	Ph. Eur. 2.2.1
Color	Substance is not more intensely colored than reference solution Y ₃	Ph. Eur. 2.2.2, method I
Composition of fatty acids		
Heptanoic acid	≥99.0%	Ph. Eur. 2.4.22 method C
Hexanoic acid	≤1.0%	
Individual Unidentified Impurities	≤0.10%	
Impurities		
Glycerol	≤1.0%	Ph. Eur. 2.2.28
Monoheptanoate	≤0.5%	
Diheptanoate	≤1.5%	
Hexano-Diheptanoate	≤1.0%	
Individual Unidentified Impurities	≤0.5%	
Total Impurities	≤5.0%	
Elemental Impurities		
As	≤0.02 ppm	Ph. Eur. 2.2.57 or 2.2.58
Cd	≤0.19 ppm	
Pb	≤0.08 ppm	
Hg	≤0.12 ppm	
Acid value	Max. 0.2 mg KOH/g	Ph. Eur. 2.5.1
Water	Max. 0.2%	Ph. Eur. 2.5.12
Total ash	Max. 0.1%	Ph. Eur. 2.4.16
ASSAY		
Triheptanoate	95% to 103%	Ph. Eur. 2.2.28

In certain specific embodiments, the triheptanoin oil has the characteristics listed in Table 4:

Parameter	Example A	Example B
Triheptanoate	Comparable to reference	Comparable to reference
Triheptanoate Impurities:	99.0%	98.2%
Glycerol	not detectable	not detectable
Monoheptanoate	not detectable	not detectable
Diheptanoate	0.66%	0.62%
Hexano-Diheptanoate	0.39%	0.40%
Individual Unidentified Impurities	0.20%	0.18%
Impurities	(RRT* 1.258)	(RRT 1.258)
Total Impurities	1.25%	1.20%
Elemental Impurities		
As	not detected	not detected
Cd	not detected	not detected

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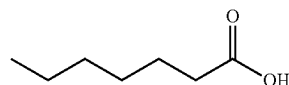
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Parameter	Example A	Example B
Pb	not detected	not detected
Hg	not detected	not detected

*RRT: relative retention time.

Triheptanoin Oil Precursors and Manufacture of Triheptanoin Oil

Also provided herein are precursors to triheptanoin oil. In some embodiments, a precursor is n-heptanoic acid. n-Heptanoic acid has the following structure:



n-Heptanoic acid as described herein is of a purity or grade greater than food grade n-heptanoic acid. In some embodiments, purity of food grade n-heptanoic acid is between 90% and 98%. In some embodiments, purity of food grade n-heptanoic acid is between 95% and 98%. In some embodiments, purity of food grade n-heptanoic acid is between 95% and 97.5%. In some embodiments, purity of food grade n-heptanoic acid has a maximum purity of 98, 97.9, 97.6, 97.5, 97.4, 97.3, 97.2, 97.1, 97.0, 96.9, 96.8, 96.7, 96.6, 96.5, 96.4, 96.3%, 96.2, 96.1, 96.0, 95.9, 95.8, 95.7, 95.6, 95.5, 95.4, 95.3, 95.2, or 95.1%.

In some embodiments, n-heptanoic acid as described herein is considered ultrapure pharmaceutical grade n-heptanoic acid.

Typically, the total concentration of impurities in n-heptanoic acid is less than the total concentration of impurities in food grade n-heptanoic acid. In some embodiments, total impurities in n-heptanoic acid amount to less than 4.0% w/w. In some embodiments, total impurities amount to less than about 4.0, 3.9, 3.8, 3.7, 3.6, 3.5, 3.4, 3.3, 3.2, 3.1, 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.09, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02, or 0.01% w/w or less, or any range derivable therein.

In one embodiment, n-heptanoic acid comprises less than 2% w/w of triglycerides esterified with acids other than C7 acids (e.g., C2, C3, C4, C5, C6, C8, or C9 acids, or others, or combinations thereof ("non-C7 triglycerides")). In some embodiments, n-heptanoic acid comprises less than 2, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2 or 0.1% w/w or less of non-C7 triglycerides, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises a sum of C7 carboxylic acids having greater than 97% purity. In some embodiments, the purity is greater than 97, 97.1, 97.2, 97.3, 97.4, 97.5, 97.6, 97.7, 97.8, 97.9, 98, 98.1, 98.2, 98.3, 98.4, 98.5, 98.6, 98.7, 98.8, 98.9, 99, 99.1, 99.2, 99.3, 99.4, 99.5, 99.6, 99.7, 99.8 or 99.9%, or more, or any range derivable therein. Compositions, such as pharmaceutical compositions, comprising n-heptanoic acid are also provided. Further, any composition comprising n-heptanoic acid can optionally be further defined as a pharmaceutical composition.

Triheptanoin oil can be prepared from n-heptanoic acid as described herein or a composition comprising n-heptanoic acid. For example, n-heptanoic acid can be esterified with glycerol to produce triheptanoin oil, such as in the presence of a basic catalyst and heat.

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In one embodiment, an n-heptanoic acid composition comprises less than 3.0% w/w of 2-methylhexanoic acid. In some embodiments, the composition comprises less than 3.0, 2.9, 2.8, 2.7, 2.6, 2.5, 2.4, 2.3, 2.2, 2.1, 2.0, 1.9, 1.8, 1.7, 1.6, 1.5, 1.4, 1.3, 1.2, 1.1, 1.0, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, or 0.1% w/w 2-methylhexanoic acid, or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises less than 0.1% w/w water. In some embodiments, a composition comprises less than 0.1, 0.09, 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, 0.02, or 0.01% w/w of water or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises a carbonyl content of less than 0.2% w/w. Carbonyl content measurements are well-known in the art. In some embodiments, the carbonyl content is less than 0.15, 0.1, 0.05, or 0.01% w/w, or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises an iodine number of less than 0.5%. Iodine number measurements are well-known in the art. In some embodiments, the iodine number is less than 0.5, 0.4, 0.3, 0.2, or 0.1% or less, or any range derivable therein.

In one embodiment, an n-heptanoic acid composition comprises less than 0.2 mg/kg iron. In some embodiments, a composition comprises less than 0.2, 0.15, 0.1, 0.05, or 0.01 mg/kg iron, or any range derivable therein.

In one embodiment, a composition comprising n-heptanoic acid has a platinum/cobalt color number of less than 0.25. Determination of the platinum/cobalt color number is well-known in the art. In some embodiments, the platinum/cobalt color number is less than 0.25, 0.2, 0.15, 0.1, or 0.05 or less, or any range derivable therein.

Therapeutic Treatments

Provided herein in some embodiments are methods of using the present solid composition for treatment. The solid composition can be administered to a subject in need thereof in an amount effective to treat the subject.

For example, a subject may suffer from any one or more of the following: a fatty acid oxidation disorder or deficiency; adult polyglucosan body disease; a mitochondrial fat oxidation defect (e.g., relating to carnitine palmitoyl transferase I, carnitine palmitoyl transferase II, carnitine acylcarnitine translocase, very long chain acyl-CoA dehydrogenase, trifunctional protein, long chain hydroxyacyl-CoA dehydrogenase, multiple acyl-CoA dehydrogenase, short chain acyl CoA dehydrogenase, alpha glucosidase, brancher enzyme, debrancher enzyme, myophosphorylase, or phosphofructokinase); a glycogen storage disease (e.g., glycogen storage disease Type II); glucose transporter type 1 (GLUT1) deficiency syndrome; or a mitochondrial myopathy. In some embodiments, triheptanoin oil as described herein or a pharmaceutical composition comprising triheptanoin oil can be used to treat any condition described in U.S. Pat. No. 8,106,093.

In another embodiment, a subject may suffer from urea cycle disorders (UCD), which include several inherited deficiencies of enzymes or transporters necessary for the synthesis of urea from ammonia, including enzymes involved in the urea cycle; or hepatic encephalopathy (HE), which a spectrum of neurologic signs and symptoms believed to result from hyperammonemia, which frequently occur in subjects with cirrhosis or certain other types of liver disease. In some embodiments, glycerol phenylbutyrate as described herein or a pharmaceutical composition comprising glycerol phenylbutyrate can be used to treat these conditions. Further details of methods of treatment using

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glycerol phenylbutyrate can be found in U.S. Pat. Nos. 8,404,215 and 8,642,012, the contents of which are hereby incorporated by reference in entirety for all purpose.

In one embodiment, the solid compositions can be administered to a patient directly in various dosage forms (e.g., powders, granules, tablets, capsules, and the like) without mixing with food, drink, or other comestible compositions. In another embodiment, the solid compositions, such as powders or granules, can be taken together with food, drink, or other comestible compositions by mixing the solid compositions into the food, drink, or comestible compositions. The food, drink, or comestible compositions can be in any form including liquid, solid, semi-solid, foamy material, paste, gel, cream, lotion, and combinations thereof. The solid compositions can provide therapeutic effects while minimize gastric upset and gastrin release, and co-administration with food, drink, or other comestible compositions can enhance such tolerability benefit. In addition, co-administration of food, drink, or other comestible compositions with the solid composition can provide delayed and/or extended release of the active ingredients and thus allow less frequent dosing, longer term of energy exposure, and potentially higher efficiency in absorption from the GI tract. The delayed and/or extended release characteristic of the solid composition can also be independently achieved by including one or more sustained release polymers in the solid compositions as discussed above.

As used herein, the term “effective” (e.g., “an effective amount”) means adequate to accomplish a desired, expected, or intended result. An effective amount can be a therapeutically effective amount. A “therapeutically effective amount” refers to the amount of an active ingredient that, when administered to a subject, is sufficient to effect such treatment of a particular disease or condition. The “therapeutically effective amount” will vary depending on, e.g., the disease or condition, the severity of the disease or condition, and the age, weight, etc., of the subject to be treated.

In general, “treating” or “treatment” of any condition, disease or disorder refers, in some embodiments, to ameliorating the condition, disease or disorder (i.e., arresting or reducing the development of the disease or at least one of the clinical symptoms thereof). In some embodiments “treating” or “treatment” refers to ameliorating at least one physical parameter, which may not be discernible by the subject. In some embodiments, “treating” or “treatment” refers to inhibiting the condition, disease or disorder, either physically, (e.g., stabilization of a discernible symptom), physiologically, (e.g., stabilization of a physical parameter) or both. In some embodiments, “treating” or “treatment” refers to delaying the onset of a condition, disease, or disorder.

EXAMPLES

Example 1

Table 5 shows the high purity of exemplary triheptanoin oil samples and solid samples comprising triheptanoin oil in accordance with the present invention.

Further, the stability of an exemplary solid composition (solid sample 2 in Table 5) comprising triheptanoin oil was tested. Measurements were conducted after storage at 25° C./60% RH (relative humidity) in double LDPE (low density polyethylene) bags in a 60 cc induction sealed HDPE (high density polyethylene) bottle and results are shown in Table 6. Measurements were also conducted after storage at 40° C./75% RH in double LDPE bags in a 60 cc induction sealed HDPE bottle and results are shown in Table 7.

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TABLE 5

Initial Purity Test Results						
Impurities	Specification	Parameter				
		OIL Sample 1	OIL Sample 2	SOLID Sample 1	SOLID Sample 2	SOLID Sample 3
Glycerol	≤1.0%	ND	ND	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND	ND	ND
Diheptanoate	≤1.5%	1.2%	0.7%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%	0.4%	0.4%
Individual	≤0.5%	RRT 0.91: 0.06%	RRT 1.26: 0.20%	RRT 0.91: 0.06%	RRT 0.91: 0.06%	RRT 0.91: 0.06%
Unidentified		RRT 1.05: 0.16%		RRT 1.05: 0.16%	RRT 1.05: 0.16%	RRT 1.05: 0.16%
Impurities:		RRT 1.06: 0.17%		RRT 1.06: 0.17%	RRT 1.06: 0.17%	RRT 1.06: 0.17%
RRT* (%)		RRT 1.07: 0.11%		RRT 1.07: 0.11%	RRT 1.07: 0.11%	RRT 1.07: 0.11%
		RRT 1.25: 0.06%		RRT 1.25: 0.06%	RRT 1.25: 0.06%	RRT 1.25: 0.06%
		RRT 1.26: 0.23%		RRT 1.26: 0.23%	RRT 1.26: 0.23%	RRT 1.26: 0.24%
		RRT 1.30: 0.10%		RRT 1.30: 0.10%	RRT 1.30: 0.10%	RRT 1.30: 0.10%
		RRT 1.36: 0.11%		RRT 1.36: 0.11%	RRT 1.36: 0.11%	RRT 1.36: 0.11%
Total Impurities	≤5.0%	2.6%	1.3%	2.6%	2.6%	2.6%

RRT = relative retention time.

ND = not detected.

TABLE 6

Stability Study Results for an exemplary solid composition stored at 25° C./60% RH:				
Parameter	Specification	Time points (months)		
		Freshly Made (Five months before)	0	1
Appearance	White to off-white powder	Conforms	Conforms	Conforms
Impurities				
Glycerol	≤1.0%	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND
Diheptanoate	≤1.5%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%
Individual	≤0.5%	RRT 0.91: 0.06%	RRT 0.91: 0.06%	RRT 0.91: 0.05%
Unidentified		RRT 1.05: 0.16%	RRT 1.05: 0.16%	RRT 1.05: 0.16%
Impurities: RRT (%)		RRT 1.06: 0.17%	RRT 1.06: 0.17%	RRT 1.06: 0.17%
		RRT 1.07: 0.11%	RRT 1.07: 0.11%	RRT 1.07: 0.11%
		RRT 1.25: 0.06%	RRT 1.25: 0.06%	RRT 1.25: 0.06%
		RRT 1.26: 0.23%	RRT 1.26: 0.24%	RRT 1.26: 0.23%
		RRT 1.30: 0.10%	RRT 1.30: 0.10%	RRT 1.30: 0.09%
		RRT 1.36: 0.11%	RRT 1.36: 0.11%	RRT 1.36: 0.09%
Total Impurities	≤5.0%	2.6%	2.6%	2.5%
Water	NMT 5%	NS	0.29%	0.35%
Triheptanoate	90%-110%	99%	101%	100%
Acetone	≤200 ppm	<100 ppm	NS	NS

RRT = relative retention time.

NMT = not more than.

ND = not detected.

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TABLE 7

Stability Study Results for an exemplary solid composition stored at 40° C./75% RH:				
Parameter	Specification	Time points (months)		
		Freshly Made (Five months before)	0	1
Appearance	White to off-white powder	Conforms	Conforms	Conforms
Impurities				
Glycerol	≤1.0%	ND	ND	ND
Monoheptanoate	≤0.5%	ND	ND	ND
Diheptanoate	≤1.5%	1.1%	1.1%	1.1%
Hexano-Diheptanoate	≤1.0%	0.4%	0.4%	0.4%
Individual Unidentified Impurities: RRT (%)	≤0.5%	RRT 0.91: 0.06% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.23% RRT 1.30: 0.10% RRT 1.36: 0.11%	RRT 0.91: 0.06% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.24% RRT 1.30: 0.10% RRT 1.36: 0.11%	RRT 0.91: 0.05% RRT 1.05: 0.16% RRT 1.06: 0.17% RRT 1.07: 0.11% RRT 1.25: 0.06% RRT 1.26: 0.23% RRT 1.30: 0.09% RRT 1.36: 0.10%
Total Impurities	≤5.0%	2.6%	2.6%	2.5%
Water	NMT 5%	NS	0.29%	0.45%
Triheptanoate	90%-110%	99%	101%	98%
Acetone	≤200 ppm	<100 ppm	NS	NS

RRT = relative retention time.

NMT = not more than.

ND = not detected.

Example 2

Single dose study of each arm with full pharmacokinetics (PK) profile through 48 hours post dose was performed to 35 determined when metabolites return to baseline (one week washout between each arm) as well as if acute release of gastrin and cholestykinin (CCK) hormones cause spasmodic stomach contractions. Blood samples were collected from each arm 0-90 min plus anytime outside this window 40 if gastric distress observed.

More specifically, animals (n=3/sex) were fasted prior to dose and fed 4 hrs post dose. Blood samples for PK analysis of triheptanoin and metabolites were collected pre-dose through 48 hrs post dose. As seen in the multiple dose study (Example 3), there were no major differences between males and females thus data was combined for all metabolites.

In FIGS. 1-21, animals were administered a single oral gavage dose level of a triheptanoin oil sample (i.e., the oil) or exemplary solid samples comprising triheptanoin oil in accordance with the present invention. The two exemplary solid compositions comprise 1) 60%:20%:20% of triheptanoin oil:Cab-O-Sil:cellulose acetate (i.e., the Powder-ER); and 2) 80%:20% of triheptanoin oil:Cab-O-Sil (i.e., the Powder), respectively.

Triheptanoin and Heptanoic Acid

Triheptanoin is metabolized to heptanoic acid. The metabolite PK profiles for the three samples are shown in FIGS. 1-4. Bio-modal peaks were observed for heptanoic acid on three arms. It is contemplated that: 1) the first peak may be when the triheptanoin hits the stomach and the second peak may be when the triheptanoin hits the duodenum; 2) fat soluble materials can be dissolved in the stomach but free heptanoin is absorbed in the duodenum (higher 2nd peak for the powders); and 3) the cellulose acetate in the Powder-ER formulation delays delivery by ~one hour. 65

Energy Metabolites: Ketones

As seen in FIGS. 5-9, we observed comparable levels of ketones with oil treatment and powder-ER treatment. Delivery of C4 ketones were faster with powders but also blunted by the powders. The ratio C4 ketone:C5 ketone indicates the state of feeding; as this ratio increases it indicates that triheptanoin is being used as a source of calories.

Energy Metabolites: Aspartate

The levels of aspartate of the three samples are shown in FIGS. 10-12. We observed sustained release with the Powder-ER. It also showed a more favorable effect on gluconeogenic precursors because of the delay in release. The levels of aspartate of the Oil and the powder return to baseline at 8 hrs, while that of the powder-ER returns to baseline at 24 hrs post dose.

Energy Metabolites: Glutamate

The levels of glutamate of the three samples are shown in FIGS. 13-15. When glutamate increases, the trend indicates that triheptanoin protects against hypoglycemia. Triheptanoin increases the chemicals that can be made into glucose. Livers in minipigs store gluconeogenic precursors in the form of glutamate based on presence in the blood.

Alternative Metabolites

The metabolite PK profiles of alternative metabolites including pimelic acid, 3-hydroxypropionate and propionyl glycine are shown in FIGS. 16-18. Pimelic acid and 3-hydroxypropionate production was not increased with the Powder-ER when compared to the oil. The levels were lower for the powder. 3-hydroxypropionate is a normal acid made by gut bacteria, but very dependent on the individual animal.

We have increased co-enzyme A activity by giving triheptanoin. In addition, the levels of average energy and alternative metabolites are shown in FIGS. 19-21.

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Example 3

Multiple dose study was performed on the oil, the Powder and the Powder-ER as used in Example 2 and the results are shown below.

Powder Arm:

Day 1: Animal #6501 vomited ~1 hour post dose

Day 3: Animal #6502 vomited

Day 3: Animal #6510 gastric distressed observed

Oil Arm:

Day 1: Animal #6501 vomited ~3 hours post dose

Day 1: Animal #6516 vomited ~3 hours post dose

Day 5: Animal #6509 gastric distressed observed

Powder ER Arm:

No vomiting or gastric distress observed

Heptanoic Acid Across Dose Groups

Overall, results as expected, the triheptanoin releases heptanoic acid. Referring to FIG. 22, triheptanoin was metabolized primarily in liver to C7 fatty acids and ketone bodies which distribute via circulation to other tissues to provide an energy source. There were no triheptanoin values in oil arm. All triheptanoin oil was converted to C7.

Trace amounts of triheptanoin were observed in powder arm. The powder matrix may have caused a delay in the lipase action, this delay is slowing the release of C7. The Powder ER arm released slightly higher levels of heptanoic acid than the powder arm but not as high as the oil.

Energy Metabolites

As seen in FIGS. 23 and 24, beta hydroxy levels rose 10-fold above resting state while ketone bodies did not increase from Day 1 to Day 7, which indicated no chance for ketosis.

Referring to FIGS. 25 and 26, we observed glutamic acid and aspartic acid increased and the Powder had higher levels of both metabolites than the powder ER. Glutamic acid and aspartic acid are gluconeogenic amino acids, and can represent a novel source of energy.

FIGS. 27-29 shows PK profiles of additional metabolites: A) pimelic acid, B) 3-hydroxypropionate, and C) propionylglycine following dosing on Day 1 and Day 7. The oil had the highest amount of pimelic acid compared to the powders. 3-Hydroxypropionate was increasing in all 3 groups but the powder-ER had the smallest increase and the lowest levels. Propionylglycine was essentially zero for the oil and the powder-ER but the powder dosing caused an increase. Overall, the Powder ER produced the least amount of excess metabolites.

The PK profiles of all metabolites are shown in FIG. 30. The metabolites include Triheptanoin, C4 (Hydroxy and Keto), C5 (Hydroxy and Keto), Heptanoic acid, Pimelic acid, Glutaric acid, Aspartic acid, Glutamic acid, Alanine and 3-Hydroxypropionic acid.

The results of the powder and the powder-ER as compared to that of the oil are shown in FIGS. 31-40. The Powder ER raised the C5-hydroxy levels the most of all the treatments, which indicates it's the most effective at delivering fatty acids to the cells. We observed differences between Days 1 and Day 7 for the powders. Day 7 levels were lower than Day 1, spike at 240 min on Day 1, which indicates that PK sampling can be carried out longer. The Powder-ER dosing resulted in lowest levels of pimelic acid. The Powder had a higher release of aspartic acid than the Powder-ER. The Powder had higher levels of glutamic acid and alanine than the Powder-ER. The Powder also had higher levels of 3-Hydroxypropionate than the Powder-ER.

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Following long-standing patent law convention, the terms "a", "an", and "the" refer to "one or more" when used in this application, including the claims.

The use of the term "or" in the claims is used to mean "and/or" unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive. It is specifically contemplated that any listing of items using the term "or" means that any of those listed items may also be specifically excluded from the related embodiment.

Throughout this application, the term "about" is used to indicate that a value includes the standard deviation of error for the device or method being employed to determine the value.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which the present application belongs. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present application, representative methods and materials are herein described.

The patents listed herein describe the general skill in the art and are hereby incorporated by reference in their entireties for all purposes and to the same extent as if each was specifically and individually indicated to be incorporated by reference. In the case of any conflict between a cited reference and this specification, the specification shall control. In describing embodiments of the present application, specific terminology is employed for the sake of clarity. However, the invention is not intended to be limited to the specific terminology so selected. Nothing in this specification should be considered as limiting the scope of the present invention. All examples presented are representative and non-limiting. The above-described embodiments may be modified or varied, without departing from the invention, as appreciated by those skilled in the art in light of the above teachings.

What is claimed is:

1. A pharmaceutical composition comprising an active ingredient, diheptanoate, and hexano-diheptanoate, wherein the active ingredient is in the form of triheptanoin oil; wherein the triheptanoin oil has a purity greater than 98%; wherein the composition comprises 0.01% to 1.5% diheptanoate by weight; wherein the composition comprises 0.01% to 1.0% hexano-diheptanoate by weight; wherein glycerol is undetectable in the composition; wherein monoheptanoate is undetectable in the composition; wherein the composition comprises no more than 0.2% water by weight; and wherein the composition is a liquid composition for oral administration to a human.
2. The pharmaceutical composition of claim 1, wherein the composition comprises 0.05% to 0.7% diheptanoate by weight.
3. The pharmaceutical composition of claim 1, wherein the triheptanoin oil has a purity greater than 98.5%.
4. The pharmaceutical composition of claim 1, wherein the total amount of diheptanoate and hexano-diheptanoate in the composition is no more than 1.3% by weight.
5. The pharmaceutical composition of claim 1, wherein the composition has an assay value of the triheptanoin oil between 95% to 103% as measured by Ph. Eur. 2.2.28.

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